

THE GERMAN FLEET

By ARCHIBALD HURD

Author of "The Fleets at War," Etc.

The exact make-up of the German fleet, both ships and men, the policy of Von Tirpitz, the romantic rise of the German navy during the last few years, told completely and authoritatively. Of especial interest to America is the account, by a member of the German General Staff, of how Germany would invade America.

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FLEET ***

THE GERMAN FLEET

*BEING THE COMPANION VOLUME TO "THE FLEETS AT WAR"
AND "FROM HELIGOLAND TO KEELING ISLAND."*

BY

ARCHIBALD HURD

AUTHOR (JOINT) OF "GERMAN SEA-POWER, ITS RISE, PROGRESS
AND ECONOMIC BASIS."

HODDER AND STOUGHTON

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INTRODUCTION

In the history of nations there is probably no chapter more fascinating and arresting than that which records the rise and fall and subsequent resurrection of German sea-power.

In our insular pride, conscious of our glorious naval heritage, we are apt to forget that Germany had a maritime past, and that long before the German Empire existed the German people attained pre-eminence in oversea commerce and created for its protection fleets which exercised commanding influence in northern waters.

It is an error, therefore, to regard Germany as an up-start naval Power. The creation of her modern navy represented the revival of ancient hopes and aspirations. To those ambitions, in their unaggressive form, her neighbours would have taken little exception; Germany had become a great commercial Power with colonies overseas, and it was natural that she should desire to possess a navy corresponding to her growing maritime interests and the place which she had already won for herself in the sun.

The more closely the history of German sea-power is studied the more apparent it must become, that it was not so much Germany's Navy Acts, as the propaganda by which they were supported and the new and aggressive spirit which her naval organisation brought into maritime affairs that caused uneasiness throughout the world and eventually created that feeling of antagonism which found expression after the opening of war in August, 1914.

In the early part of 1913 I wrote, in collaboration with a friend who possessed intimate knowledge of the foundations and the strength of the German Empire, a history of the German naval movement,^[1] particular emphasis being laid on its economic basis. In the preparation of the present volume I have drawn upon this former work. It has been impossible, however, in the necessarily limited compass of one of the *Daily Telegraph* War Books, to deal with the economic basis upon which the German Navy has been created. I believe that the chapters in "German Sea-Power" with reference to this aspect of German progress—for which my collaborator was responsible and of which, therefore, I can speak without reserve—still

constitute a unique presentation of the condition of Germany on the eve of the outbreak of war.

Much misconception exists as to the staying power of Germany. The German Empire as an economic unit is not of mushroom growth. Those readers who are sufficiently interested in the subject of the basis of German vitality, will realise vividly by reference to "German Sea-Power" the deep and well-laid foundations upon which not only the German Navy, but the German Empire rest.

Whether this history should be regarded as the romance of the German Navy or the tragedy of the German Navy must for the present remain an open question. In everyday life many romances culminate in tragedy, and the course of events in the present war suggest that the time may be at hand when the German people will realise the series of errors committed by their rulers in the upbuilding of German sea-power. Within the past fifteen years it is calculated that about £300,000,000 has been spent in the maintenance and expansion of the German Fleet, the improvement of its bases, and the enlargement of the Kiel Canal. Much of this money has been raised by loans. Those loans are still unpaid; it was believed by a large section of the German people that Great Britain, hampered by party politics and effete in all warlike pursuits, would, after defeat, repay them. That hope must now be dead.

The German people, as the memorandum which accompanied the Navy Act of 1900 reveals, were led to anticipate that the Fleet, created by the sacrifice of so much treasure, would not only guarantee their shores against aggression, but would give absolute protection to their maritime and colonial interests, and would, eventually, pay for itself. The time will come when they will recognise that from the first they have been hoodwinked and deceived by those in authority over them. It may be that German statesmen, and the Emperor himself, were themselves deceived by the very brilliance of the dreams of world power which they entertained and by the conception which they had formed of the lack of virility, sagacity and prescience of those responsible for the fortunes of other countries, and of Great Britain in particular.

German Navy Acts were passed in full confidence that during the period when they were being carried into effect the rest of the world would stand

still, lost in admiration of Germany's culture and Germany's power. The mass of the German people were unwilling converts to the new gospel. They had to be convinced of the wisdom of the new policy. For this purpose a Press Bureau was established. Throughout the German States this organisation fostered, through the official and semi-official Press, feelings of antagonism and hatred towards other countries, and towards England and the United States especially, because these two countries were Germany's most serious rivals in the commercial markets of the world, and also possessed sea-power superior to her own.

It is interesting to recall in proof of this dual aim of German policy the remarks of von Edelsheim, a member of the German General Staff, in a pamphlet entitled "Operationen Ubersee."^[2]

The author, after first pointing out the possibility of invading England, turned his attention to the United States.^[3] His remarks are so interesting in view of the activity of German agents on the other side of the Atlantic after the outbreak of war, that it is perhaps excusable to quote at some length this explanation by a member of the German General Staff of how the German Fleet was to be used against the United States as an extension of the power of the huge German Army.

"The possibility must be taken into account that the fleet of the United States will at first not venture into battle, but that it will withdraw into fortified harbours, in order to wait for a favourable opportunity of achieving minor successes. Therefore it is clear that naval action alone will not be decisive against the United States, but that combined action of army and navy will be required. Considering the great extent of the United States, the conquest of the country by an army of invasion is not possible. But there is every reason to believe that victorious enterprises on the Atlantic coast, and the conquest of the most important arteries through which imports and exports pass, will create such an unbearable state of affairs in the whole country that the Government will readily offer acceptable conditions in order to obtain peace.

"If Germany begins preparing a fleet of transports and troops for landing purposes at the moment when the battle fleet steams out of our harbours, we may conclude that operations on the American soil can

begin after about four weeks, and it cannot be doubted that the United States will not be able to oppose to us within that time an army equivalent to our own.

"At present the regular army of the United States amounts to 65,000 men, of whom only 30,000 could be disposed of. Of these at least 10,000 are required for watching the Indian territories and for guarding the fortifications on the sea coast. Therefore only about 20,000 men of the regular army are ready for war. Besides, about 100,000 militia are in existence, of whom the larger part did not come up when they were called out during the last war. Lastly, the militia is not efficient; it is partly armed with muzzle-loaders, and its training is worse than its armament.

"As an operation by surprise against America is impossible, on account of the length of time during which transports are on the way, only the landing can be affected by surprise. Nevertheless, stress must be laid on the fact that the rapidity of the invasion will considerably facilitate victory against the United States, owing to the absence of methodical preparation for mobilization, owing to the inexperience of the personnel, and owing to the weakness of the regular army.

"In order to occupy permanently a considerable part of the United States, and to protect our lines of operation so as to enable us to fight successfully against all the forces which that country, in the course of time, can oppose to us, considerable forces would be required. Such an operation would be greatly hampered by the fact that it would require a second passage of the transport fleet in order to ship the necessary troops that long distance. However, it seems questionable whether it would be advantageous to occupy a great stretch of country for a considerable time. The Americans will not feel inclined to conclude peace because one or two provinces are occupied by an army of invasion, but because of the enormous, material losses which the whole country will suffer if the Atlantic harbour towns, in which the threads of the whole prosperity of the United States are concentrated, are torn away from them one after the other.

"Therefore the task of the fleet would be to undertake a series of large landing operations, through which we are able to take several of these

important and wealthy towns within a brief space of time. By interrupting their communications, by destroying all buildings serving the State commerce and the defence, by taking away all material for war and transport, and, lastly, by levying heavy contributions, we should be able to inflict damage on the United States.

"For such enterprises a smaller military force will suffice. Nevertheless, the American defence will find it difficult to undertake a successful enterprise against that kind of warfare. Though an extremely well-developed railway system enables them to concentrate troops within a short time on the different points on the coast, the concentration of the troops and the time which is lost until it is recognised which of the many threatened points of landing will really be utilised will, as a rule, make it possible for the army of invasion to carry out its operation with success under the co-operation of the fleet at the point chosen. The corps landed can either take the offensive against gathering hostile forces or withdraw to the transports in order to land at another place."

These declarations of German naval and military policy are of interest as illustrating the character of the propaganda by which the naval movement was encouraged. *The Navy was to give world-wide length of reach to the supreme German Army, and enable Germany to dictate peace to each and every nation, however distantly situated.* An appeal was made to the lowest instincts of the German people. They were counselled to create a great naval force on the understanding that the money expended would by aggressive wars be repaid with interest and that, as a result of combined naval and military operations, they would extend the world power of the German Empire, and incidentally promote Germany's maritime interests in all the oceans of the world.

Those who were responsible for the inflammatory speeches and articles by which the interest of the German people in the naval movement was excited, forgot the influence which these ebullitions would have upon the policy of other Powers and upon their defensive preparations. It was only after hostilities had broken out that the German people realised what small results all their sacrifices had produced. By the words, more than the acts, of those responsible for German naval policy, the other Powers of the world

had been forced to expand and reorganise their naval forces. Germany had at great cost won for herself the position of second greatest naval Power in the world, but in doing so she had unconsciously forced up the strength of the British Fleet and dragged in her path the United States, France, Italy, Japan, Russia, and to a limited extent, but only to a limited extent, her ally, Austria-Hungary. During the years of agitation the other Powers of the world had not stood still, as it was assumed in Germany they would do. First, the British people increased their naval expenditure and more ships were built and more officers and men were entered; and then the German Navy Act of 1912 was passed.

It had been the practice of the naval Powers to keep about one-half only of their ships in full sea-going commission. The armed peace, before Germany began to give expression to her maritime ambitions, was a yoke which rested easily upon the navies of the world. As a British naval officer has remarked:—

"Up to the end of the last century our Navy enjoyed a peace routine. We maintained squadrons all over the world, and the pick of our personnel was to be found anywhere but in home waters. The Mediterranean claimed the pick of both our ships and men. Here naval life was one long holiday. The routine was to lay in harbour for nine months out of the year. About July the whole fleet would congregate at Malta for the summer's cruise. Sometimes it would be east of Malta, taking in the Grecian Archipelago and the Holy Land; at others it would be west, visiting the French and Italian ports, paying a visit to the Rock, and then home to Malta for another long rest.

"Preparation for war was never thought of. Why should it be? The French Navy had no aggressive designs, and was much below our own, both in material strength and in personnel, while the Russian Navy was partly confined in the Black Sea, the other part being in the Baltic. And so we, both officers and men, set out to have a good time. Our ships were kept up to yacht-like perfection as regards their paintwork, while their bright work shone like gold, and the road to promotion lay not through professional efficiency, but the state of cleanliness and splendour of one's ship. All kinds of drills and evolutions were devised, not because of their war value, but because

they had a competitive value, and so ship could be pitted against ship and an element of sport introduced.

"There was nothing really wrong in all this. The British Navy was there to maintain for us our title of 'Mistress of the Seas,' and as no other nation apparently wished to challenge our title, there was nothing to do but pass away the time as pleasantly as possible; when the Navy was called on to perform any task it carried it through with vigour, valour, and efficiency, and immediately settled down again."^[4]

This regime came to an end soon after Grand Admiral von Tirpitz became German Naval Secretary towards the end of the nineteenth century. He set the navies of the world a new model. He determined to take advantage of the easy-going spirit which animated the pleasant relations then existing between the great fleets. There was to be nothing pleasant about the German Fleet. It was to be a strenuous agent of Germany's aggressive aims. In the organisation of German sea-power new principles found expression. In home waters and abroad the German Navy was always ready instantly for war. The screw was applied gradually stage by stage. Under the German Navy Act of 1912 this aggressive sea policy found its ultimate expression: it was proposed to keep always on a war footing nearly four-fifths of the ships in northern waters, while at the same time the squadrons abroad were to be greatly increased in strength. Happily, owing to Lord Fisher's foresight and strategical ability, the British Navy was enabled step by step to respond to each and every measure taken by Germany. He created for us a Grand Fleet and when hostilities broke out that fleet took up its war stations and denied to the main forces of Germany the use of any and every sea.

German policy operated as a tonic, though not to the same extent, on the other great fleets of the world. In the summer of 1914 Germany discovered that every anticipation upon which her foreign, naval and military policies had been based had been falsified by events. In particular, in adding to her strength at sea and on land, she had rendered herself weak by creating enemies east and west. Her navy, which was to have engaged in a victorious campaign against the greatest naval power of the world in isolation—the rest of the world watching the inevitable downfall of the Mistress of the Seas with approval—found arrayed against it not the British fleet only, but

the fleets of France and Russia in Europe and the Navy of Japan in the Far East.

In studying, therefore, the history of the naval development of Germany, and contrasting the high hopes which inspired the naval movement with the events which occurred on the outbreak of war, and in subsequent months, one is led to wonder whether, after all, the romance of the German Navy will not be regarded in the future, by the German people at least, rather as a great and costly tragedy.

FOOTNOTES:

[1] "German Sea-Power, Its Rise, Progress and Economic Basis," by Archibald Hurd and Henry Castle (1913, London, John Murray, 10s. 6d.).

[2] "Modern Germany" (Smith Elder, 1912).

[3] Germans always assumed that they could attack the United States without intervention on our part, just as they assumed that they could engage in war with us without becoming involved with the United States. They believed that Germany would fight both countries in turn—and victoriously.

[4] "The British Navy from Within" by "Ex-Royal Navy" (Hodder & Stoughton).

THE GERMAN FLEET

CHAPTER I

PAST ASCENDENCY

Like the foundations of the Empire in 1870, the formation of the modern German Fleet is the result of a movement that had its origin among the people and not among the Princes of the country. And this naval movement sprang up and reached its greatest vigour in those sea-board districts that still sedulously keep alive the splendid tradition of the Hanseatic League, which, as the strongest maritime Power of its day, for centuries almost monopolized the trade of Northern and Western Europe, and with the word "sterling," a corruption of "Easterling," the name popularly given to its

members, has left on Great Britain the indelible stamp of its former mercantile domination. For the coin of the Hanse towns, by reason of its unimpeachable quality, was once universally sought after in England, and thus became the standard of monetary excellence.

The memories of the Hansa are the "historical foundation" on which have been based Germany's claims to a leading place among the maritime nations, and they have played a prominent part in every agitation for the increase of her fleet. Why, it was asked, should she not again assume upon the seas that dominating position which she once undoubtedly held? Why, with her expanding population, trade, and wealth, should she not reclaim that maritime ascendancy which she forfeited to Holland in the seventeenth century, and which a hundred years later passed to Great Britain? Why should she not realize that dream which was in the mind of Friedrich List when he wrote: "How easy it would have been for the Hanse towns, in the epoch of their rule over the sea, to attain national unity through the instrumentality of the imperial power, to unite the whole littoral from Dunkirk to Riga under one nationality, and thus to win and maintain for the German nation supremacy in industry, trade, and sea-power!"

It is, moreover, not without significance that the Hansa itself was, in a sense, democratic, and that, at a time when Germany, as a national unit, was rendered impotent in the world by her superabundance of Princes, her citizens were able, on their own initiative, and by their own energies, to assert their power and capacity as a maritime people.

The story of the Hansa is full of strange anomalies and antitheses. Historians differ by centuries as to the date at which the existence of the League commenced, and just as it never had a definite beginning, so it has never had a formal end, for to this day two of the Hanse towns—Hamburg and Bremen—have certain institutions in common, such as their supreme law courts and their diplomatic representation in Prussia. For hundreds of years the Confederation acted, and was treated by foreign Governments, as an independent State and a great Power, but its composition was never certain and always fluctuating. From first to last the names of no fewer than ninety cities and towns were entered upon its rolls, but it is impossible to say of each of them how often and when it joined or left the League. Foreign rulers, and especially the English monarchs, made repeated attempts to obtain from the Hansa an official list of its members, but

compliance with their demands was systematically evaded on one pretext or another. The League's policy was, as far as possible, to assert the claims of its members, and to disown responsibility for those made against them. This policy is pretty clearly expressed in the following answer returned by the League in 1473 to complaints put forward on behalf of English merchantmen who had suffered through the depredations of the Dantzic privateer or pirate, Paul Beneke: "The towns of the Hansa are a corpus in the possession of the privileges they hold in any realms, lands, or lordships, and when their privileges are infringed, they are accustomed to meet and consult, and then to issue for all of them ordinances against all goods from the countries in which their privileges have been infringed, that they shall not be suffered in the commonalty of towns. But they were not making war against England; only some of the towns of the Hansa, which had been injured by England, had determined upon it at their own venture, win or lose, which did not take place in the name of the Hanse commonalty." The theory of the Federation was, in fact, that it existed for the purpose only of taking, and not of giving, and it refused to imply a corporate responsibility by publishing its membership rolls.

It is impossible, in the space available, to tell in any detail the fascinating story of the rise of the Hansa to the position of a great power, with its guild halls and factories in foreign lands, of which the oldest and most important was the Steelyard, in London. The history of this institution is believed to go back to the latter days of the Roman occupation. When the Hanseatic League was at the height of its power—from the last quarter of the fourteenth to the first half of the sixteenth centuries, the Steelyard, in London, closely resembled a state within a larger state. It occupied a site now covered by Cannon Street Station, extending from Thames Street to the river, and bounded to the east and west respectively by All Hallows and Cousins Lane. The Steelyard had something of the appearance of a fortress and was stoutly defended against attack. The community within its precincts was governed with monastic severity. Their affairs were administered by an alderman with the assistance of two adjuncts and nine counsellors who took part in all the State and civic pageants of London as a Corporation.

This great German commercial institution on British soil, and the other houses established in other countries, reflected the great power which was

wielded by the Hanseatic League in commerce. These German traders, however, realised that their increasing trade on the seas required adequate defence. Mainly at the instigation of the merchants of Lübeck, a considerable navy was created, this German city being dependent for its prosperity mainly upon the herring fishing and curing industries of Europe. In process of time the Germans succeeded in driving away English, French and Spanish rivals, and created a great monopoly of the herring fisheries of northern Europe, from which they drew immense wealth and on which depended a number of other industries.

It was mainly for the protection of the Sound herrings that the Hansa undertook against the Scandinavian States the numerous campaigns by which it won the keys of the Baltic. The war which culminated with the peace of Spralsunde in 1370 raised the League to the rank of a first-class sea Power. Encouraged by its success in crushing and humiliating Denmark, the Hansa had little hesitation in measuring itself against England. The towns became associated through the Victualling Brothers with an active form of corsair warfare on English shipping.

By its triumph over the Danes, the Hansa secured a practical monopoly of the shipping and trade of the Baltic and North Sea, which it held almost unimpaired for nearly two hundred years. In the words of Gustav Wasa, "the three good (Scandinavian) Crowns remained small wares of the Hansa up to the sixteenth century," and as long as this was so the commercial and maritime supremacy of the League was practically unchallengeable. The manner in which the Easterlings availed themselves of the ascendancy they had now acquired is a classic example of the ruthless and unscrupulous exploitation of political power for the purposes of purely material gain, for they were actuated by no national or ideal aims, but solely by the desire to enrich themselves. Favoured by the confusion and chaos prevailing in the lands of their potential rivals, they became the exclusive brokers through whose mediation the spices of the Orient, the wines of France, the cloth of Flanders, the tin, wool, hides, and tallow of England, were exchanged for the dried cod of Norway, the ores of Sweden, the wheat of Prussia, the honey and wax of Poland, the furs of Russia, and the myriads of herrings which every summer were caught in the Sound, and salted and packed on the coast of Scania. What they aimed at, and what for long years they substantially obtained, was the disappearance of all flags but their own from

the North Sea and the Baltic. Moreover, a great part of the carrying trade between England and France also fell to their lot.

The conditions were such as rendered warlike operations between England and the Teutonic order inevitable. It is impossible to trace in any detail the guerilla tactics which were adopted on both sides. It is only necessary for our present purpose to convey some idea of the sea power which the Hansa exercised in order that we may better understand the ambitions of Germany to which the Emperor William the Second and Grand Admiral von Tirpitz gave expression in the early years of the twentieth century. At the outset of its career, its warships were manned by the burghers themselves, but as the fleet increased in size—it was quadrupled during the first half of the fifteenth century—recourse to mercenaries became more and more general. The commanders of the ships were invariably citizens of the towns which had equipped them, and were frequently members of the governing council, while the admiral of a fleet was always a councillor, and usually a burgomaster. The officers of the land forces, which were raised as occasion demanded, were principally drawn from the impoverished nobility, whose members welcomed any opportunity of repairing their shattered fortunes by martial adventure. Of the naval resources of the League, some idea can be formed from the fact that, in the war against the Scandinavian Kingdoms in 1426, it sent out a fleet of 260 ships, manned by 12,000 sailors and fighting men. For the exhausting, if not inglorious, seven years' war against Gustav Wasa's successor, Lübeck alone fitted out 18 men-of-war, of which one, the *Adler*, carried 400 sailors, 500 fighting men, and 150 "constables." Her armament consisted of 8 carthouns, 6 demi-carthouns, 26 culverins, and many smaller pieces of ordnance. Among her munitions were 6,000 cannon-balls and 300 hundredweight of powder.

CHAPTER II

THE FIRST GERMAN FLEET

In one of the window niches on the ground floor of the Military Museum (Zeughaus) at Berlin lies an old and dilapidated 8-pounder gun. In its deep

and disfiguring coat of rust it is an inconspicuous object, and, amid that rich and varied collection of artillery from all the ages, the eye of the casual visitor will not rest upon it for more than a disparaging moment. And yet few of the treasures of the museum have a more interesting history to tell, for it is the sole remaining relic of the first serious experiment in naval and colonial policy ever made by a German ruler. On an elevation rising from the beach of Cape Three Points, on the Gold Coast, now British territory, are still to be seen the crumbling ruins of the fort of Gross-Friedrichsburg, built there by the Elector of Brandenburg in 1681, and when the German corvette *Sophie* visited the spot, with pious purpose, in 1884, this corroded gun was unearthed from beneath the weeds and brushwood that have overgrown the decayed ramparts.

Frederick William, the Great Elector, has been exemplary for many of his successors. Frederick the Great rightly considered him the most able of the previous Princes of the house of Hohenzollern, while the present German Emperor has made a special cult of his memory, and assuredly had a symbolic intention when he appeared at a fancy-dress ball disguised as the first of his ancestors who equipped a fleet and founded a colony.

When Frederick William was called to the Brandenburg throne in 1641 at the age of twenty, Germany was still in the throes of the Thirty Years' War, and no part of the Empire had suffered more than his Electorate from the consequences of that unspeakable calamity. Of all the causes which have contributed to impede the normal development of the painstaking and industrious German race, none had so malign an influence as that stupendous conflict. It not merely delayed civilization, but over vast tracts of country positively exterminated it. At the close of the war many once flourishing towns had absolutely disappeared from the face of the earth, and where formerly a numerous peasantry had tilled its fertile fields a howling wilderness extended in all directions as far as the eye could reach. In North Germany to-day an apparently purposeless pond, or a detached clump of venerable trees, still shows where once a village stood, and bears mute witness to the ruthless barbarity with which the religious partition of Central Europe was brought about.

When an end was put to the bloodshed and rapine by the Peace of Westphalia (1648), the population of Germany had been reduced to one half—in some districts to one tenth—of its former dimensions. Many portions

of the Empire are even to-day not so thickly inhabited as they were before the war. Industry and commerce had migrated to England, France, and Holland; and Leipzig and Frankfort were the only German towns that had retained any trade worthy of mention. The Hansa, with its fleets of warships and merchantmen, was but a memory of the past. Königsberg had no longer a ship of its own; the trade of Dantzic and Stettin was almost entirely carried in foreign bottoms; and even Hamburg, which directly had been but comparatively little touched by the thirty years of chaos and turmoil, and had benefited from its exceptional connection with England, was left commercially crippled. At a Hanse Parliament held in 1630, only Hamburg, Lübeck, and Bremen were represented. Germany had been so drained of money that barter had generally taken the place of purchase by coin; wages were paid in the products of labour, grain, ore, and manufactured goods, and even state officials in some cases received their salaries in kind.

Even before the war broke out, Brandenburg, a country of barren soil and few natural resources, had stood far below the rest of Germany both materially and intellectually. In 1600 the twin towns, Berlin and Cöln, which faced one another from opposite banks of the Spree, and have since been merged to form the colossal capital of the new Empire, contained together no more than 14,000 souls. Brandenburg and Frankfort-on-Oder each had a population of 10,000. Only two other towns, Stendal and Salzwedel, could boast more than 5,000 inhabitants. And it was of the mere ruins of this country that Frederick William formed the foundation-stone of the Prussian Kingdom and of the German Empire of to-day.

If the Thirty Years' War had produced any form of national consolidation, if it had increased the authority of the Empire or resulted in the absorption of the smaller States by the larger, that would at least have been some compensation to Germany for its long and terrible ordeal. But exactly the opposite was the case. The war ceased simply because no one had the will or the strength to continue it, and a miserable compromise was the result. The only gainers were the Princes, who, as the wielders of the armed forces, had been able to enhance their power, and now acquired a larger measure of independence in their relationships to the Emperor. Their number remained legion. In the Germany mapped out by the Westphalian negotiators there were eight electors, sixty-nine spiritual and ninety-six temporal Princes,

sixty-one imperial towns, and a multitude of Counts and Barons exercising various degrees of sovereign power.

Frederick William's claim to the title "Great," which was bestowed upon him by his own generation, has been contested, but may be allowed to pass. As military leader, diplomatist, organizer, and administrator, he certainly had unusual gifts. Above all, he excelled in duplicity and treachery. The most eminent living German historian has said of him that "both in internal and external politics he acted with an unscrupulousness so manifest that it cannot be palliated," and can find no better excuse for his many deeds of "faithlessness" and "double-dealing" than that, in this respect, he was merely "the master of the diplomatic art of his day." The Elector was actuated solely by his own personal and dynastic interests, and was utterly devoid of "German" patriotism, for in return for the liberal subsidies on which he prospered, he undertook, in a secret treaty, to support the candidature of the French King or Dauphin for the Imperial German throne, and he was mainly responsible for the truce which left Strasburg in French hands for nearly two centuries. During the incessant wars which filled up most of his reign he fought both with and against every other belligerent. His sword was always at the disposal of the highest bidder, either of hard cash or of territorial extension, and by adroit choice of the moment for changing sides he generally made a profitable bargain. True, he was obliged to restore the western portion of Pomerania which he had conquered from the Swedes, but he obtained a much more important acquisition—the recognition of his full sovereignty in what is now East Prussia.

That region had been wrested from the Slavs by the German orders of chivalry, founded at the time of the Crusades, and had subsequently become an evangelical duchy, ruled by a junior branch of the house of Hohenzollern, as a fief of the Kingdom of Poland. On the extinction of the ducal line, it had reverted to the rulers of Brandenburg, and by a timely sale of his military assistance, first to the Swedes and then to the Poles, the Great Elector induced both to admit his unrestricted and unqualified rights of sovereignty in the duchy. His successor persuaded the Emperor to agree to his assumption of the kingly title for this territory, and it is an interesting fact—especially in view of the last development of the German Empire, which in its present constitutional form and in much else is dependent upon Catholic support—that this elevation was largely brought about by the

intervention of two Jesuit fathers. It was from the Kingdom of Prussia which was thus established, and which was a completely independent State altogether outside the competencies of the Holy Roman Empire, that arose the Hohenzollern ascendancy in Germany, and round it that the new German Empire crystallized. For this reason the episode is quite germane to our present purpose.

The Germans excel as diligent pupils and patient imitators, and the Great Elector was no exception to this rule. From his fourteenth to his eighteenth year he had been educated under the care of Frederick Henry, the Statthalter of Holland, then the chief Sea-Power of the world, from whom he had imbibed many ideas as to the importance of navies, colonies, and sea-borne trade. His connection with the Netherlands was maintained and strengthened by his marriage with an Orange Princess, the aunt of William III. of England, and many Dutchmen entered his service. Among them was an ex-admiral, Gijsels by name, who assiduously kept alive the dreams of sea-power which the Elector had brought back with him from Holland. It was on his prompting that, in 1659, when Frederick William was embroiled with the Swedes, and found his operations hampered by the lack of a fleet, an enquiry as to the possibility of remedying this deficiency was ordered by the Elector. The investigation resulted, for the time being, only in the compilation of a memorandum as to a "Brandenburg-Imperial admiralty," and some fruitless attempts to obtain ships in the Netherlands.

But Gijsels' projects went far beyond a mere fleet. All the world was then discussing the colonizing activity of the western European States, and Frederick William's predecessor on the Electoral throne had conceived abortive plans for founding an East Indian trading company. What the ex-admiral proposed to the Elector in 1660 was, that Brandenburg, Austria, and Spain should combine for the purpose of securing a colonial ascendancy, which was to be arrived at by playing off England, France, and Holland against one another. Negotiations to this end seem actually to have been commenced, but they broke down over the jealous suspicions of the diplomatists approached, and the perpetual turning of the European kaleidoscope.

During the next fifteen years the idea of a Brandenburg navy appears to have been allowed to sleep. In the meantime a very remarkable book had been published, which should be mentioned here because it contains the

essential elements of the programme of the most modern naval agitation in Germany. The author was Johann Becher, by profession a chemist, but in his leisure a political seer of the type of Friedrich List, whose great forerunner he was. His work, "Political Discourse on the Causes of the Rise and Decline of Towns and Countries," was published in 1667. Becher had travelled much, and he wrote:

"In Germany there is hardly any longer trade or commerce; all business is going to ruin; no money is to be found with either great or small; on the other hand look at Holland, how rich she is and how she grows richer every day; that could not be if she feared the sea as much as our nation of High Germany."

Becher then addressed to his countrymen the following impassioned exhortation:

"Up, then, brave German; act so that on the map, besides New Spain, New France, New England, there shall in the future be found also New Germany. You are as little lacking as other nations in the intelligence and resolution to do such things; yea, you have all that is necessary; you are soldiers and peasants, alert, laborious, diligent, and indefatigable."

Becher had held positions at various German Courts, and it is not improbable that his appeal fell upon sympathetic ears among the entourage of the Great Elector. But however that may be, the war of Denmark and Brandenburg against Sweden, which broke out in 1675, did actually, for the first time in history, witness a fleet at the disposal of a member of the dynasty that now occupies the imperial throne in Germany. True, it was not yet the actual property of the Elector, but of Benjamin Raule, an enterprising Dutch merchant, who had migrated to Denmark, and now laid a naval project before the Brandenburg sovereign. His proposals were readily acceded to, and he received permission to fit out a flotilla of two frigates and ten smaller vessels, and to operate with them under the Brandenburg flag against the Swedes. The Elector merely stipulated that he should receive 6 per cent. of the value of all prizes captured. Raule's vessels rendered substantial service in the capture of Stettin, and of that much-

coveted strip of the Pomeranian coast which was so essential to the realisation of Frederick William's maritime aspirations.

The Elector's hopes were disappointed by the Treaty of St. Germain, under which he was compelled to restore this precious booty to the intrusive Scandinavians, but in the meantime his naval plans had taken a wider scope in fresh contracts with the resourceful Dutchman. In the first of these, Raule undertook, for a monthly subsidy of 5,000 thalers,^[5] to maintain a fleet of eight frigates and a fire-ship, mounting altogether 182 guns. Shortly afterwards the terms of the agreement were extended, and at the commencement of the year 1680, twenty-eight ships of war, with a total of 502 guns, were flying the red eagle of Brandenburg.

Though robbed by the peace of the coast-line and seaports on which he had counted as the base of his maritime power and the recruiting ground for his fleet, the Elector did not allow himself to be discouraged, and he very soon found fresh work for his little flotilla to do. The greatest master of German mercenaries at that date, he had, a few years previously, hired a portion of his army to Spain for use against the French. As repeated applications for the price of this support had proved unavailing, he now determined to collect the debt, which amounted to 1,800,000 thalers, by forcible distraint.

Accordingly six ships, which were followed at an interval of some months by three others, were sent out to attempt to intercept the silver fleet on its way to the Spanish Netherlands. The vessels were almost without exception commanded by Dutchmen, but were mainly manned by Germans, though the crews included many English, Dutch, Danish and Norwegian sailors. Naturally the soldiers carried on board were drawn from the Brandenburg army; and orders were given that they should be trained in ship's work "because we are disposed to use the same permanently for the navy."

Though the flotilla did not fulfil either its immediate or its ultimate purpose, the expedition was notable for two reasons. In the first place, a large Spanish warship, the *Carolus Secundus*, with a valuable cargo of lace on board, was captured, and so became the first war vessel that was actually the property of a Hohenzollern State. In the second place, the quest of the Spanish silver resulted in a sea-fight, which, in respect both of the force engaged and the losses sustained, still heads the record of naval warfare under a Hohenzollern flag.

A detachment of four ships, cruising in the neighbourhood of Cape St. Vincent, sighted a fleet of a dozen Spanish frigates, which had put out for the special purpose of chasing the Germans from the sea. The Brandenburg commander, thinking that this was the anxiously-expected silver flotilla, bore down upon it, and did not realise his mistake till it was too late to avoid something of a conflict. Before he could succeed in man[oe]uvring his ships out of range of his overwhelmingly superior enemy, he had lost ten men killed and thirty wounded; and since that day Germany had fought no more terrible battle on the sea until the war broke out in 1914.

Another section of the Elector's fleet cruised for several months in West Indian waters without achieving much result, while the retaliatory measures adopted by the Spaniards secured a safe passage for the silver ships and rendered it prudent for Frederick William to abandon his daring and risky enterprise.

Meanwhile the Elector had allotted his infant navy a task of a different character. Soon after entering the service of Brandenburg, Raule had drawn up plans of colonization, and in the same year in which the fruitless search for the silver convoy began, he obtained permission to try his luck on the Gold Coast, and got together a syndicate to finance the undertaking. The Elector was wary, and declined to risk pecuniary participation, but he ordered that "twenty good healthy musketeers, together with two non-commissioned officers," should be placed under Raule's command. One of the principal objects of the expedition was to secure a share in the profitable trade in slaves which was then carried on between the West Coast of Africa and North America, but modern German historians for the most part ignore this feature of the enterprise.

The two vessels despatched on this errand reached the Gold Coast in safety, but aroused the resentment of the Dutch already settled there, who confiscated one of them, and compelled the other to quit African waters. However, the leader of the expedition had by that time managed to conclude what served the purposes of a treaty with certain native chiefs, who thereby placed themselves under the suzerainty of the Elector, and consented to the erection of a fort in the district under their control.

On the strength of this questionable document, an "African Company" for the "improvement of shipping and commerce wherein the best prosperity of

a country consists," was called into existence in the year 1682. In the charter of incorporation, the Elector promised to protect the Company against "all and everyone who may undertake to trouble, incommode, or to any extent injure the same in its actions in free places on the coasts of Guinea and Angola"; but both the naval and the military commanders were charged to keep at a respectful distance from "all Dutch Company fortresses, as well as those of other potentates, such as England, France, Denmark, etc." The capital of the Company was the modest sum of 50,000 thalers. Of this Frederick William contributed only 8,000, and the Electoral Prince 2,000 thalers, while almost half of the total was supplied by Raule, who had by now become "Director-General of the Brandenburg Navy."

The two frigates in which the second Gold Coast expedition shipped cast anchor off Cape Three Points on December 27th, 1682, but some difficulty was experienced in finding the chiefs who had "signed" the provisional treaty and who were each to have received a ratification engrossed in letters of gold, "a silver-gilt cup, and a portrait of his Electoral Highness." Frederick William had also issued instructions that his black allies and their wives were to be entertained on board the warships.

After a great deal of trouble, some other chieftains of the "Moors," as they are called in the official correspondence relating to this matter, were hunted out and induced to contract a second and definite treaty; and on January 1st, 1683, with due ceremony and much beating of drums, blowing of trumpets, and firing of guns, the Brandenburg flag was hoisted over "the first German colony." The flagstaff had been planted on a little eminence, which was subsequently, with all speed, transformed into the fort Gross-Friedrichsburg, and no doubt the rusty cannon now in the Zeughaus at Berlin is one of the half-dozen which had been mounted on the hill on the previous day in preparation for the great occasion.

In the following year the headquarters of the African Company was removed from Pillau to Emden. This latter town was not situated on Brandenburg soil, and the manner in which the Elector secured a footing in it is both instructive and characteristic of his easy methods of intervening and making a good bargain wherever an opportunity presented itself. It chanced that at that time the Estates of East Frisia were at loggerheads with their ruler, and they appealed to Frederick William for assistance. Nothing loth, he landed a force by night, and by a surprise attack seized the castle of

Greetsiel, which thus became his naval base. By an agreement with the town of Emden he subsequently acquired the right to station within its walls a "compagnie de marine" for the service of the African Corporation. This force, which was gradually increased to three, and temporarily to four, companies, and ultimately received the name of the "Marine Battalion," was drawn upon to man both the ships and the forts in Africa.

The transfer to Emden brought other advantages besides an ice-free port, a base on the North Sea, and an abbreviation of the route to Gross-Friedrichsburg, for the East Frisian Estates and the Elector of Cologne were both persuaded to invest largely in the African Company in consequence of the change.

In the year of the Emden agreement, the Brandenburg Navy was formally founded by the establishment of an "Admiralty" at Berlin. The Cabinet order by which this institution was created shows that the fleet then in full possession of the State comprised 10 ships, with 240 guns, while Raule was still under contract to provide 17 further vessels. The permanent personnel consisted of 1 vice-commodore, 5 naval captains, 3 officers of Marines, 12 mates, and 120 seamen. In 1686, the Elector took the Company entirely into his own hands, and simultaneously acquired a station on the island of St. Thomas, in the West Indies, as a place of call for the ships engaged in the slave traffic. He had also at that time made preparations for forming an East Indian trading company (at a much earlier date he had unsuccessfully attempted to acquire Tranquebar, on the Coromandel Coast, from the Danes) and for fitting out an expedition to China and Japan. These schemes, however, came to nothing.

The settlement at Cape Three Points had by no means an easy existence. Fever made fearful ravages among the garrison, which, when the first reliefs arrived, after an interval of nearly a year and threequarters, had been reduced by sickness from ninety to sixteen men. Everything that was needed for the construction of the fort, even building-stone, had to be brought thousands of miles across the sea from Germany. The Dutch traders in the neighbourhood had at once raised objections to the new colony, and, as their protests were unheeded, stirred up the natives against its members. It was only after prolonged negotiations at The Hague that the Elector secured a full recognition of his right to the settlement. And none the less the Dutch West India Company continued to harass the German colonists,

appropriating their ships, and turning them out of a couple of subsidiary fortifications which they had erected at other points along the coast. Gross-Friedrichsburg and Taccroma, another of the four Brandenburg stations on the Guinea littoral, for several years maintained themselves only by the menace of their guns. These untoward events are believed to have preyed upon the mind of the Great Elector, and to have hastened his end. At the time of his death, in April, 1688, Brandenburg and Holland were on the brink of war over the Gold Coast affair.

His successor on the Electoral throne in one very important respect reaped what Frederick William had sown, for he obtained the title of King of Prussia, by virtue of which, far more than from any specifically imperial prerogatives, William II. holds his present power in Germany. Frederick I. was a vain man, who was more interested in appearances than in realities, and cared more for the pomp and ceremonies of Court life than for the solid business of colonisation and slave-trading. As a source of revenue, with which to defray the cost of his empty extravagances, the African undertaking was feebly encouraged to continue its work; but, deprived of the directing brain and the stimulating enthusiasm of its founder, it soon sickened and languished. Accada and Taccarary, the two settlements which had been seized by the Dutch, were delivered up after a lengthy squabble, but the fortifications of the latter had been destroyed, and they were not rebuilt.

At first the trade of the colony, which had called into existence a flourishing shipyard at Havelberg, near the junction of the Navel and the Elbe, was fairly satisfactory, and the spirit of the Brandenburg Navy was raised by the successful operations of a couple of its frigates against French merchantmen, but in 1697 the Company fell upon evil days. It suffered pecuniary loss, both through the capture of some of its ships by the French and through the peculations of several officials, whose multiple dishonesty hints at a scandalous laxity of control. The invaluable Raule, too, fell into disfavour, and spent four years in gaol, though he was reinstated in his position on being liberated. At last the Company was no longer able to send out ships of its own, and for eight years, during the War of the Spanish Succession, the garrison of Gross-Friedrichsburg was left entirely to itself. For a considerable portion of that time five large Brandenburg ships of war were rotting in the harbours of Emden and Hamburg, when they might have

been much more profitably employed in attempting to keep up communications with the perishing colonists. When at last reliefs reached Gross-Friedrichsburg only seven men out of an original force of 1,700 were fit for duty.

What little credit attaches to the last days of the first German colony is the due of Jan Cuny, a native chief, who had placed himself under Brandenburg protection, apparently for the purpose of obtaining support against the English and Dutch settlements of the vicinity, with both of which he was at feud. It is characteristic of the period that, while Prussians were fighting shoulder to shoulder with English and Dutch on the continent of Europe, they were in open conflict with them on the West Coast of Africa. Frederick I. at one time thought it necessary to protest, through his Minister at London, against the difficulties which the English were causing him on the Gold Coast.

All the trouble seems to have arisen out of the demand made by a Dutch official at Axim for the surrender of a female relative of Cuny whom he claimed as a slave. Jan was evidently a man of considerable parts. He led his army with great discretion and resourcefulness, and no doubt the Prussians at Gross-Friedrichsburg thought it to their advantage to be on good terms with so formidable a warrior, especially as he was the sworn foe of their jealous European neighbours. At any rate, the relations between Cuny and the fort became both cordial and confiding, and when the last Governor of Gross-Friedrichsburg, Du Bois, discouraged by the indifference and neglect of the home authorities, sailed for Emden to enter remonstrances, he entrusted the protection of the colony to his black ally.

Du Bois arrived in Europe only to find that the doom of Gross-Friedrichsburg was already irrevocably sealed. The parsimonious Frederick William I., the father of Frederick the Great, had ascended the Prussian throne, and his careful mind, completely absorbed by plans of immediate economy, was incapable of taking such flights into the distance and the future as were necessary for the appreciation of the value of colonial policy. The African settlements had been doing badly and had become unremunerative, and his only thought was to dispose of them as speedily as possible for hard cash, which could be either hoarded or spent on his solitary extravagance—seven-foot grenadiers. Immediately after his accession, he instructed his representative in London that he was prepared

to "transfer his forts on the coast of Guinea to anyone else upon easy conditions." He was not long in finding a purchaser in that very Dutch West India Company which had from the outset been a thorn in the side of the Great Elector's colonial enterprise. On November 22nd, 1717, Gross-Friedrichsburg and its dependent territory passed from Hohenzollern rule for the sum of 6,000 ducats and twelve negro boys, of whom it was stipulated that six should be adorned with golden chains.

The signing of the contract and its execution were, however, two very different things. The redoubtable Jan Cuny had not been reckoned with, and when two Dutch vessels arrived to take over the fort they found him in possession and flying the Prussian flag. The order for the transfer of the fort was shown to his emissaries, who, after a good deal of delay, were sent on board the ships, but this he flatly refused to recognise, declaring that he would yield up his trust only to a vessel belonging to the King of Prussia. The commander of the Dutch expedition, Captain van der Hoeven, thought he would make short work of this insolent chieftain, and landed a body of fifty men to take the fort by storm. But Cuny once again showed the generalship which had raised him to the eminence of a Prussian deputy-governor. A force of 1,800 natives fusilladed the landing party from an ambushade and killed nearly every one of them. Hoeven was only able to save himself by swimming back to his ship, with three bullets in his body, and retired to the nearest Dutch settlement to excogitate a fresh plan of campaign.

Cuny, however, was flushed by his success, and not at all inclined to give up the prestige which he derived from a fortress bristling with guns and well furnished with small arms and ammunition. For seven long years he held out, repulsing the repeated attacks of the Dutch, and it was only when his supplies were exhausted and an overwhelming force had been put into the field against him, that he withdrew from his defences and vanished into the jungle from which he had come.

Simultaneously with Gross-Friedrichsburg, there was transferred from the Prussian King to the Dutch Company yet another African colony, of which mention has yet to be made. This was the island of Arguin, which lies off the coast of what is now French territory to the south of Cape Blanco, and in some maps is given the ominous name of Agadir. The islet, which was one of the principal centres of the gum trade, had been first occupied by the

Portuguese in 1441, but had passed by conquest to Holland, and from the latter to France. After the peace of Nymegen, in 1678, however, the French Senegal Company found itself unable to maintain a garrison in Arguin, and obtained permission from Louis XIV. to blow up the fort which had been erected there. The island then fell into the hands of the native ruler of Arguin, on the mainland, and remained subject to him till two ships of the Great Elector appeared off its coasts in October, 1685.

On the strength of a treaty concluded by the commander of the expedition with the King of Arguin, Frederick William seems to have claimed jurisdiction right along the coast of Africa from the Canary Isles to the Senegal River. These pretensions were not allowed to pass undisputed, and, towards the end of 1687, a couple of French vessels appeared off the fort and demanded its evacuation by the Germans. As this was refused they made an attempt to seize it by force, but, meeting with a stubborn resistance, abandoned the attack, and, after an unsuccessful endeavour to assert their rights during the peace negotiations at Ryswick, the French seemed to reconcile themselves to the new situation, for they even proposed commercial co-operation with the occupants of the Arguin fort.

After the death of the Great Elector, Arguin suffered, like Gross-Friedrichsburg, through the indifference of his successor, and the difficulty of communication arising from the War of the Spanish Succession. When a relief ship arrived in 1714, it found that the Governor had been captured by the natives, with whom he had quarrelled; and the remnant of the Arguin garrison was in so deplorable a condition, that "in a few days they must have perished of hunger."

The transfer of Arguin to the Dutch proved as difficult as that of Gross-Friedrichsburg. In 1717 the French had renewed their claims to the island, and, a few years later, the Senegal Company, landing 700 men and heavy guns, laid siege to the fort. After holding out for a few weeks, the commander, Jan Wynen, a Dutchman, withdrew secretly by night with his force in order to escape the humiliation of a formal surrender, and when its new owners at last arrived to take possession of it the colony was actually in French hands. It was in both cases a foreigner who last kept the flag flying over what were to be the only German colonies established till the final quarter of the nineteenth century. With the colonies disappeared the

force with which they had been won, the fleet, and it too had to wait long, though not quite so long, before it experienced a revival.

It is interesting to reflect how the history of the world might have been changed if the Great Elector's two immediate successors had united to his far-reaching schemes of "world-policy" his determination in carrying them out, and had bequeathed to the greater Frederick prosperous colonial possessions and a formidable navy. As it was, the naval episodes of the reign of this gifted monarch only show how pitifully and completely the dawning sea-power of his grandfather had passed away.

In the Seven Years' War, the shores of Prussia were continually ravaged by Swedish frigates, and as nothing could be effected by the armed fishing boats and coasting vessels which were all that could be pitted against them, Field-Marshal Lehwald, to whom the protection of that part of Prussia had been entrusted, appealed for help to the corporation of merchants at Stettin. That body responded with energy and promptitude, and, with great haste, a flotilla of four galliots, four large fishing boats, and four coasting vessels were transformed into "ships of war." In August, 1759, this improvised fleet ventured out of the Oder to attack the Swedes, but it was so completely overthrown after several days' fighting that the experiment was never repeated.

In the meanwhile Frederick had been inveigled into another maritime adventure, which was to prove just as barren of positive results. Early in the war several Englishmen communicated to the King their readiness to fit out privateers to prey on the commerce of Austria and Sweden, both of which countries had seized Prussian merchantmen. They protested in all cases that their principal motive was a desire to serve the cause of a monarch whom they admired and revered, and who was, as a matter of fact, at that time the ally of England. But at the same time they promised him "prodigious profits" from the enterprise, and it was admittedly the latter consideration which induced the King to listen to their proposals. Though his own Ministers expressed strong doubts, and the English Government urged that he would run the risk of embroiling himself with neutral States, he issued a number of letters of marque. The advice which had been given him proved to have been only too well founded. Not only were there no "prodigious profits," but the blunders of the royal officials and the indiscretions of the

ships under his flag involved the King in voluminous diplomatic correspondence and long and fruitless litigation.

To accelerate the process of destroying the enemy's trade, a number of blank letters of marque, ministerially signed and stamped with the royal seal, were sent out to the Prussian Minister in London, and he somewhat imprudently lent a couple of these to an interesting adventurer, named Erskine Douglas, who said that he wished to show them to shipowners with whom he was in treaty for the equipment of privateers. Douglas claimed to be a relative of the Prussian Field-Marshal Keith, who was of Scottish origin, and he brought letters of introduction from well-known members of the English nobility, so the Minister may perhaps be excused for entrusting the documents to him. But his confidence was gravely abused, for Douglas, having come to an agreement with the firm of Dunbar and Eyre, filled in the forms on his own responsibility, and two privateers were sent out with these fraudulent credentials.

Shortly afterwards, one of these ships, the *Lissa*, put into Emden with a rich Swedish prize. Lying in the harbour was an English man-of-war, and the captain of this ship, declaring that the English sailors on board the *Lissa* were all either deserters or men who had bound themselves to serve in the British Navy, required that they should be given up to him. As compliance was refused, he went on board the *Lissa* with an armed escort, and, disregarding all the protests of its captain, took away with him twenty-six members of the crew. This action was regarded by Frederick as an infraction of Prussian rights of sovereignty, and representations to that effect were made in London before it was discovered in how irregular a manner the *Lissa* had become possessed of her papers. The matter was then discreetly allowed to drop. The Swedes, for their part, contested the legality of the capture, but the Prussian Government ruled that the letter of marque was valid, although it had not actually been issued by royal authority. At the same time Prussia advanced the strange view that, in the event of the owners of the *Lissa* having had cognizance of the deception which had been practised, King Frederick was entitled to the whole value of the prize. Instructions were, however, given that the *Lissa* should be deprived of her charter, but before they could be executed she had sailed for England.

Another of Douglas's privateersmen, the *Prince Ferdinand*, under a Captain Merryfield, had betaken herself to the Mediterranean, where, in a nine-

months' cruise, she captured thirteen prizes, but caused so much confusion that the King thought it wiser to put a stop to the whole undertaking. The immediate ground for this step was the complaints of the Ottoman Government, with which Frederick was negotiating with a view to obtaining its support in the prosecution of the war. The appropriation of a couple of female negro slaves belonging to a pasha, who were on board one of the ships captured by Merryfield, seems to have had at least as much weight in the Turkish grievance as the more substantial losses of the merchants of Salonika. As Prussia had no territory and very little diplomatic representation on the shores of the Mediterranean, Merryfield was obliged to take his prizes into neutral harbours and place them in the custody of the English Consuls. They were the subjects of endless law suits, tedious international wrangling, and practically no profits. Merryfield's wild career was terminated by a charge of secretly selling neutral goods from one of his prizes to his own advantage. At the instance of the Prussian Government he was flung into gaol at Malta. He remained in prison five years, and even at the end of that term would not have regained his liberty if the Grand Master of the Maltese Knights had not refused to pay for his maintenance any longer.

Hardly less chequered were the fortunes of Captain Wake, the only regularly accredited Prussian privateer of whom anything is known. The operations of his ship, the *Embden*, in the Mediterranean also resulted in ceaseless bickerings, and he was delayed in Cagliari for two years by disputes of one sort or another. At last, growing weary, he set off to Berlin to prosecute his claims to a Swedish ship which he had seized, but of which the authorities at Cagliari would not permit him to dispose. Four and a half years after the capture, she was adjudged his good prize; but before he could enter into possession of her she was sunk at her moorings by a violent storm.

The total gain of the Prussian Government from the activity of these three privateers was quite negligible; while, on the other hand, the trouble and annoyance caused by them was immeasurable. The anticipations that the seas would be swept of Austrian and Swedish commerce by a swarm of vessels under the Prussian flag proved to have been quite illusory, and it was a particular disappointment to Frederick that the German shipowners

looked askance at the whole business, and in no single instance applied for letters of marque.

A noteworthy feature of the episode is that Frederick's Government, reversing the practice of the Hansa, laid down for its privateers the rule that a neutral flag covered the enemy's goods, and that neutral goods were safe from capture even when under the enemy's flag. This, it is maintained, has ever since been Prussian tradition.

A final word is due to the "Société de Commerce Maritime"—now under the name "Seehandlung," the State bank of the Kingdom of Prussia—which was established by Frederick the Great in 1772, "to carry on shipping under the Prussian flag, and trade with the ports of Spain and all other places where reasonable and certain prospects of substantial profits from imports and exports are to be found." It was vessels of this corporation which, towards the close of the first half of the nineteenth century, bore a German flag for the first time round the world, and its foundation shows that the Great Elector's ideas were only dormant and not dead.

Frederick's immediate purpose was to open up the markets of South America to Silesian linen, but, in consequence of the rigid protectionist policy of Spain, it was only possible to do this by transshipment at Spanish ports. The original capital of the company was 1,200,000 thalers, in shares of 500 thalers each, and of these 2,100 were the property of the King. The Société was granted the exclusive right of trading in English, French, and Spanish salt, and in Polish wax, and was also endowed with many other privileges. It did not at first prove a very profitable venture, and its early days were also clouded over by the defalcations of one of its managers. In course of time it became little more than a branch of the Royal Treasury and the negotiator of State loans, but in the thirties of last century it passed under the control of a man who determined to restore to it something of its original character, and laid out a considerable capital in English-built ships. At that period German merchantmen seldom ventured beyond Bordeaux and Lisbon; but the vessels of the Seehandlung repeatedly encircled the globe, showed their flag in the remotest harbours of Orient and Occident, and established directly that export to South America of the wares of the Riesengebirge which Frederick the Great had in his mind when he called the company into existence.

FOOTNOTES:

[5] Thaler then = about 4s. 6d.

CHAPTER III

GERMANY'S FLEET IN THE LAST CENTURY

Though the sword of Napoleon completed the destruction of the Holy Roman Empire, which had done so much to hamper the development of the Teutonic race, the Vienna Congress, rearranging the map of Europe after his overthrow, left Germany still divided into thirty-nine different states. There were four kingdoms, one electorate, seven grand duchies, ten duchies, ten principalities, one landgraviate, and the four free towns—Hamburg, Bremen, Lübeck, and Frankfort-on-Main. These states were loosely united in the German Confederacy.

The people of Germany, and especially those who had risen against Napoleon, had expected a more complete unity on a democratic basis, and the disappointment of their hopes was one of the chief causes of the revolution which, in 1848, broke out simultaneously in nearly every one of the federal capitals. This movement took the Governments by surprise, and so overwhelming was the popular demand for unity, that they offered but little opposition to the convening of a National Assembly, which met at Frankfort-on-Main on May 18th, 1848, and appointed the Austrian Archduke Johann provisional "Administrator of the Empire." It is generally asserted that the failure of this serious attempt to weld Germany together was an inevitable consequence of the jealousy existing between Austria and Prussia, but none can say with certainty what the sequel might not have been, had not Frederick William IV., the grand-uncle of the present German Emperor, refused the imperial crown when it was offered to him by the National Assembly. It is very well conceivable that, if that monarch had been less fully persuaded of the divine rights of kings and of the incompetence of popular representatives to bestow crowns, the work which

Bismarck did in the next twenty years, with so grievous an expenditure of blood and iron, might have been accomplished by peaceable means, and that the world might to-day have been confronted with the problem of a much larger, much richer, and much more united Germany. Those who would not regard German domination in Europe as an unmixed blessing have reason to be thankful for Frederick William's archaic theories on the relationships of Princes to their peoples.

And those who care to amuse themselves by following up the grand alternatives of history must not forget that 1848 saw the birth of the modern German Fleet, which was the fruit of a purely popular movement. Indeed, the patriots of the Frankfort Parliament found in the "imperial fleet," which they actually founded, the necessary symbol of that national unity which was the goal of their aspirations.

Strong, spontaneous, and almost universal as was the German naval movement of 1848, it did not attain its actual dimensions without an effective external stimulus. In the very month in which the revolutionaries were defending their barricades in the streets of Berlin and other German capitals, Frederick VII. had declared his intention of incorporating Schleswig in Denmark; and, while an informal convention was arranging the preliminaries for the National Assembly, the Danish fleet was blockading the coasts of Prussia in retaliation for the military support afforded by that Kingdom, as the mandatory of the German Confederation, to the rebellious duchies. Nothing was better calculated than an incident of this sort to bring home to the German mind the importance of sea-power. That the ships of a little country like Denmark should be able, with impunity, to forbid the sea to a great military Power, seemed to every German who reflected upon it a grotesque inversion of the natural order of events.

Though the National Assembly, at one of its first sittings, appointed a permanent committee to grapple with the naval question, the impatient interest of the public displayed itself in schemes and suggestions which poured in from every side. In many places committees were formed to help to raise the funds necessary for the equipment of a fleet. It is significant of the widespread nature of the movement that the raftsmen of Gernsbach, in the Black Forest, offered to transport down the River Murg free of cost the timber required for the building of Germany's war ships. The seaports,

which felt most keenly the insulting pressure of the Danish blockade, took the leading part in the agitation. A congress of delegates from the German coast towns came together at Hamburg and nominated a "naval commission," on which, in addition to the Governments most immediately concerned, a number of private committees were represented. This body wasted no time in talk, but set to work with feverish activity. As warships were not to be had ready-made, several merchant vessels were purchased and hastily armed with guns furnished by Hanover; and at the beginning of July, the Federal Government was notified that these extemporized men-of-war were ready to put out and attack the enemy. But at the moment the negotiations with Denmark for a truce had already begun, and for the time being the squadron remained peacefully at its moorings.

Meanwhile, even before an Imperial Executive had been got together, the Frankfort Parliament had voted for naval purposes a sum of 6,000,000 thalers,^[6] half of which was to be spent immediately and the remainder as necessity might arise. Part of the money was to be taken from the fortress fund of the old Confederacy, and the remainder raised by levies in due proportion on the various states of the union. The question of these "matricular contributions," which in some cases were altogether refused, and in others only paid after much hesitation and vacillation, was one of the chief reasons for the ultimate dissolution of the first "German" Navy.

In November an imperial naval authority was constituted under the control of the Minister of Commerce, who was at the same time deputy for Bremen. An advisory commission of experts was also appointed, and the chair in this body was, at the personal request of the Archduke-Administrator, taken by the man who, in one sense, may be regarded as the father of the present German Fleet, Prince Adalbert of Prussia, and to whom, for this reason, more detailed reference must be made hereafter. The commission submitted a scheme, in which it was recommended that Germany should, for the present, make no attempt to gain a place in the ranks of the first-class naval Powers, but content herself with the protection of her Baltic and North Sea coasts and her sea-borne trade. These purposes, it was held, could be fulfilled by a fleet of fifteen sixty-gun sailing frigates—if possible with auxiliary engines—five steam frigates, twenty steam corvettes, ten despatch-boats, five schooners, and thirty gun-sloops.

During the winter, officials were despatched to England to purchase and order ships, and to America to induce the United States Government to allow some of its naval officers to enter temporarily into the German service. These latter negotiations at first promised success, but in the end the Government at Washington declared itself unable to entertain the request. With the purchase of material the German emissaries had better luck, and when the truce with Denmark expired in the spring of 1849, the Navy List already contained the names of twelve vessels, though, it is true, hardly one of them was yet fit for action. A Commander-in-Chief had also been found in the person of Karl Bromme, a native of Leipzig, whose name had been permanently anglicized into "Brommy" while he was learning seafaring in the American merchant service. This man, "the first German Admiral," had followed Cochrane to Greece, where he was successively Flag Captain to Admiral Miaulis, organizer in the Ministry of Marine, and Commandant of the Military School at the Piræus. From there he was tempted away to become "Imperial Commissioner" to the incipient German Navy, and after taking part in the sittings of the commission of experts, he was sent in that capacity to Bremerhaven to supervise the formation of the fleet and to found a naval arsenal.

On June 4th Brommy, with a steam frigate and two steam corvettes, attacked a Danish frigate which was lying becalmed off Heligoland. Hardly, however, had the engagement commenced before a signal shot from the island warned the belligerents that they were within British territorial waters, and must suspend hostilities. Soon afterwards the Danish blockading squadron approached the scene, and the German ships hurried back to their harbour. This was the only opportunity the German Fleet had of showing its quality. Brommy was promoted to Rear-Admiral later in the year.

Insignificant as the Heligoland skirmish was in itself, it had a sequel which has played a great part in all subsequent movements for increasing the German Fleet. Brommy's ships had fought under the black-red-and-gold that were to be the colours of the new Empire. But this Empire had then no legal existence, and, as a matter of fact, never did have one, and no doubt Palmerston was only giving expression to recognised principles of international law when he wrote that vessels committing acts of belligerency under the black-red-and-gold flag would render themselves

liable to be treated as "pirates." The Frankfort Government, a product of excitement and inexperience, made many mistakes which the ripe tradition of an old-established administration would have avoided, and, in its haste to assert itself on the seas, doubtless did not give sufficient thought to the restrictions imposed upon it by its own anomalous status. The hoisting of the black-red-and-gold on a flotilla or warships was undeniably a questionable proceeding, and one which justified the view propounded by the British Foreign Minister. At the same time, his words belong to the category of things which had better have been left unsaid. The word "pirate" rankled then, and has ever since continued to rankle, and the Palmerstonian note has been cited ten thousand times, and is still cited, as the supreme example of the tyrannous arrogance with which Britain rules the waves.

A fortnight after Brommy's one exploit as a German naval commander, the remnant of the National Assembly was dispersed by military force at Stuttgart, where it had taken refuge, and Germany relapsed into the condition of a loosely-jointed federation of mutually jealous and suspicious Princes, whose rival claims had to be settled on the battlefield before the great work of unification could be accomplished. The infant navy, which had been the work of a popular movement and a popular Parliament, proved a source of dissension and embarrassment to the Confederacy Governments. Several of the inland states were altogether opposed to the idea that Germany needed a navy. A strong party advocated that one fleet should be provided by Austria for the Adriatic, a second by Prussia for the Baltic, and a third by the remaining German states for the North Sea. The last point of this project was the subject of special negotiations, and at one time there seemed some chance of Hanover assuming the office of "Federal Admiral."

In the end, however, divergent interests and irreconcilable rivalries produced the only possible result, and, in February, 1852, the Confederated Governments decided to cut the Gordian knot. The promising German Navy was dissolved, Admiral Brommy received his discharge (he was subsequently employed for some time as Chief of the Technical Department of the Austrian Admiralty), and an Oldenburg official, whose unforgettable name has helped to brand his memory with the whole infamy of a transaction for which he was in nowise responsible, was appointed

"Commissioner of the Germanic Confederation charged with the regulation of naval affairs." This, at least, is the designation appended to his signature on the advertisement which, in the German, English, and French languages, announced to all the world that the German Navy was forthwith to be knocked down to the highest bidder. It was the form rather than the fact of the sale which was taken so ill in Privy Councillor Hannibal Fischer, but it is difficult to see what else he could have done. He made efforts to dispose of the ships by private treaty, and actually sold some of them to Prussia and others to English firms, but a residue remained for which no purchaser could be found in this way, and there was nothing for it but to put them up to public auction. There thus came under the hammer two steam frigates, six steam corvettes, a sailing frigate, and twenty-seven gunboats propelled by oars. Of the eight steamers three had been built at Bristol, and one each at Glasgow, Leith, New York, Hamburg, and Bremen. Except in the case of the American vessel, the engines were all of British make.

Concurrently with the abortive efforts to found a German Navy, Prussia had taken independent action, and laid the real foundation of the great fleet which now aspires to contest the British mastery of the seas. At that time there was not even the slenderest basis for the kingdom to work upon. The task had to be undertaken from the very beginning. During the first half of the nineteenth century, it is true, the advisability of building a navy had more than once been exhaustively discussed by the Prussian Government. In the general resettlement of 1815, the island of Rügen and the strip of Pomeranian coast opposite to it had passed from Sweden to Prussia, and included in the transfer were six gun-sloops and a Swedish officer, Captain Christian Lange, who was summoned to Berlin to report to the War Ministry on the utility of the little flotilla. As the result of his representations, he was commissioned to submit plans and estimates for a war schooner, and for an armed rowing boat for use on the rivers. These vessels were eventually built, with the express idea that they were to serve as experiments and models for the construction of a regular fleet. In great haste prescriptions as to a naval uniform were issued, and the questions of dockyards and harbour works were also deliberated. But the only issue of all this work was the conviction that the national resources were not yet equal to the financial strain which would have been entailed by the creation of a navy. Similar investigations and discussions in the years 1825 and 1832 were, for the same reason, equally fruitless. At the commencement of the

revolutionary year, the only vessels in the possession of the Prussian Government were a corvette, which was employed as a Navigation School, a paddle steamer, which conveyed the mails between Stettin and St. Petersburg, and which, under the terms of the contract for its construction, was to be adaptable to the purposes of an "auxiliary cruiser," and a couple of armed yawls.

By the autumn of 1848 a Prussian flotilla of ten sloops and yawls, three of which had been built with the funds collected by private committees, was ready for operations against the Danes. It was placed under the command of a Dutch ex-naval captain named Schröder. The crews provided for him—465 men in all—were a strange medley of active soldiers, reservists, and seamen from the merchant service. For various reasons, not the least weighty of which was the doubtful status of the black-red-and-gold flag, the squadron sailed under the Prussian colours. While it was fitting out, the first steps were taken towards the establishment of a naval organization and the training of a corps of officers.

By the following summer the Prussian fleet could already boast two steamers, one sailing corvette, and twenty-one gun sloops, with a total complement of thirty-seven officers and 1,521 men, and mounting in all sixty-seven guns. But only once did this primitive navy have the satisfaction of taking part in a pitched naval engagement. This was a duel between a Prussian steamer and a Danish brig, which fought for five hours off the island of Rügen. The encounter was terminated by the fall of darkness, and before day broke again another Danish corvette arrived on the scene and put the Prussians to flight. But, in spite of a lack of fighting, the presence of Commodore Schröder's force along the coast undoubtedly did much to relieve the pressure of the blockade.

The peace with Denmark in 1850 ushered in a period of assiduous and systematic labour at the task of building up a Prussian fleet. Throughout this important period, the moving spirit was the man who has already been described as the father of the German Navy, Prince Adalbert of Prussia. This enthusiastic and indefatigable sailor was a first cousin of King Frederick William IV., who refused the imperial crown as a democratic gift, and of the Emperor William I., who finally won it on the battlefields of France. In his boyhood, Prince Adalbert had had the doctrine of the vital importance of sea-power implanted in his mind by a veteran soldier, Field-

Marshal Gneisenau, and he never forgot the lesson. At the age of twenty-one he paid a visit of two months' duration to England, where he was cordially welcomed into naval circles, and where his passion for the sea was inflamed by the conversation of men who had fought under Nelson at Trafalgar. He lost no opportunity of inspecting war vessels, shipyards, and docks, and returned to Germany with note-books crammed with information as to all he had seen and heard. A British admiral is said to have declared that the Prince knew more about the warships of Great Britain than many of their own officers, and one of the last acts of this sailor Hohenzollern was to pay a visit to the English dockyards to familiarize himself with the latest novelties in naval construction.

Four years after his first journey to England, one of those naval enquiries already alluded to was held at Berlin, and a commission was appointed to advise as to the types of vessels to be chosen for the fleet which the Prussian Government contemplated building at some indefinite future date. Prince Adalbert was a member of this body, but when asked for his views on the subject he satisfied himself with laying before his colleagues the opinion of his friend, Captain Mingaye, a British naval officer, who advised that the triumph of steam over sails and oars presented Prussia with a splendid opportunity to create sea-power which should be "mighty" from the outset. Curiously enough, the War Minister, von Rauch, inferred from this suggestion that naval construction was passing through a transition stage of doubtful issue, and it was used by him as a pretext for postponing the consideration of the whole question; for, he argued, Prussia could not afford to squander money on uncertain experiments. In the succeeding years, the Prince cruised the Mediterranean in an Austrian ship with his friend the Archduke Johann, afterwards the Imperial Administrator, and made in Sardinian and British war vessels several longer voyages, during which he devoted himself with a whole heart to the study of seamanship and navigation. He also added materially to his knowledge while on board one of the ships of the British Mediterranean Squadron, which at the time was engaged in man[oe]uvres. On his return home from these experiences, he secured the appointment of Schröder to the Navigation School ship *Amazon*, always with the idea that the vessel would be the training-ground of the officers' corps of a future Prussian Navy. As we have seen, the Prince was chosen as chairman of the Frankfort advisory committee on naval questions. Some months previously he had addressed to the National

Assembly a "Memorandum as to the Formation of a German Fleet." This document, which was printed and published, not only is a remarkable testimony to the author's insight into the true nature of naval problems, but also contains a clear enunciation of the principles which have since guided Germany's naval policy. Pointing to the humiliation of the Danish blockade he wrote:

"And this Germany—united Germany—must calmly submit to, precisely at the great moment when, after long years, it once more feels itself a whole, a Power of forty millions of people. But the Fatherland recognises the oppressive nature of its situation; it demands a remedy all the more speedy because after these events, it foresees with certainty how much more painful its position might some day be if it were pitted against one of the great Sea Powers, a Power against which the German ships would not be secure even in their own harbours, a fleet which could menace our coasts with debarkations on a much more extensive scale than is possible to our present foe. United Germany, however, wishes to see her territories energetically protected, her flag respected, her trade once more flourishing, and in the future to have some influence on the sea."

Prince Adalbert then weighed the three alternatives: (a) Defensive coast protection; (b) offensive coast protection; and (c) an independent German sea-power; and finally reached the conclusion:

"Germany must either build no battleships or at once build so many that she can act towards her neighbours as an independent Sea-Power. Anything intermediate would be a useless expense, an empty pretension, and would arouse in the nation expectations which, in the moment of danger, our sea-power would not be able to fulfil.

"If we now ask what would be the smallest number of battleships which would allow us to act in European waters as an independent fleet, especially against the ever-ready Russian Baltic fleet, I think we must take twenty battleships as the minimum that would be able to measure itself with it. But such a fleet would make Germany fourth among the Sea-Powers of first rank, and place her incontestably in a position to play a great rôle on the sea, a rôle which would be worthy

of her position in Europe. For with her twenty battleships she would be able to throw an enormous weight into the scales, turn the balance by her adherence to an alliance, and consequently be as much sought after as an ally on account of her sea-power as on account of her land-power."

The Prince accordingly proposed that the German building programme should include 20 battleships with auxiliary screws, 10 frigates, 30 steam cruisers, 40 gunboats, and 80 gun-sloops; and that the construction of these vessels should be spread over a period of ten years. In this project we have that same principle of the gradual working up to a fixed standard of strength which has characterised all modern German naval legislation.

However, the Prince did not manage to persuade the Frankfort technical commission to adopt his scheme in its entirety, though the programme approved went a long way towards meeting his views. Why this programme was never carried out has already been seen. In the Memorandum just quoted from, Prince Adalbert had written: "The entire nation unanimously demands a German war fleet, for German, absolutely German, it must be, a true representative of the new-born unity of the Fatherland"; and it must have been with a heavy heart that he saw his vision melt away, and went back to Berlin to employ his gifts in a more restricted and less promising field.

The difficulties which opposed themselves to the realisation of the Prince's ideas will be appreciated, when it is stated that the man who built the first warship of any size which had been launched from a German yard since the days of the Hanseatic League is still alive. Wilhelm Schwarm, now ninety-four years of age, was employed as a young man in Klawitter's shipyard at Dantzig, and at the time when the air was filled with talk of a future German Navy, the firm very shrewdly sent him over to the works of Robinson and Russell, on the Thames, to learn the art of constructing vessels of larger size than were then built on the Baltic. He brought back with him the plans for a paddle corvette, which was built under his supervision on the Klawitter slips, fitted with English engines, and, under the name of *Dantzig*, was an important addition to the Prussian fleet.

At the time of the Crimean War this vessel showed the Prussian flag at Constantinople for the first time in history, and it was also with her that

Prince Adalbert experienced a rather grotesque adventure in the Mediterranean in 1856. In the previous year a German ship had been plundered by the Riff pirates, and the Prince, happening to be in those parts with the *Dantzig*, made a reconnaissance, in one of the ship's boats, of the coast of Cape Tree Forcas, where the outrage had occurred. The natives, as was their custom, fired on the party from the shore. Annoyed by this molestation, Prince Adalbert determined to teach the Arabs a severe lesson. Having manned and armed all his boats, he stormed the steep and rocky shore and planted the Prussian flag on the summit of the cliffs. His triumph was, however, a very brief one, for the enemy immediately returned to the attack, and drove the landing party back to the boats with the loss of seven killed and twenty-three wounded. Official panegyrists extol this rash escapade as an "heroic deed," and declare that it did much to raise the confidence of the young Prussian Navy. As the Riff pirates were no doubt also exultant over their victory, the affair must have been one of those rare encounters with the issue of which both sides were equally satisfied. The *Dantzig* was sold a few years later in England, in the belief that her timbers were unsound, and was then passed on to Japan, where she was run ashore and burnt by her own crew during an engagement in the civil war.

The problem of obtaining properly qualified personnel for the corps of naval officers was not less difficult to solve than that of building efficient warships. England would have been the natural source on which to draw for instructors, but for political reasons it was decided not to seek assistance from that most competent of all quarters, and the services of three officers of the Swedish Navy were secured. For similar reasons a Swedish naval constructor was engaged. A few years later, however, permission was asked and obtained for a number of cadets to learn their profession on British men-of-war.

The year 1852 brought an event of the utmost importance for the development of the Prussian Navy—the acquisition of Wilhelmshaven as a North Sea base. At that time Prussia did not possess an inch of coast-line on the North Sea, and could obtain access to it only through the Belt and the Sound, then under the control of the superior naval power of Denmark. Among the innumerable projects with which the National Assembly had been deluged, was the scheme of three citizens of Rendsburg for the construction of a water-way pretty much along the line subsequently

followed by the Kaiser Wilhelm Canal. This plan was, however, based on the false assumption that Schleswig-Holstein would at once become, and ever afterwards remain, German territory. It had also been proposed to the Frankfort Government by an Oldenburg official that the Jade Bay should be chosen as the North Sea base for the fleet, and this suggestion seems to have fixed the attention of Prince Adalbert on the inlet which is now the chief naval headquarters of the German Empire. The Grand Duke of Oldenburg was approached, and he consented to cede to Prussia the piece of marshy land which has since been covered by the harbours, docks, shipyards, workshops, barracks, and fortifications of Wilhelmshaven. Prussia paid a sum of 500,000 thalers for this invaluable possession, and at the same time took upon herself the protection of the coast and sea-trade of the duchy.

Herculean efforts and inexhaustible patience were required to adapt Prussia's acquisition on the Jade to its destined purposes. Years had to be spent in a careful survey of the bed of the harbour, in order to ascertain how far the channel was affected by the movements of sand and mud under the influence of the tide. Further years were consumed by the task of sinking piles in the treacherous peaty soil to obtain a solid foundation for dock and harbour walls. Frequently a storm or a spring tide destroyed in a few hours the fruits of months of strenuous labour. As Hanover refused to allow the construction of a railway across her territory, which lay between Prussia and Oldenburg, it was necessary to convey all the building materials to the spot by the long and tedious sea-route. At first not even drinking-water was to be had on the desolate site, and prolonged and costly exertions were needful before it could be procured in sufficient quantities. Sixteen years elapsed before the new harbour was formally declared open by the Prussian King, afterwards the Emperor William I., in the presence of British ships, the officers of which probably regarded the works with indulgent curiosity and little guessed the significance which Wilhelmshaven would one day possess for their own country.

When the second war with Denmark broke out in 1864, Prussia's fleet was still absurdly inadequate to deal with the naval force opposed to it. The ship establishment at the close of 1863 was composed as under:

Steamships with Fighting Value.

3 corvettes, mounting 27 or 28 guns each.
1 corvette, mounting 17 guns.

Steamships with little Fighting Value.

4 first-class gunboats, mounting 3 guns each.
17 second-class gunboats, mounting 2 guns each.
3 despatch-boats, mounting together 8 guns.

Steamship without Fighting Value.

1 corvette, mounting 9 guns.

Sailing Ships with little or no Fighting Value.

3 frigates, mounting a total of 112 guns.
3 brigs, mounting a total of 4 guns.
2 schooners, mounting a total of 4 guns.

Also without Fighting Value.

40 rowing-boats, mounting a total of 76 guns.

Denmark, on the other hand, had 31 steam war vessels, among which were 1 battleship, 5 frigates, 3 corvettes, and 4 armoured craft. Even with the assistance of a number of Austrian ships, which arrived in the North Sea from the Mediterranean, the Prussian fleet could contribute nothing decisive towards the issue of the war. At the most it prevented the Danish blockade of the German coast-line from being effective. The Prussian Government attempted to reduce its inferiority by hiring merchant vessels, and hurriedly purchased warships in France and England. One of these latter, the monitor *Arminius*, which was of English build, was almost entirely paid for with the voluntary contributions which had continued to flow in. This fact shows how steady and keen the interest of a large section of the population in the development of the navy already was, and how erroneous it is to ascribe the naval enthusiasm in Germany of recent years entirely to the official agitation. Peace was concluded before the new ships could be made ready for sea.

The war of 1864 was one of the great cross-roads of British history. Difficult as it is to "overlook the cards of Providence," as Bismarck puts it, there can be little doubt that we took the wrong turning. The great German Chancellor candidly admitted that the possession of Kiel and a strategic

canal through Holstein were two of the principal objects which Prussia had in view when she drew the sword. The two leading members of the British Cabinet were in favour of backing up Denmark; and one of them, Palmerston, used language in Parliament which might well have led that country to count upon our support. A strong body of English public opinion also warmly espoused the Danish cause. But Queen Victoria, largely influenced by the sympathy for Germany which she had imbibed from the Prince Consort, threw all the weight of the Crown into the opposite scale.

There are few more agitated passages to be found in the records of diplomacy than those letters to Lord Granville in which she argued, threatened, entreated, and, finally falling back on the last strength of woman, her weakness, complained that she was "completely exhausted by anxiety and suspense," and "so tired and unwell she can hardly hold up her head or hold her pen." Her will prevailed in the end, and she was able to congratulate herself that, "owing to the determined stand she had made against her two principal ministers, she had saved the country from an unnecessary war." When Prussia, completely reversing her attitude, made those very claims of the Danish King which she had contested by force of arms her pretext for annexing the two duchies under the "rights of conquest," the Queen suffered a bitter disillusionment, and, on her instructions General Grey wrote to Lord Granville, that "Prussia should at least be made aware of what she and her Government and every honest man in Europe must think of the gross and unblushing violation of every assurance and pledge that she had given which Prussia had been guilty of." It will hardly be contended now that a war which should have left Schleswig-Holstein in the hands of Denmark would have been anything but exceedingly advantageous, economical, and opportune for Great Britain.

Even before, in the formal division of the spoils, Prussia had obtained Austrian recognition of her right to Kiel, she had occupied that port and transferred her naval headquarters thither from Dantzig. The construction of the North Sea Baltic Canal was delayed many years, mainly by the opposition of Count Moltke, who argued that its cost would be so great that it would, on the whole, be cheaper to build a second fleet with the money. He further urged that the canal would be navigable only in the summer, and that in the event of a war the army would be weakened by the necessity of providing for its defence. But for the doubts and jealousies of the sister

service, the German Navy might years ago have enjoyed the benefits of that prolongation of the Canal, contemplated by Bismarck, which would have allowed its ironclads to steam from Kiel to Wilhelmshaven without putting out into the open sea.

In the hope that the lessons of the war would have produced the desirable effect on the public mind, the Prussian Government, in 1865, laid before its Parliament a bill that may be considered as the definite inauguration of the naval policy which Germany has ever since pursued. In the Memorandum submitted to the House with the measure, it was contended that the time had come for Prussia to join the ranks of the Sea-Powers, in order that she might be in a position to protect her own and the other German coasts and maritime trade, and, for all future time, to assert her European position as against such States as were accessible only by water. "For the present," it was stated, "she is unable to enter into rivalry with the first-class naval Powers, but she must occupy a position commanding esteem among those of the second class."

Accordingly, the Government asked for authority to build 10 armoured frigates of the highest efficiency, an equal number of armoured vessels of the cupola or turret type for coast defence, 16 corvettes for the protection of sea-borne trade, 6 despatch-boats, and at least 4 transports. It was calculated that ten years would be necessary for the execution of this plan, but rather for the training of the personnel and the provision of the indispensable harbour works than for the actual construction of the ships. The cost of the proposed fleet was estimated at 34,500,000 thalers, that of its annual maintenance at about 5,000,000 thalers. In recommending the scheme to the Diet, Bismarck used the following words, which contain very noteworthy implications: "During the last twenty years no question has so unanimously interested public opinion in Germany as precisely the naval question. We have seen associations, the Press, and the Diets give expression to their sympathy, and this sympathy exercised itself in the collection of comparatively important sums. The Government and the Conservative party have been reproached with the slowness and parsimony with which action has been taken in this direction. It was particularly the Liberal parties which carried on this agitation. We believe, therefore, that we are doing you a great pleasure with this Bill."

But the Liberal majority, then exclusively preoccupied with the constitutional struggle against the masterful and autocratic Minister-President, threw out the Bill, and modified naval estimates were given the force of law by royal decree. The attitude of the Prussian Liberals of that epoch was very similar to that of the Socialists in recent years.

In the brief war of 1866, the Austrian fleet was tied down to the Mediterranean by the superior sea-power of Italy, and the operations of the Prussian ships were confined to a few cheap victories over the antiquated coast and river fortifications of Hanover. As the result of the war, Prussia was rounded off by the incorporation of the Kingdom of Hanover, Electoral Hesse, Nassau, and the old imperial town of Frankfort-on-Main. She thus secured for herself the entire German North Sea littoral, with the exception of the coast-line of Oldenburg, which by treaty was already hers in fact if not in law. Immediately after the conclusion of peace, all the States to the north of the Main were closely welded together in the North German Confederation, the first decisive step towards the creation of the Empire. An article in the Federal Constitution ran: "The Federal Navy is one and indivisible under the command of Prussia. Its organization and composition fall to His Majesty the King of Prussia, who nominates the officials and officers of the navy, and to whom they, as well as the crews, must take the oath of fealty. Kiel harbour and the Jade harbour are federal war harbours. The expenses necessary for the establishment and maintenance of the fleet and the institutions connected therewith will be borne by the Federal Treasury."

Two years later a fresh naval programme was submitted to, and approved by, the North German Reichstag. It laid down that within ten years the fleet should be brought up to the subjoined strength:

- 16 large and small armoured ships.
- 20 corvettes.
- 8 despatch-boats.
- 3 transports.
- 22 steam gunboats.
- 7 school-ships.

The new vessels actually needed for the attainment of this establishment were 12 armoured ships, 12 corvettes, 6 despatch-boats, 2 transports, and 1

school-ship. As native ship-builders had so far had no experience in the construction of ironclads, only one vessel of this type was placed in Germany, the State yard at Dantzic being experimentally entrusted with the work, while the rest were purchased or ordered in England or France. No fact could illustrate more vividly the tremendous progress which Germany has since made in this respect.

Oddly enough, the great war with France was succeeded by a marked cooling-off of the popular enthusiasm for the navy in Germany. The reasons for this appear to have been disappointment with what the fleet actually accomplished and the complete overthrow of the enemy without its assistance. Even if all the federal ships had been in perfect trim and manned by thoroughly trained crews, they were confronted by so overwhelming a superiority of force that at best they could have achieved little or nothing. But the outbreak of hostilities coincided with a series of accidents which temporarily disabled several of Germany's best war vessels, and at that time there was not a single dock in the country in which they could be repaired. Officers and crews were, too, imperfectly trained and insufficiently familiar with both engines and guns, the harbour equipments were inadequate, and, in fact, everything was in a state of unpreparedness.

That the French, with their great naval superiority, effected so little, and did not even make a determined attempt to force the Jade and destroy the works at Wilhelmshaven, can only be ascribed to their lack of initiative and the paralyzing operation of their crushing defeats on land. The only regular engagement fought at sea during the war was an encounter of uncertain issue between a small German gunboat and a French despatch-boat off the coast of Cuba. But in spite of the odds against the federal fleet, public opinion in Germany protested that it should have shown more dash and enterprise, and in some way have crowned itself with laurels. Even more prejudicial to the popularity of an ambitious naval policy was the patent fact that the hereditary and most formidable foe had been thoroughly and rapidly humbled by a purely land campaign, and that his superiority on the sea had availed him practically nothing. To such considerations must be attributed a large share of the indifference with which many Germans regarded their navy during the next thirty years.

The prevalent views were reflected in the Memorandum with which, in 1872, the Minister of Marine, Lieutenant-General von Stosch, ushered in

the first naval programme of the new German Empire. This document stated that in a long war Germany must leave the offensive to her land force, and that the proper task of her navy was to assert the power of the Empire where smaller interests were at stake in places to which the army could not penetrate. An increase in the fleet was, however, stated to be necessary on the ground of the growth of German sea-borne trade, and it was proposed that the following vessels should be available by the year 1882:

- 8 armoured frigates.
- 6 armoured corvettes.
- 7 armoured monitors.
- 2 armoured batteries.
- 20 cruisers.
- 6 despatch-boats.
- 18 gunboats.
- 28 torpedo-boats.
- 5 school-ships.

The cost of these vessels was estimated at 73,000,000 thalers, that of their maintenance in the year 1882 at 1,300,000 thalers. The plan, which was much more modest in its pretensions than its predecessors, and in principle constituted a retirement from the position formerly taken up, was approved by the first Parliament of the new Germany.

The first royal review of the German Fleet took place in the Warnemünde roads in 1875. The ships present were four ironclads, a despatch-boat, and four school-ships; their total complements 2,862 officers and men.

When the year 1883 arrived, General von Stosch published a Memorandum on the execution of his plan. It is significant of the change that had come over public opinion that the Government had not dared to ask the Reichstag for a substitute for the armoured frigate *Grosser Kurfürst*, which was lost in collision off Folkestone, and that consequently one of the eight vessels of her type was lacking. The last of the six armoured corvettes had yet to be built, and instead of five monitors thirteen armoured gunboats had been constructed, because it was thought that the latter class of ship was better suited for the defence of the Jade, Weser, and Elbe. It had been decided not to build the floating batteries, which would have been an easy prey to the "fish" torpedo, introduced as a weapon of naval warfare since they were

projected. One out of the twenty corvettes, and eight large and nine small torpedo craft were also still wanting. German national vanity had, however, secured a questionable triumph: the Empire was now entirely independent of foreigners so far as its warships were concerned. But if Germany had continued to purchase some of her warships in England while she was still but a tyro in the art of naval architecture, she would have saved much money, and made more rapid progress.

General von Stosch simultaneously presented another Memorandum, dealing with the future development of the navy. In it he laid stress on the reasons which could be adduced against the principle hitherto followed, and since readopted, of fixing the building programme in advance for a longer period, and advised that it was inexpedient to look farther ahead than three or four years. While admitting that "the seas are ever more ceasing to separate the nations," and that "the course of history seems ever more to indicate that a State cannot withdraw from the sea if it is striving to maintain for itself a position in the world beyond the immediate future," he laid down the axiom that "naval battles alone seldom decide the destinies of States, and for immeasurable time the decision of every war will for Germany lie with her land army." Thus, though he admitted the desirability of "a concentrated high sea fleet always ready for action," he considered it best to defer the construction of battleships till further experience had shown whether their functions could not be equally well performed by vessels of a smaller type. The conclusion reached by the Memorandum was that it was necessary to add without delay to only one class of vessel—namely, that which served the purposes of coast defence. In this connection the following words were used:

"Here it is the torpedo-boat, which, especially when used in large numbers at night, will render the carrying through of a blockade almost impossible. Every night the blockading ships would be compelled to withdraw to a distance under steam. Their coal consumption would thereby be much increased, the tension of the crews, in consequence of the need for unremitting vigilance, would become intolerable, and at night the blockaded harbours would be accessible. Even when in motion, the blockading ships would not be safe at night. The torpedo-boats would follow them and recognize their aim by the lights which the enemy would not be able to do without

when steaming in squadron formation. The torpedo-boat is a weapon which is of special advantage to the weaker on the sea. A few States already possess a considerable strength in torpedo craft. For the German Navy 150 torpedo-boats are considered necessary, and of these thirty-five will be ready for service shortly."

It was while the German Fleet was still impotent for all serious purposes that the Empire acquired the mass of its colonies: South-West Africa, Togo, the Cameroons, German New Guinea, the Bismarck Archipelago, and the Marshall Islands were all annexed in 1884. The decisive step towards the acquisition of German East Africa was taken in the following year.

William I. lived just long enough to lay the foundation-stone of the Kiel Canal, which had been one of the dreams of the Frankfort patriots forty years earlier. His death was followed after an interval of three months by that of Frederick I., and with the accession of William II., in 1888, the latest era of German naval policy may be said to have commenced. Until, however, Admiral Tirpitz was put in charge of the Ministry of Marine, in 1897, practically nothing was done to add to the fighting strength of the fleet. Any progress which was made in connection with the navy was confined to developments of organisation, and to the exchange of German rights in Zanzibar and Witu for the islet of Heligoland. This transaction was scoffed at by Bismarck, then in retirement, who, however, only contemplated the possibility of a naval war with France, and it was bitterly resented by German public opinion, and especially by that heated section of it which poses as the pioneer on the path of militarism, navalism, and colonism. Only during the last three or four years has the conviction gradually begun to gain ground, that perhaps, after all, Germany did not make such a bad bargain, and Heligoland has simultaneously taken an ever more and more prominent place in the speculations of political prophets as to the probable outcome of an Anglo-German war.

The keen interest of the Emperor William, and his ambition to play the leading part on the stage of the world, would not, in themselves, have sufficed to bring about the change which has been wrought during the past seventeen years. The decisive personal factors here have been the fixed purpose, the steady will, the unflagging energy, the inexhaustible patience, the profound political insight, and the rare diplomatic skill of Admiral

Tirpitz, the nearest approach to a really great man that Germany has produced since Bismarck. He is the true creator of the German fleet.

FOOTNOTES:

[6] Thaler = about 3s.

CHAPTER IV

BRITISH INFLUENCE ON THE GERMAN NAVY

In a very special sense the German Navy is the child of the British Navy, which is the mother of all the great naval forces of the world to-day. From the very first it has been no secret that the German Fleet was definitely planned on the model furnished by the many centuries' development of the British Navy, and the Emperor William has been one of the principal agencies through which this formative influence has been exerted in more recent years. He came to the throne at a moment when naval sentiment in Germany was at its lowest point, and he assisted in the initial revival which occurred before Grand Admiral von Tirpitz came on the scene.

Old residents of Portsmouth still remember a boy whom they occasionally saw walking about the dockyard looking at the ships with admiration and rapt attention. His greatest delight seemed to be to watch the great ironclads moving in and out of Spithead. Sometimes he would find his way on board vessels of the Royal Navy. This lad was none other than the present German Emperor. As a grandson of Queen Victoria, he was a frequent visitor in his boyhood and early manhood to his grandmother during the summer months when she was in residence at Osborne, and on one occasion his father and mother, then Crown Prince and Princess of Germany, rented Norris Castle, on the outskirts of Cowes, and lived there for several months with their children. Prince William, who was a great favourite of the late Queen, thus not only became an eager spectator of the naval pageants in the Solent directly under the windows of Osborne House and Norris Castle, but

watched with interest the gay assemblage in Cowes roadstead for the regatta from year to year.

At this time the newly-created German Empire had practically no fleet. During the Franco-Prussian War the few ships which flew the flag of the North German Confederation were so weak that they could take no part in the conflict. The memory of these recent events was still fresh in the mind of the future Emperor when he visited England and watched the activities of the British Navy, whose far-flung squadrons performed the triple task of protecting the Motherland from fear of invasion, safeguarding all her oversea possessions, and defending British ocean-borne commerce. He determined that he, too, would have a great fleet when he succeeded to the throne of the German Empire.

This is no imaginary picture of the ideas which were taking root in the mind of the ruler of the German Empire to-day. Years afterwards—in fact, in 1904—addressing King Edward, on the occasion of His Majesty's visit to the Kiel Regatta, the Emperor paid a tribute to the power and traditions of the British Navy, with which, he added, he became acquainted as a youth during visits which he paid to England. He recalled that he had had many a sail in the *Dolphin* and *Alberta*, old British yachts, and had seen mighty ironclads constructed which had since served their time and disappeared from the Navy List. "When I came to the throne I attempted to reproduce on a scale commensurate with the resources and interests of my own country that which had made such a deep impression on my mind when I saw it as a young man in England."

When he first advocated the construction of a big navy, the German people viewed his dreams with indifference and distrust. Shackled by a system of conscription in order to provide the Empire with its huge army, they asked what it would profit them if to the burden of a great army they added the vast expense of a fleet capable not merely of defending their coasts, but of operating on the offensive in distant seas. At first the Emperor made little progress in educating public opinion; but he still nursed those dreams of sea-power—very moderate dreams at that date, before Admiral von Tirpitz came on the scene—which had first taken shape in his mind when he wandered about Portsmouth Dockyard, and viewed from the grounds of Osborne House the coming and going of mighty British warships. In the early days of the present century he referred with some pride to the

persistency with which he had pursued his aims in spite of popular disfavour. At the launch of the *Kaiser Karl der Grosse* he said: "If the increase in the navy which I had demanded with urgent prayers had not been consistently refused me during the first eight years of my reign—I did not even escape derision and mocking at the time—in how different a manner should we now be able to promote our prosperous commerce and our interests overseas!" He had to wait for many years before he saw his dreams reaching fruition.

As the British Parliament is the mother of all popular representative institutions, so the British Navy is the mother of navies. If the records of most of the great fleets of the world are searched, it will be found that in greater or less degree they owe their birth to the more or less direct assistance of British naval officers, oft times acting with the direct authority of the British Admiralty; while in almost every fleet in the world even to-day may be found ships designed by British brains and constructed of British material by the skilled craftsmen of these islands. It was to England that Peter the Great came to watch the shipbuilding on the Thames, and it was with a large body of British mechanics that he returned to Russia to create a fleet with which to defend his empire and extend its borders at the point of the gun. The prestige of the Russian Navy in the seventeenth century was due entirely to the skill and daring of Scotsmen. The Greigs of four generations, Admiral Elphinstone, Lord Duffus Gordon, and a number of other Scotsmen entered the Navy of the Czar and did splendid service; and some of the descendants of these pioneers of the Russian Navy may still be traced in the fleet of to-day. The American Navy was, of course, of distinctly British origin; so were the fleets of many of the South American Republics; while, as everybody knows, the seeds of sea-power of Japan were sown by British naval officers, including first and foremost Admiral Sir Archibald Douglas, and her ships were mainly built in England. The excellence to which the Chinese Navy once attained was also due to British instruction under another Scotsman, Admiral Lang; and one of the principal shipyards of Italy, as well as her gun factory, is of British origin, and is still linked with its British parent. The Spanish Navy is now being recreated under British supervision; Turkey never was so nearly a sea power as when she had British naval officers in her service; and under Admiral Mark Kerr the glories of the Greek Navy are being revived.

In the case of the modern German Fleet the British Admiralty had little part in its upbuilding, but British naval power fired the imagination of the Emperor, and it was a kindly present made years before by King William IV. to the then King of Prussia which first directed his Majesty's thoughts towards the sea. When the present Emperor was a boy, one of his favourite recreations was to sail a beautiful model of about 20 tons of a British frigate on the Havel lakes near Potsdam. This little ship, of excellent workmanship, was sent as a present to the then ruler of Prussia early in the last century by our sailor King, and was a never-failing source of pleasure to the present German Emperor as a youth. From his earliest years at home and in England the future ruler's aspirations were always towards the sea, and we can now see that his dreams of later years, which have taken such tangible shape, were largely due to those vivid impressions of sea-power which he obtained during his visits to England, and which reached their climax in 1889, when Queen Victoria, on the occasion of his visit to the Cowes Regatta, conferred on him, a foreign monarch, the, then, unique rank of Admiral of the Fleet.

Though other foreign princes and monarchs have since been made honorary officers of the British Navy, the German Emperor remained for some years the only person of foreign birth holding supreme rank. The commission conferred upon the Kaiser was of course purely honorary, but his Majesty never concealed the pride with which he donned the British uniform with its deep gold cuffs and cocked hat, and he could claim that he was the only ruler of a foreign State who ever commanded the British Navy in modern times.

Great Britain has boasted of her "splendid isolation," and the German Emperor's is the only alien hand which has controlled any of her fleets. In times gone by a British squadron was placed under the orders of Peter the Great. This incident occurred during the Czar's operations against Sweden, when he received the assistance of a squadron from these islands and hoisted his flag in command of the allied forces. Between that date and the year when the German Emperor became an Admiral of the Fleet the British Navy maintained its absolute independence, and British officers were not even permitted to accept foreign decorations. But soon after receiving the honorary rank from Queen Victoria, the Emperor seized the opportunity to emulate the example of Peter the Great, and he afterwards confessed in a

speech he delivered on board the British battleship, *Royal Sovereign*, that the incident had left an indelible impression upon his mind. "One of the best days of my life," he remarked, "which I shall never forget as long as I live, was the day when I inspected the Mediterranean Fleet when I was on board the *Dreadnought*,^[7] and my flag was hoisted for the first time."

The Emperor at this time was making a cruise in the Mediterranean, and visited the Piræus to attend the wedding of his sister to the present King of Greece. Sir Anthony Hoskins, who was then only a Vice-Admiral, was in command of the British Fleet which had assembled in honour of the royal marriage. The German Emperor decided that in his new rôle as a British officer he would exercise command, and consequently the emblem of an Admiral of the Fleet, which consists of the Union flag, was broken at the main on board the old battleship *Dreadnought*, and Sir Anthony Hoskins, being a junior officer, was forthwith relieved of the control of the British men-of-war, and nominally, though not of course actually, the German Emperor, during the time that his flag was flown, was in command of the greatest of all the fighting squadrons of the British Empire.

On a subsequent occasion, at Malta, his Majesty again visited the British Fleet. Arriving at this great naval base he announced that on the following day he would inspect one of the men-of-war. Accordingly, he proceeded on board, and his flag was forthwith hoisted. It was thought that his Majesty would formally walk round the decks and then take some light refreshments and return to his yacht. This was not the case, however. No sooner did the Emperor reach the quarter-deck, where he was received with naval honours by all the officers, than he took off his coat and intimated that he was ready to go over the ship. His Majesty went everywhere, from the turrets to the engine and boiler-rooms, and kept the Captain fully occupied in answering a multitude of questions as to the design and equipment of the vessel. With all the impetuosity of his nature he dived into every hole and corner and saw everything, and the Captain was kept so busy that he forgot his duty as host and the wines he had laid in for the occasion. At last the inspection ended, the questions ceased, and his Majesty prepared, after complimenting the Captain on the smartness of his ship, to go down the companion ladder to his lunch. As he did so, he turned to this commanding officer and said: "Yours must be the longest ship in the British Navy." "I think not, your Majesty," replied the Captain, "it's only 420 feet long." "Oh, you surely are

mistaken," added the Emperor, and then the Captain remembered the naval slang as to "long-ships in the navy"—namely, those with long intervals between refreshments. He forthwith apologised profusely for the oversight, and implored the Emperor to return to the cabin. His Majesty would not, however, do so, but added: "January 27th is my birthday, and my orders are that on that day you entertain all your brother captains to dinner and drink my health." He then left, pleased at the result of the incident.

When the day arrived, the dinner was duly held, and the guests enjoyed themselves immensely. During the evening they despatched the following message to the Emperor: "The orders of our Admiral of the Fleet have been carried out, and we have drunk your Majesty's good health. But there is one point on which we cannot agree with your Majesty, and that is as to the length of H.M.S. ———." From this the Emperor, who is familiar with the language of the navy, was able consequently to infer that on that evening there had been no lack of hospitality.

After the lapse of many years during which the progress of the German Navy became ever more and more the preoccupation of the British people, it is difficult to realize that when the movement for naval expansion on the other side of the North Sea first began to take shape it was regarded with sympathy by the British nation, and the German Emperor, wearing his uniform as an honorary British officer, was, of all monarchs, the most popular in this country. The two countries were on terms of growing cordiality when the Emperor succeeded his father in 1888. The absence of any reference by the new Emperor in his proclamation either to England or to France caused momentary anxiety, but that feeling quickly passed away, and in the following summer the new Emperor was the central figure in the great naval pageant at Spithead.

For the first time in the history of the British Fleet naval man[oe]uvres had been held in 1885, and in the year after William II.'s accession the young ruler witnessed the greatest display of British sea-power which had ever been organised. The assembly of 1889 far exceeded in numbers and in the suggestion of power the Naval Review which had marked the celebration of Queen Victoria's Jubilee. It was the most powerful fleet ever brought together in time of peace. The Naval Defence Act, the culmination of a long and vigorous agitation, had been passed in the spring, and it was thought appropriate to mobilize the fleet as a demonstration in the eyes of the world.

The German Emperor determined to visit this country for the special purpose of joining in this festival of British sea-power. In those days the act of mobilization occupied considerable time; though the ships in reserve were manned in the middle of July it was not until August 1st that the fleet assembled at Spithead. It included 20 battleships, 6 coast defence vessels, 29 cruisers, 3 gun vessels, 14 gunboats, and 38 torpedo boats. The great anchorage presented a brave appearance when, on the following day, the Emperor arrived, escorted by a squadron of his small navy. This force consisted of the battleships *Friedrich der Grosse*, *Preussen*, *Deutschland*, *Kaiser*, *Sachsen*, *Baden*, and *Oldenburg*, together with the despatch-vessels *Zieten* and *Wacht*; while the training ship for German naval cadets, the *Niobe*, was also present together with the corvette *Irene*, commanded by the Emperor's brother. The German Emperor and his ships received an enthusiastic welcome as he passed through the British Fleet on board his yacht, the *Hohenzollern*. The spectacle was one of the most brilliant and imposing ever witnessed in waters which had often been the scene of naval displays. On the following Monday, when the Prince of Wales, representing Queen Victoria, inspected the ships, his Royal Highness was accompanied by his Majesty, to whom, subsequently, all the principal officers were presented on board the *Victoria and Albert*. Early on the following day the fleet proceeded to sea, steaming past the German Emperor, who watched the evolution from the deck of the *Osborne*, moored in Sandown Bay.

Thus did the new ruler of Germany, on whom Queen Victoria had just conferred the honorary rank of Admiral of the Fleet in the British service, gain a unique knowledge of the size and efficiency of the British Navy normally maintained on a peace footing in home waters. The contrast in organization and in administration between the British Navy and the German Army can hardly have failed to impress the young Emperor, who had devoted himself with unremitting persistency to the study of the military machine of his own country. Looking back with the knowledge which we now possess of the rapidity with which a navy can be raised from a peace footing to a war footing as exemplified by the modern German Navy, we can imagine the impression which the British mobilization made upon his Majesty. And then, when the time came for the ships to pass out of the anchorage into the channel, the delays and confusion which occurred must have suggested to the young ruler, familiar with the standard of efficiency attained by the German Army, that something was lacking.

A contemporary account of this evolution records that: "It was at half-past three in the morning that the fleet began to unmoor preparatory to proceeding to sea, but it was not until nearly eleven that Sir George Tryon—the Admiral in supreme command—was able to give the signal for his squadron to weigh anchor. Nearly all the delay was caused by trouble and mishaps connected with the anchoring gear of various ships. There is no part of the equipment of a man-of-war which requires more management and experience in handling than the ground tackle. Every vessel has peculiarities of her own in this respect, therefore it is due, probably, to the crews being in most cases quite strange to their ships, and to the officers not yet having got the hang of things, that so many shortcomings were made apparent. Soon after ten o'clock Admiral Baird, in command of the other section of the fleet, got impatient of further delay, for it was manifest that if he did not start speedily another review might have to be postponed. So he signalled the ships of his squadron to proceed to sea as soon as ready, and shortly afterwards they began filing out eastward in a long single line. But some ships could not obey the order; and amongst these were the *Anson*, *Collingwood*, and *Inflexible*, still engaged in getting up their anchors."

This same writer concluded his account of the spectacle with the remark that "A grander, a more magnificent demonstration of England's Fleet it would indeed be difficult to imagine." But behind the seeming of things there stood revealed an organization which, though it had recently been greatly improved, still left much to be desired in rapid and efficient action. Moreover, at this time even in the Channel Fleet, which then consisted of five ships, and was the only fully commissioned force in home waters, the main purpose of sea-power, to shoot straight, was certainly not kept in view. In his interesting book of reminiscences, "The Navy as I have Known It," Admiral the Hon. Sir Edmund Fremantle, describing the conditions which existed in the Jubilee year, records: "We had large crews and, as all the ships were masted, there was a fair amount of sail drill, while I fear gunnery was little attended to."

There is no record of the impressions which the German Emperor carried home with him from Spithead, but it is more than probable that, while his Majesty was impressed by the great display of ships and men, he was not less impressed by the failure to utilize these resources to the best possible advantage.

The British Navy was living on its past achievements. Though it possessed a mass of material and a large personnel, neither was well organized for war. The available resources exceeded anything belonging to any other nation, but the fleet still basked, content, in the glow of the triumphs achieved in the early years of the nineteenth century. The Navy was unreformed. Steam had taken the place of sails, wood had been superseded, first by iron and then by steel, but the routine of the squadrons, the training of officers and men had undergone little change. The conditions of naval warfare had altered, but the British Fleet remained faithful to the old regime, holding fast to the belief that when war occurred there would be a sufficient interval to allow it to complete its arrangements, elaborate its plans, and place all its resources on a war footing. As the British Navy in its influence on world policy inspired German ambitions, so German thoroughness in organization, when applied to the growing German Fleet, reacted upon the British Navy and gave it a new and vigorous life.

FOOTNOTES:

[7] This ship was, of course, the predecessor of the present *Dreadnought*.

CHAPTER V

THE GERMAN NAVY ACTS

Among the political developments of the last quarter of a century there is none more remarkable than the evolution of German naval ambitions as revealed in the legislation passed since 1898.

One of the first acts after the Emperor ascended the throne was the reorganization of the central Navy administration, which had hitherto been presided over by a general officer of the army. This fact in itself indicates the subordinate position which the navy had hitherto occupied in the defensive machinery of the German Empire. The fleet itself was of extremely modest proportions. It consisted only of a few small battleships of heavy gun-power, but limited radius of action, whose rôle was the defence of the coasts of Germany, and more particularly the Baltic littoral, for at this period few men-of-war under any flag cruised in the North Sea. The spearhead of the British Navy was exposed in the Mediterranean, where the latest and most powerful ships were stationed, and the small Channel Fleet spent most of its time not in the Channel, but ringing the changes on Vigo and other Spanish ports—Lisbon, Lagos, Gibraltar, Madeira, and Port Mahon. This squadron consisted of five obsolescent ships, and the only British vessels permanently in home waters—so complete was the domination of the situation in southern waters—were a number of port and coast-guard ships, half manned and distributed round the coast, and the unmanned vessels in reserve in the dockyards. The distribution of the French Fleet was on much the same lines, the majority of the modern ships being concentrated in the Mediterranean, while a small force was based upon Brest. Russia alone was represented in northern waters, and it was consequently in the Baltic that the German Fleet, such as

it was, was trained and drilled. Except for a few gunboats, the German naval ensign was entirely unrepresented in distant seas, and public opinion showed no desire to increase the naval votes in order to enable German influence to be exercised beyond home waters.

After the Emperor's accession to the throne in June, 1888, and after the reorganization of naval administration, an effort was made to obtain an increased grant from the Reichstag, but only with partial success. From 1874 to 1889-90 the naval expenditure had increased gradually from £1,950,000 to about £2,750,000. In 1890-91 the Estimates had advanced to nearly £3,600,000, and in the following year they rose still further to £4,750,000, and then they began to fall once more under the pressure of the Reichstag, which viewed with no sympathy the new naval ambitions which were finding expression in the Press. During these years the Reichstag repeatedly reduced the votes put forward by Admiral von Hollmann, the Minister of Marine. Throughout his period of office, from 1890 to 1897, he failed signally to inoculate the Parliamentary majority with the new ideas and the new enthusiasm which dominated the Marineamt; and at last in 1897, after being repulsed, first by the Budget Committee and then by the Reichstag itself, the Marine Minister, whose ambitions were really extremely modest, retired from the scene, compelled to admit defeat. He was a sailor and neither a statesman nor an administrator, and his blunt methods were not to the liking of the politicians. No surprise consequently was felt when three months after this final humiliation the Admiral resigned his office. One of the pioneers of German sea-power, Admiral von Hollmann began, under the inspiration of the Emperor, the naval movement which, a few years later, under the impulse that the Boer War imparted to public opinion, and with the help of an elaborate Press Bureau, was carried to such lengths by his successor.

On the resignation of Admiral von Hollmann, the Emperor appointed as Naval Secretary a comparatively unknown naval officer named Tirpitz. Born on March 19th, 1849, at Cüstrin, and the son of a judge, Alfred Tirpitz became a naval cadet in 1865, and was afterwards at the Naval Academy from 1874 to 1876. He subsequently devoted much attention to the torpedo branch of the service, and was mainly responsible for the torpedo organization and the tactical use of torpedoes in the German Navy—a work which British officers regard with admiration. Subsequently he became

Inspector of Torpedo Service, and was the first Flotilla Chief of the Torpedo Flotillas. Later he was appointed Chief of the Staff of the naval station in the Baltic and of the Supreme Command of the German Fleet. During these earlier years of his sea career Admiral Tirpitz made several long voyages. He is regarded as an eminent tactician, and is the author of the rules for German naval tactics as now in use in the Navy. In 1895 he was promoted to the rank of Rear-Admiral, and became Vice-Admiral in 1899. During 1896 and 1897 he commanded the cruiser squadron in East Asia, and was appointed Secretary of State of the Imperial Navy Office in January, 1897. In the following year he was made a Minister of State, and in 1901 received the hereditary rank of nobility, entitling him to the use of the prefix "von" before his name.

With the advent of this officer as Marine Minister, German naval affairs at once underwent a change. His predecessor, who entertained very modest theories as to the size of the fleet which Germany should possess, had attempted to browbeat the politicians, thumping the table in irritation when he could not get his way. The new Minister from the first adopted other methods. He devoted himself to the education of the people by means of an elaborate Press Bureau, and was soon the undisputed master of German naval policy. He met opposition in the Reichstag with a smiling reasonableness, and set himself to win the support of opponents by good-tempered argument. In fact, Admiral von Tirpitz from the first revealed himself as a politician and diplomatist, and from the time that he took office, though now and again slight checks were experienced, naval policy in Germany made rapid—indeed, astonishing—progress.

In the year after Admiral von Tirpitz went to the Marine Office, a Navy Bill, far more ambitious in its terms than any proposal that had been put forward by Admiral von Hollmann, was accepted by the Reichstag.

This measure was believed to embody at any rate the beginnings of a scheme which he had submitted to the Emperor some time prior to his appointment. At any rate, it enunciated a new and vital principle. As has been seen, the Government, whether Prussian or German, had on previous occasions drafted extensive naval programmes for the carrying out of which a period of ten or twelve years was required. Not once, however, had the establishment of ships and personnel been fixed by law; and the Parliament in each case committed itself to the entire scheme only to the extent of

passing the first annual instalment considered necessary by the Government as the initial step towards the desired goal. In this way neither Diet nor Reichstag bound itself or its successors for the future, but left both free to deal with the annual naval estimates as they thought fit. And in practice it had been found that very liberal use was made of the budgetary prerogatives, that standards once approved were not considered binding, and that the fate of the naval estimates depended to a considerable extent on the relations which happened for the moment to exist between the Government and the majority parties on questions totally unconnected with the naval requirements of the Empire.

Another disadvantage of the practice of leaving the Reichstag free to determine annually the number of vessels which should be laid down in a given year, was, that it gave the shipbuilding, armour-plate, and ordnance industries no sure basis for their plans for the future. The rule that Germany must build, engine, arm, and equip her own war vessels had been generally accepted, but the industries which should enable her to do this were still in their infancy, and were almost entirely dependent upon the orders of the home Government. If they were ever to be able to supply the demands of a powerful fleet, it was necessary that slips should be multiplied, plant increased, and workshops extended. But so long as the naval policy of the Empire was indefinite and subject to violent fluctuations, ship-builders and manufacturers would not endanger their businesses by locking up large amounts of capital in appliances which could be used for the building and arming of warships and for no other purposes. If German industry was ever to be in a position to satisfy the demands of a large and efficient fleet, some guarantee of steady and remunerative orders must, it was urged, be afforded to the trades concerned. And apart altogether from its own needs, the Government also hoped that, some day, Germany would be able to claim a share in those large profits which Great Britain appropriated to herself as the world's shipbuilder.

It was by such arguments that Admiral von Tirpitz justified his demand that the strength of the fleet, the date at which it should attain that strength, and the age at which each ship should be automatically replaced by a new one, should be fixed by legal enactment. No portion of his Bill was more hotly contested than this. It was objected that, by accepting it, the Reichstag would be depriving itself of a considerable portion of that power of the

purse which constituted the only effective bulwark of its rights. But in the end the smiling and imperturbable patience of Admiral von Tirpitz gained the day, and the Reichstag satisfied itself with the formal right of drawing the absolutely unavoidable conclusions from its own enactment and passing every year the naval estimates, which could not be rejected without an infraction of the law. The repeated sections in the Act of 1898 which appear to reserve the Chamber's Budget rights, are, in reality, meaningless and valueless—except as a monument to the folly of those who believed they had a meaning and a value. Admiral von Tirpitz apparently drew from his first legislative experience the perfectly correct conclusion that the Reichstag can be made to do almost anything if one only treats it in the right way.

In the explanatory Memorandum attached to the Bill, Admiral von Tirpitz was able to adduce two convincing reasons why the fleet should at once be considerably augmented. One of these was the fact that Germany's naval strength had in recent years actually diminished. In case of mobilization, it was pointed out, she would have had only seven efficient battleships, whereas she had once had fourteen. Of the armoured cruisers which had been adopted in other navies for foreign service in times of peace, she did not possess a single example, and their work had to be done by three antiquated battleships. Moreover, to the tasks allotted to the fleet in the Memorandum of 1873, another of great importance had been added—namely, the defence of Germany's newly-acquired colonial empire. Further, it was contended that the growth of the Empire's population, trade, and industry, the development of her sea-fisheries, and the increasing investment of German capital abroad, had all added to the possibilities of her becoming involved in quarrels with other nations. The fleet which Admiral von Tirpitz considered necessary to fulfil the old and the new sea requirements of the Empire was as under:

THE BATTLE FLEET.

- 19 battleships (2 as material reserve).
- 8 armoured coast-defence vessels.
- 6 large cruisers.
- 16 small cruisers.

FOREIGN SERVICE FLEET.

Large Cruisers.

For East Asia	2
For Central and South America	1
Material reserve	3
	—
Total	6

Small Cruisers.

For East Asia	3
For Central and South America	3
For East Africa	2
For the South Seas	2
Material reserve	4
	—
Total	14

1 station ship.

The period proposed for the gradual attainment of this strength was seven years, but the Reichstag shortened it by a year, and thus it became known as the "Sexennat." It was pronounced inexpedient to attempt to fix for some years in advance the Empire's requirements in torpedo craft, school-ships, and training ships.

That this scheme was intended by its author to be merely a beginning has been shown by the sequel, but Admiral von Tirpitz himself little dreamed that he would so soon be able to take the next and decisive step, which should bring him to within measurable distance of his goal. Early in 1899 he said in the Budget Committee: "I declare expressly that in no quarter has the intention to submit a new navy plan in any way been manifested; that, on the contrary, in all quarters concerned, the firmest intention exists to carry out the Navy Law, and to observe the limits therein laid down." In other words, the law was to run its six years' course.

Nevertheless, before the year was at an end, the Bill which was to become the Navy Law of 1900 had already been announced by the Government.

In the light of the vast development of Germany's colonial and commercial interests the Navy Act of 1898 was of an unambitious character. The

German Fleet was at this time still the weakest possessed by any of the Great Powers of Europe, except Austria-Hungary, which then had no naval pretensions. If only as a matter of historical interest it is interesting to record that, at the moment when this effort towards expansion was made, Germany kept in commission only four ships which could be dignified with the description of battleships, together with four smaller armoured vessels. The only modern ships of the line under the German ensign consisted of these four battleships of the *Worth* class, vessels of 9,874 tons displacement in comparison with ships of 15,000 tons which had already been incorporated in the British Fleet. The German ships, though nominally battleships, were really only coast-defence vessels, heavily gunned and thickly armoured, but with storage for only 680 tons of coal; whereas contemporary British ships of the *Majestic* class possessed a capacity of 1,850 tons.^[8] The four vessels of this class, in addition to the *Worth*, were the *Weisenburg*, *Kurfürst Friedrich Wilhelm*, and *Brandenburg*. They marked a notable advance on the little armoured ships of three to four thousand tons of the *Siegfried* class, which had been built during the early 'nineties, but, owing to their limited fuel capacity, their radius of action was extremely restricted, and they were, in fact, only very powerful coast-defence ships, with a speed on trial of between sixteen and seventeen knots. The design of every armoured ship is a compromise between armament, armour, speed, and coal capacity, and in this German design a predominance then unprecedented in any navy in the world was given to the two first-named characteristics. On paper these ships were vessels of great offensive power, as is revealed by the contrast given on p. 103 between them and the contemporary battleships of the *Majestic* class of the British Fleet, which displaced about 15,000 tons and attained a speed of eighteen knots, with 1,850 tons of coal on board.

These few details reveal the fundamental differences between the character of the British and German Navies at this time and the policy which they represented. The British Government, in accordance with precedent, was providing a fleet of the high-seas type, while the German Government was content with a small force built specifically for the purpose of coast-defence. These four large German coast-defence ships were at this time supported by the four vessels of the *Sachsen* type of 7,283 tons, already obsolescent; by six old ships—one dating back to 1868; by eight little armoured vessels of the *Beowulf* class, of about 3,500 tons, which had been

constructed during the early 'nineties; and the tail of the list was brought up by eleven armoured gunboats each of 1,000 tons displacement. This enumeration of the naval forces of Germany indicates conclusively the modest ambitions which hitherto had animated her naval administration.

The German Fleet, except for the purposes of coast defence, and specifically for the protection of her Baltic shores, was a negligible quantity, having no

	Majestic Class.	Worth Class.
Length	Over all, 413 ft. (390 ft. at water-line.)	380 ft. 6 in. (354 ft. 3 in. at water-line.)
Beam	75 ft.	65 ft. 6 in.
Mean draught	27 ft. 6 in.	24 ft. 4 in.
Armour	Partial 9-in. Harveyed belt, 16 ft. broad, and 220 ft. long; bulkheads, 14 in. (max.); barbettes, 14 in.; barbette-shields, 10 in.; casemates (12), 6 in.; protected deck 2·5 to 4 in.; forward conning tower, 14 in.; after conning tower, 3 in.	Complete belt, 11·8 to 15·7 in. (compound in earlier, steel in later, ships); barbettes and conning-tower, 11·8 in.; ammunition hoists, 11·8 in.; gun-hoods, 5 in.; cellulose cofferdam belt; casemate for 4·1-in. guns, 3 in.; steel deck, 3 in., flat on top of belt.
Armament	4 12-in. 46-ton (wire-wound) breech-loading; 12 6-in. quickfirers in casemates; 16 12-pounder quickfirers; 2 12-pounder boat-guns; 12 3-pounder quickfiring; 2 Maxims; 5 torpedo-	6 11-in. Krupp breech-loading, 2 in each barbette; 8 4·1 in. quickfiring of 30 calibres in a casemate forward of the centre barbettes; 8 3·4-in. quickfirers of 30 calibres; 2·23 in. breech-loading boat or

tubes (18-in.), 4
submerged, 1 above
water astern.

field guns; 12 1-
pounder quickfirers; 8
machine; 3 torpedo
tubes, 2 submerged.

influence either upon European or world policy. The truth of this statement is conclusively proved by the following table showing the relative strength of the only five navies of the world which were, at that time, of appreciable importance, the fleets of Japan and of the United States being then still in their infancy:

	Britain.	France.	Russia.	Italy.	Germany.
Battleships:					
First-class	29	14	6	8	4
Second-class	7	8	4	2	4
Third-class	18	7	5	3	6
Total battleships	54	29	15	13	14
Coast-defence ships	14	16	13		18
Cruisers:					
First-class	23	8	6		1
Second-class	47	13	3	5	3
Third-class	34	9	1	9	9

Total cruisers	104	30	10	14	13
Torpedo gunboats	34	19	8	15	4

It must be confessed that at this time the German Fleet bore no reasonable relation to Germany's growing trade and oversea interests. But the mass of the people of the German Empire were still unconscious of any deficiency, and, blinded by the success of their armies during the war with France and the small influence which naval power exerted in that struggle, they had refused for many years to take upon themselves the burden which the new naval ambitions represented.

But with the passage of the Navy Act of 1898, and the widespread agitation carried on by the Navy League, under the highest patronage, and—even more important—by the Press Bureau under Admiral von Tirpitz, a change immediately occurred; and the success with which the British forces were enabled to conduct their military operations in South Africa, while Europe was forced to stand by inactive, owing to the supreme control which the British Fleet possessed of sea communications, produced a revulsion of feeling. The current of European events, and the reception with which the Emperor's speeches met, convinced the Government, within a comparatively few months of the passage of the Act of 1898, that they might safely abandon this modest measure and replace it by a new Bill.

What had happened in the meantime? This: the outbreak of the Boer War had generated in Germany an absolutely unprecedented hostility to Great Britain, which was afterwards roused to white heat by the seizure of the mail steamer *Bundesrat* and other German vessels on the African coast. Admiral von Tirpitz had a unique opportunity such as was never likely to present itself to him again. He made prompt and full use of it, and while Great Britain was in the thick of the embarrassments of the early stages of the South African War, the great Navy Bill of 1900 was passed into law.

The seizure of the German vessels was admitted by the British Government to have been a blunder. An apology was tendered to Germany on account of it, and promises made that similar incidents should not recur. The action of

the British warships did nothing but harm, and would certainly never have been taken if the Foreign Office in London had been properly informed on the situation in Germany by its representatives in Berlin, and had itself kept the Admiralty fully posted.

Consequently, in the spring of 1900, the Act of 1898 was replaced by a new one, in face of all Admiral von Tirpitz's protestations of two years before. *This measure set up an establishment of almost twice the size of the former one, and embraced ships intended for battle purposes on the high seas.* During the discussion of the measure in the Reichstag the Centre Party compelled the Government to modify their original scheme, and to drop five large and five small cruisers for service on foreign stations, while the reserve of cruisers was reduced by one large and two small vessels. In the course of the debate the Naval Secretary announced that, while the Government were compelled to agree to the amendment of their proposals, they still insisted upon the necessity of providing the original number of ships for duty in foreign seas, but would agree to postpone the final settlement of the question until a subsequent date.

In its final form, as it received the approval of the Reichstag and of the Emperor, and as it was published in the *Imperial German Gazette* of June 20th, 1900, the Bill set up the following establishment for the Fleet:

THE BATTLE FLEET.

- 2 fleet flagships.
- 4 squadrons, each of 8 battleships.
- 8 large cruisers for scouting purposes.
- 24 small cruisers for scouting purposes.

FOREIGN FLEET.

- 3 large cruisers.
- 10 small cruisers.

RESERVE.

- 4 battleships.
- 3 large cruisers.
- 4 small cruisers.

The new Act was based upon the same calculation of the effective life of ships as the one of 1898, and provided that, except in the case of total loss, battleships were to be replaced after twenty-five years and cruisers after twenty years. It was provided that the age of ships was to be reckoned from the grant of the first instalment in payment for the ship to be replaced to the passing of the first instalment in payment for the ship to be built as "substitute" (Ersatzschiff). It was proposed to keep half the battle squadrons—the First and Second—fully manned on a war footing, together with one-half of the torpedo craft and all the school-ships and auxiliary vessels. The Third and Fourth battle squadrons were to form the Reserve Fleet, half the ships of which were to be kept in permanent commission. The Act also made provision for nucleus crews for the second half of the torpedo-boats, for the requirements of ships serving abroad, and for the needs of the shore establishments.

More remarkable, perhaps, than the actual terms of the Navy Act was the character of the explanatory Memorandum put forward by the Navy Department.^[9] In this notable document occurs the following statement of the new naval policy of the German Empire:

"To protect Germany's sea trade and colonies, in the existing circumstances, there is only one means: Germany must have a battle fleet so strong that, even for the adversary with the greatest sea-power, a war against it would involve such dangers as to imperil his position in the world.

"For this purpose it is not absolutely necessary that the German battle fleet should be as strong as that of the greatest naval Power, because a great naval Power will not, as a rule, be in a position to concentrate all its striking forces against us. But even if it should succeed in meeting us with considerable superiority of strength, the defeat of a strong German fleet would so substantially weaken the enemy that, in spite of a victory he might have obtained, his own position in the world would no longer be secured by an adequate fleet."

The Memorandum well repays study in the light of subsequent events. Almost at the moment of its publication Admiral von der Goltz, a former Chief of the Admiralstab, gave a less reserved exposition of German policy,

thus reflecting the opinions held by the naval officers responsible for the character of the proposed expansion of the German Fleet.

"Let us consider," he said, "the case of a war against England. In spite of what many people think, there is nothing improbable in such a war, owing to the animosity which exists in our country towards England, and, on the other side, to the sentiments of the British nation towards all Continental Powers, and in particular against Germany. These are not Chauvinistic exaggerations, but the opinion of the whole of the people of Great Britain, who are jealous of our commercial development. If England should ever lose her mercantile supremacy on the seas, the decline of her naval dominion would only be a question of time, and she realizes the fact instinctively. Of course the British Government will make every effort to prevent the violent explosion of these sentiments, preferring peaceful competition to war. But how long can that last? Violence becomes a right to a people which fears for its existence.

"The opinion is generally held in this country that any resistance against England at sea would be impossible, and that all our naval preparations are but wasted efforts. It is time that this childish fear, which would put a stop to all our progress, should be pulled up by the roots and destroyed.

"At this moment (1900) we are almost defenceless against England at sea, but already we possess the beginnings of a weapon which statesmanship can put to a good use, and our chances of success in a war against England grow more favourable day by day.

"The maritime superiority of Great Britain, overwhelming now, will certainly remain considerable in the future; but she is compelled to scatter her forces all over the world. In the event of war in home waters, the greater part of the foreign squadrons would no doubt be recalled; but that would be a matter of time, and then all the stations oversea could not be abandoned. On the other hand, the German Fleet, though much smaller, can remain concentrated in European waters.

"With the increases about to be made it will be in a position to measure its strength with the ordinary British naval forces in home waters (then

consisting only of the small and inefficiently manned Channel squadron); but it should not be forgotten that the question of numbers is far less important at sea than on land. Numerical inferiority can be compensated by efficiency, by excellence of material, by the capacity and discipline of the men. Careful preparation permitting rapid mobilization can ensure a momentary superiority."

With the passage of the Navy Act of 1900, Germany proceeded to develop a High-Sea Fleet—a naval force capable of going anywhere and doing anything. Hitherto her ships had represented in their design the domination of a coast-defence policy. She now entered upon the construction of ships of the first class. Naval construction was regularized, and forthwith proceeded with great rapidity. During the five years—1886 to 1890—no ship even nominally of the battleship class was launched. During 1891 to 1895 only four vessels, and between 1890 and 1900 only six vessels, and these all of relatively modest fighting power, were put in the water, but in 1901 no fewer than five first-class battleships were sent afloat.

At the time when the Navy Act of 1900 was passed Germany had just completed the five ships of the old Kaiser class, with a displacement of about 11,150 tons, and mounting four 9·4-inch guns of 40 calibre as battle weapons in association with a large number of secondary guns—eighteen pieces of 5·9 inches. The technical advisers of the German Admiralty at this date pinned their faith to a storm of projectiles from quick-firing guns, and in order that weights might be kept down and the ships might be restricted to dimensions to enable them to navigate the Kiel Canal, reliance was placed upon the 9·4-inch gun at a moment when in practically all the navies of the world a 12-inch weapon was being mounted.

The type of battleship design which was introduced with the passage of the Act of 1898, and which was yet in hand when the measure of 1900 was prepared, still combined a weak main armament of four 9·4-inch guns with an exceedingly heavy secondary armament and a complete armoured belt. Whereas British ships at this time, such as those of the Duncan class, were being given only partial belts, and these only 7 inches thick amidships, tapering off fore and aft, the German vessels received thicker belts extending over the whole length. Of this new design—known as the Wittelsbach class—five units were building when the 1900 Act was passed.

They had a maximum coal capacity of 1,770 tons of coal, with 200 tons of oil, and were capable of steaming at a speed of about eighteen knots, thus reflecting the rise of German ambition for something more than a coast-defence fleet. The belts of these ships were 7·5 inches wide, with a thickness amidships of 8·9 inches, while the four 9·4-inch guns were protected with armour 9·8 inches thick, and the secondary turrets and casemates carrying the eighteen 5·9-inch guns were protected with armour 5·9 inches thick.

After the passage of the Navy Act of 1900 the 9·4-inch gun, as the battle weapon, was abandoned in favour of an 11-inch of 40 calibre, and the displacement of the new ships of the Deutschland class, as they are generically termed, although there are minor differences in the ten vessels, was nearly 13,000 tons. These ships really represented the entrance of Germany upon the high seas as a first-class naval Power, possessing vessels fit to lie in the line and to fight the men-of-war under any foreign flag. The new design may be contrasted with advantage with that of the Worth class which has already been described:

DEUTSCHLAND CLASS.

Armour.

Krupp, complete belt, about 7 feet wide,^[10] 8·9 inches amidships, tapering to 3·9 inches at ends; lower edge amidships, 6·7 inches; lower deck side amidships, 5·5 inches; main turrets and barbettes, 11 inches to 9·8 inches; secondary turrets, 6·7 inches; battery, 5·9 inches; conning-tower, 11·8 inches; s.t.—aft, 5·5 inches; deck, 2·9 inches on slopes, 1·6 inches on flat.

Armament.

Four 11 inch (40 calibre) in pairs in turrets, fore and aft; 14 6·7 inch (40 calibre), 10 in battery on main deck, 4 singly in turrets on upper deck; 12 3·4 inch (24 pounder); 4 machine; torpedo tubes, 6 (18 inch), 4 submerged, 1 bow, and 1 stern.

Simultaneously with the construction of these ten battleships, six armoured cruisers, ranging in displacement from 8,800 to about 11,000 tons, were laid

down, and in 1906 a single clause amending the Act was passed increasing the foreign fleet by five armoured cruisers and the fleet reserve by one armoured cruiser, thus fulfilling in part the original programme of the Navy Department with which the Reichstag had interfered.

At about the same date German naval opinion made a complete *volte face* in regard to the fighting value of the submarine. About the time when the Act of 1900 was passed the British Admiralty, after a careful study of the progress of submarine navigation in France and America, decided that it could no longer ignore this type of man-of-war. It was forthwith decided to buy an experimental ship from the Holland Company of the United States, which had already demonstrated the practical value of this particular type of submersible torpedo-boat. The original craft which was purchased under these circumstances was a little ship with a submerged displacement of only 120 tons, and a water-line displacement of 104 tons. She was propelled on the surface by a four-cylinder gasoline engine giving a speed of eight to eight and a half knots, while below the surface she was driven by an electric motor, and was capable of only six or seven knots.

The entrance of this little ship into the British service was hailed in Germany with something approaching derision, and in the technical papers the futility of the submarine was urged with a wealth of argument. The little Holland boat, however, was merely the foundation from which the British authorities proceeded to develop a type of craft in keeping with the offensive rôle of the British Navy, and in 1906 submarines were being built for the British Fleet mounting two torpedo tubes on a displacement of about 300 tons, and possessing a surface speed of fourteen knots in combination with a submerged speed of ten knots. When it is added that these craft possessed a full speed radius of about 3,000 miles on the surface and were estimated to be able to travel 150 miles under water, it is not surprising that German naval opinion as to the advantages of the submarine underwent a sudden and dramatic change. Henceforth the submarine was to be treated by German naval officers with respect. Without the formality of any public announcement, either in the Reichstag or in the Press, an under-water boat was laid down at the Germania Yard at Kiel in 1906, and thenceforward an energetic policy of construction was pursued, although it was not until two years later that legislative provision was made for the building of this type of warship.

A very remarkable feature of German policy has been the persistency with which cruisers have been built even at a time when other naval Powers, including Great Britain, were inactive. As a matter of course, during the period when the German Government was content to provide a fleet mainly for the purposes of coast defence, great importance was attached to the efficiency and adequacy of the cruiser squadrons. At the time of the passage of the Navy Act of 1900, for instance, there were eighteen cruisers completed and nearly a dozen others in hand. Under the Act of that year provision was made to continue this policy while attaining a higher standard of battle strength.^[11] Even when, in 1908, legislative effect was given to the ambition of the Marine Office further to expedite battleship construction, in spite of the heavy cost involved by the transition from mixed armament ships to the all-big-gun ships of the Dreadnought era, the Reichstag was asked to stereotype the cruiser programme. The Act made provision for two light cruisers to be laid down annually, and in the measure passed in 1912 an addition of two "small cruisers" was made for the period 1912-1917. A notable contrast is provided by a study of Germany's action and the policy of the British Admiralty charged with the protection of a vast oversea trade and half the shipping of the world. During the later years of the last century and the first four years of the present century a persistent policy of construction was pursued both in armoured and protected cruisers, and then for several years there was a complete cessation of this form of shipbuilding activity. Other countries, Germany only excepted, either acting on their own initiative or accepting the lead of the British authorities also desisted from cruiser construction. The advance in the size and cost of large armoured ships threw heavy burdens upon the respective Exchequers, and no doubt the saving effected was a welcome relief at a moment when under every flag naval expenditure was advancing at an unparalleled rate. The result of the persistent policy adopted by Germany became apparent in 1911, when in modern swift cruisers suitable for scouting the two fleets were practically upon an equality. It was in these circumstances, faced by evidence of German progress in cruiser construction, that the British authorities again decided to embark upon the building of new squadrons of cruisers of small size and high speed—in fact, of considerably smaller size than the ships then in hand in Germany.

But in battleship construction German policy has necessarily been less continuous and consistent. The war between Russia and Japan in the Far

East, and the lessons which it taught to the naval world were destined to upset completely the theories upon which battleship and larger cruiser design in Germany had been based in the early years of the present century. The German naval authorities had persisted in attaching primary importance to the secondary gun, still believing in the moral and material effect of a storm of projectiles from numerous quick-firing guns. They were still proceeding with the construction of ships—battleships and large cruisers—embodying these ideas when a new Board of Admiralty in London, with Admiral Sir John—now Lord—Fisher as First Sea Lord, appointed a Committee to reconsider the design of British ships in the light of the information which the gunnery tests of the fleet and the struggle in the Far East had supplied.

Thanks to the British alliance with the Japanese, British officers, and British officers only, had been permitted to be present with the Japanese Fleet during the decisive battles of the war. With the advantage of the information thus obtained the designs of British ships were reconsidered. The report of this Committee was treated as confidential. In presenting the Navy Estimates for 1905 to the House of Commons, the Earl of Selborne, the First Lord, contented himself with making the following statement as to the work of this body, and of the new programme of construction:

"I may claim that the work of the Committee will enable the Board to ensure to the Navy the immediate benefit of the experience which is to be derived from the naval warfare between Russia and Japan, and of the resultant studies of the Naval Intelligence Department. I can however hold out no hope that it will be consistent with the interests of the public service to publish either the reference to the Committee or its report.

"It is proposed to begin during the financial year 1905-06: 1 battleship, 4 armoured cruisers, 5 ocean-going destroyers, 1 ocean-going destroyer of the experimental type, 12 coastal destroyers, 11 submarines.^[12]

"His Majesty has approved that the battleship should be called the *Dreadnought*, and the first of the armoured cruisers the *Invincible*."

It was not until many months later that it gradually became known that the British Admiralty were embarking upon the construction of an entirely new type of battleship, and it was even later that information was available as to the character of the "armoured cruisers" mentioned in the First Lord's statement. In the following spring a partial revelation of the change in British design was made in the *Naval Annual*:

"The *Dreadnought*, officially laid down at Portsmouth on October 2nd, 1905, though some material had already been built into her, was launched by His Majesty on February 10th, 1906. The Admiralty announce that the period of building for armoured vessels is to be reduced to two years, but the *Dreadnought* is to be completed in February, 1907. The rapidity of her construction will therefore out-rival that of the *Majestic* and *Magnificent*, which were completed within two years from the date of the laying of their first keel plates.

"The *Dreadnought* represents a remarkable development in naval construction, which has been for some time foreshadowed, notably by Captain Cuniberti, the famous Italian naval constructor. The Russo-Japanese War, more particularly the Battle of Tsushima, established the fact that naval engagements can, and will, be fought at greater distances than were formerly considered possible. Hence the medium armament is held by many authorities to lose much of its value."

In the *Naval Annual* of that year, it was reported that the Japanese contemplated laying down a battleship with an armament of four 12-inch and ten 10-inch guns. It was then announced that the *Dreadnought* was to carry a main armament of ten 12-inch 45 calibre guns, of 50 per cent. greater power than those carried by the *Majestic*, while the medium armament was to disappear entirely.

The question of protection entered also very largely into the consideration, and *The Times*, in describing the new ship, said that it was understood that "she was to be made as nearly unsinkable as possible from the explosion of a torpedo or mine." It was even stated that there would be no openings in the watertight bulkheads, and this proved to be the fact. Moreover, this ship was the first large vessel in the world to be fitted with turbines.

It was stated unofficially that this new ship of the all-big-gun type rendered obsolescent practically all the battleships of the world with mixed armaments—that is with guns of varying size. The British naval authorities continued to maintain a discreet silence as to the character of the new vessels, and the design, as its main characteristics became known, was assailed with a good deal of criticism. The controversy was at its height when President Roosevelt called upon Commander Sims, the Inspector of Target Practice in the United States Navy, to make a report upon the advantages possessed by the all-big-gun ship of high speed and complete armour protection in view of the criticism of the British design of Admiral Mahan.^[13] Commander Sims, who had made a life-study of gunnery questions, prepared a long report describing the character of the revolution in design, and its influence upon the navies of the world. It is interesting to recall some passages from this report, which in its essential portions appeared in the proceedings of the United States Naval Institute, particularly as the British Admiralty have never considered it wise to enter upon a detailed defence of their policy. Commander Sims stated:

"Concerning the advisability of building all-big-gun ships, that is, discarding all smaller guns (except torpedo-defence guns) and designing the ships to carry the maximum number of heavy turret guns, these alone to be used in battle against other ships, I think it could be clearly shown that Captain Mahan is in error in concluding that it would add more to our naval strength to expend the same amount of money that the big ships would cost, for smaller and slower ships, carrying the usual intermediate guns (6-inch, etc.); and that, as in the question of speed, this error is due to the fact that much important information concerning the new methods of gun-fire was not considered by the author in preparing his article. (NOTE.—Unfortunately these methods of gun-fire cannot at present be specifically explained in a published article, as this would involve a discussion of our methods of controlling our ships' batteries, and bringing our ships into action with an enemy.)

"I may, however, assure the reader that, from the point of view of the efficiency of gun-fire alone, it would be unwise ever to build a man-of-war of any type whatever, having more than one calibre of gun in her main battery. In other words, it may be stated that the abandonment of

mixed-battery ships in favour of the all-big-gun, one-calibre ship was directly caused by the recognition of certain fundamental principles of naval marksmanship developed by gunnery officers.

"Therefore we have but to decide what the calibre for each class of ships should be, a decision which should present no special difficulty, provided it be first determined how we are to defeat the enemy—whether by the destruction of his ships (by sinking them or disabling their guns) or by the destruction or demoralization of their personnel.

"In this connection the following facts should first be clearly understood—namely:

"1. Turrets are now, for the first time, being designed that are practically invulnerable to all except heavy projectiles. Instead of having sighting-hoods on the turret roof, where sights, pointers, and officers are exposed to disablement (as frequently happened in the Russian ships) there will be prismatic sights, projecting laterally from the gun trunnions, through small holes in the side of the turret, and the gun-ports will be protected by 8-inch armour plates, so arranged that no fragments of shells can enter the turrets.

"2. On the proposed all-big-gun ships the heavy armour belt will be about eight feet above the water-line, and extending from end to end. The conning-tower, barbettes, etc., will be of heavy armour; and there being no intermediate battery (which could not be protected by heavy armour, on account of its extent), it follows that in battle all the gunnery personnel, except the small, single fire-control party aloft, will be behind heavy armour, and that, therefore, neither the ship or her personnel can be materially injured by small calibre guns.

"Considering, therefore, that our object in designing a battleship is that she may be able to meet those of our possible enemies upon at least equal terms, it seems evident that it would be extremely unwise to equip our new ships with a large number of small guns that are incapable of inflicting material damage upon the all-big-gun one-calibre ships of our enemies, or upon the personnel manning their guns."

In the same paper Commander Sims explained the principal tactical qualities that are desirable in a fleet—namely, compactness of the battle formation and the flexibility of the fleet as a unit—that is, its ability to change its formation in the least possible time and space with safety to its units. Proceeding to elaborate his views, Commander Sims stated:

"For example, suppose two fleets of eight vessels each, composed of ships that are alike in all respects, and suppose their personnel to be equally skilful, with the exception of the Commanders-in-Chief, whose difference in energy and ability is such that one fleet has been so drilled as to be able to man[oe]uvre with precision and safety while maintaining one-half the distance between its units that the other fleet requires.

"This is putting an extreme case, but it shows:

"1. That the short fleet, being about half the length of the other one, can complete certain important man[oe]uvres in about one-half the time and one half the space required for similar man[oe]uvres of the long fleet.

"2. That, when ranged alongside each other, the defeat of the long fleet is inevitable, since the rapidity of hitting of the individual units is assumed to be equal, and each of the four leading ships of the long fleet receives about twice as many hits as she can return, though the eighth ship of the short fleet would suffer a preponderance of gun-fire from the fifth or sixth vessel of the long fleet, the seventh and eighth being too far astern to do much damage, as would also be the case if the long fleet had several vessels astern of these.

"It is because of the principle here illustrated that the constant effort of competent flag-officers is to reduce the distance between the units of their fleets to the minimum that can be maintained with safety under battle conditions—that is, while steaming at full speed, without the aid of stadimeters, sextants, and other appliances that should be used only for preliminary drills.

"Doubtless some flag-officers, by constant competitive exercises in man[oe]uvring, may succeed in attaining an interval between ships that is less by 15 or 20 per cent. than that attained by others; but manifestly

there is hardly any possibility of much greater improvement in this respect, because the minimum practical interval between ships depends upon their lengths and man[oe]uvring qualities. For example, the German interval is 300 metres from centre to centre, while larger ships, say 400 feet long, require about 400 yards, and those between 450 and 500 feet in length require about 450 yards.

"If we accept Captain Mahan's advice and build comparatively small, low-speed battleships, while our possible enemies build large, swift, all-big-gun ships, it seems clear that we will sacrifice the enormous advantages of fleet compactness and flexibility, the superior effect of heavy-gun fire and the ability to concentrate our fire—the loss of these advantages to be fully realised twenty-five years hence, when our enemies have fleets of big ships while we still have those of our present size."

Finally, this officer added:

"If it be claimed that it would be better to reduce the speed of the large vessel to sixteen knots and put the weight saved into guns, it may be replied that the heavy turret guns cannot be mounted to advantage (so as to increase the hitting capacity of the vessel) without very considerably increasing the size of the ship, because the number of heavy turrets that can be placed to advantage is governed largely by the length of the ship—which increases slowly with the displacement. This point is fully discussed in a recent article in a German publication. I do not remember the displacement used by the author to illustrate the principle, but, supposing the ones quoted below to be correct, he shows that if it requires a displacement of 20,000 tons to obtain a broadside fire of, say, eight 12-inch turret guns, you could not advantageously mount any additional turrets on 21,000 or 22,000 tons, but would have to go to 25,000 or 26,000 tons to obtain the necessary space. And, conversely, if you design a 20,000-ton battleship for sixteen instead of twenty knots, you cannot utilise the weight saved to

increase the gun-power by adding 12-inch turrets, as you could by adding a number of intermediate guns.

"It is now hardly necessary to state that adding superimposed turrets (by which the number of guns could be doubled, if the weights permitted) does not materially increase the hitting capacity of the ship as a whole, because of the 'interference' caused by having four guns in one two-story turret, while it decreases her defensive power by adding to the vertical height of her vital targets.

"Captain Mahan characterizes the sudden inclination in all navies to increase the size of the new battleships (from about 15,000 to about 20,000 tons) as a 'wilful premature antiquating of good vessels' ... 'a growing and wanton evil.' If these words are intended in their true meaning, the statement is to me incomprehensible. I can understand an individual being wilful and wanton, but I cannot believe that the naval officers of the world could, without good cause, be suddenly and uniformly inspired in this manner. On the contrary, it seems to me that the mere fact of there being a common demand for such large vessels is conclusive evidence that there must be a common cause that is believed to justify the demand.

"This common cause is undoubtedly a common belief that the same amount of money expended for large war vessels will add more to a nation's naval power than the same amount expended for small vessels, for it cannot reasonably be assumed that the tax-ridden nations of Europe expend their great naval budgets wilfully and wantonly. Undoubtedly each nation earnestly strives to expend these sums as to derive the greatest increase of naval power. The same is true in reference to their armies. As the mechanical arts improve each nation endeavours to improve its war material. When a nation adopts new rifles, it is not a wilful premature antiquating of several million excellent ones, it is a case of *force majeure*—it must adopt them or suffer a relative loss of military efficiency, and it must make no mistake as to the relative efficiency of its weapons. In 1870 the French suffered a humiliating defeat as a direct result of the colossal conceit which rendered them incapable of accepting conclusive evidence that the German field artillery was greatly superior to theirs.

"The same law—that of necessity—governs the evolution of battleships. As might have been expected, this evolution has, as a rule, been gradual as regards increased displacement. The exception is the sudden recent increase (4,000 to 5,000 tons) in displacement. This exception therefore needs explanation.... It was due to a complete change of opinion as to the *hitting capacity* of guns of various calibres. This is now well understood by all officers who have recently been intimately associated with the new methods of gunnery training. These methods have demonstrated this point in such a manner as to leave no doubt in our minds as to the correctness of our conclusions. The rapidity of hitting of the heaviest guns has been increased several thousand per cent., and that of smaller guns about in proportion to their calibre.

" ... The inception of the epoch-making principles of the new methods of training belongs exclusively to Captain (now Rear-Admiral) Percy Scott, Director of Naval Practice of the British Navy, who has, I believe, done more in this respect to improve naval marksmanship than all of the naval officers who have given their attention to this matter since the first introduction of the rifled cannon on men-of-war; nor should we forget that this degree of improvement was rendered possible by the introduction of telescope sights, the successful application of which to naval guns was made by Commander B.A. Fiske, U.S. Navy, as early as 1892. As soon as the above facts gained general acceptance in Great Britain and the United States, the evolution of the all-big-gun one-calibre battleship became a foregone conclusion; and the reason for the great increase in displacement, as I understand it, is simply that you cannot build an efficient ship of this class on less than about 20,000 tons, because you cannot mount more than two 12-inch turrets to advantage upon a battleship of much less displacement, because the length and breadth are not sufficient."

The Dreadnought design and all that it meant threw the German Admiralty into confusion. At the moment they were still engaged in the construction of the vessels of the Deutschland class, of about 13,000 tons, in which primary importance was given to the secondary gun—fourteen 6·7-inch weapons—to the sacrifice of the big gun—four 11-inch pieces—and speed; whereas the new British design ignored the secondary gun in order to mount no

fewer than ten big guns, and develop the speed to the extent of three or four knots above battleships then building. Before the *Dreadnought* of the British programme of 1905 had been laid down at Portsmouth, two German battleships of the familiar design with mixed armament had been begun—the *Schleswig Holstein* in the Germania Yard and the *Schlesien* at Dantzig. So completely were the German authorities unprepared for the revolution initiated by the British Admiralty, that from the summer of 1905 until July, 1907, the keel of not a single further battleship was laid in Germany. In the meantime, while British yards were busy with vessels of the new type, the design of the German ships was reconsidered. After an interval of two years the keels of two vessels of the *Dreadnought* type were laid down, and two more keels were placed in position a month later—that is, in August, 1907. These four ships—the Nassau class—inaugurated the Dreadnought policy in Germany. Two were completed in May, 1910, and two in September following.

These ships embody the all-big-gun principle in association with a powerful secondary armament, consisting of a dozen 5·9-inch guns and sixteen 24-pounders. Moreover, whereas the British Dreadnought had been provided with only ten big guns, which was held by the British gunnery experts to be the maximum number which could be carried with advantage on the displacement then considered advisable, the German vessels were given twelve guns, not of the 12-inch but of the 11-inch type. Each of these ships displaces 18,600 tons, and has a nominal speed of twenty knots. Their normal coal capacity is 885 tons, with a maximum storage of 2,655 tons. On the other hand, the early British *Dreadnought*, with about the same displacement and coal-carrying capacity, attained a speed of one or two knots more, owing to the use of turbines in place of reciprocating engines. The contrast between the armour and armament of the British and German ships, comparing the four Nassaus of the German Fleet^[14] with the Superb class of the British Navy, is given in the table on p. 129.

By energetic action the British Admiralty had obtained a lead in the new type of battleship.^[15] Moreover, even after the character of the Dreadnought became known, the German authorities remained ignorant of the fact that the "armoured cruisers" of the Invincible class were really swift battleships carrying the same type of battle gun as the *Dreadnought*, in association with a speed exceeding twenty-five knots, and an armour belt not inferior to that

placed on the latest pre-Dreadnought German battleships. By this decisive move, the British authorities had depressed the value of all mixed armament battleships, in which the British Fleet was becoming weak in face of foreign—and particularly German—rivalry, and had started the competition of armaments on an entirely new basis upon terms of advantage.

No sooner was the true inwardness of the Dreadnought policy realized than the German authorities began the preparation of a new German Navy Act. It was eventually decided that the best

	Superb Class.	Nassau Class.
Armour	Krupp: Complete belt, about 16 ft. wide (narrower aft), 11 in. amidships, tapering to 6 in. forward and 4 in. aft; turrets, 8 in.; barbettes, 12 in.; forward conning-tower, 12 in.; after conning-tower, 8 in.; deck, sloping, 2·7 in.	Krupp: Complete belt, 12 in. amidships, tapering to 3·9 in. forward, and 3·9 in. aft; lower deck side, 7·9 in. amidships, 3·9 in. narrow belt at ends; turrets and barbettes, 11 in.; battery, 6·1 in.; conning-tower, 11·8 in.; deck, sloping, 2·9 in.
Armament	10 12-in. (45 calibres) in pairs in turrets, 1 forward, 1 on each beam, 2 aft on centre line; 16 4-in. (50 calibres), 2 on each turret (except No. 4), 8 in superstructure; 5 machine; torpedo tubes, 5 18-in., submerged, broadside, and stern.	12 11-in. (45 calibres) in pairs in turrets, 1 forward, 1 aft, and 2 on each beam; 12 5·9 in. (45 calibres) in battery; 16 3·4 in. (24-pounder); torpedo tubes, 6 18-in., submerged, bow, stern, and broadside.

means of accomplishing the end in view—namely, the construction of a larger number of ships of the armoured classes in the next few years than was provided in the Act of 1900, was to reduce the nominal effective age, and legislate for the replacement of all battleships and large cruisers within

twenty years. Accordingly, attached to the new Act passed early in 1908, which was over two years after the laying down of the Dreadnought, was a schedule setting forth that four large armoured ships should be laid down annually between 1908 and 1911, both inclusive, and that in 1911 onwards to 1917, two keels annually should be placed in position. By means of this single clause measure, which became law on April 6th, 1908, the construction of ships of the Dreadnought type was accelerated, and whereas the British Admiralty had definitely abandoned the construction of large cruisers of the armoured class—as the German authorities knew by this time—the Marine Office decided that each of the "large cruisers" specified in the Act of 1900 should be swift Dreadnoughts.

This point is an important one. Between 1897 and 1904, Great Britain laid down 27 battleships and 35 armoured cruisers—a total of 62 armoured ships in eight years, or an average of 7·75 ships a year. In this period Germany built 16 battleships and 5 armoured cruisers, or 21 armoured ships—equal to an average of 2·62 ships a year. In 1905 the Admiralty determined to cease building armoured cruisers. In that year they laid down 4 "capital ships"—all of them Dreadnoughts; in the next two years 3 annually, and in 1908, 2 ships only. While the British authorities abandoned the building of armoured cruisers, Germany decided to accelerate her battleship construction, and she also decided that all the "large cruisers" specified in her Law should be swift Dreadnoughts, and thus from 38 battleships and 20 armoured cruisers, she rose to an establishment of 58 battleships.

At the end of 1911, when it was imagined that the German programme would fall from 4 large ships annually to 2 ships, a new Navy Bill was produced.^[16] Incidentally this measure added to the establishment 3 battleships and 2 unarmoured cruisers, and made provision for the construction of a maximum of 72 submarines.

The significance of the successive changes in shipbuilding policy in Germany, reflecting in an ascending scale the naval ambitions of the Marineamt, may be realised from the following summary, showing the establishment of large armoured ships fixed under successive measures:

	Establishment of Ships
Act.	
Adopted.	

	Battleships.	Large Cruisers.
1898	17	8
1900	38	14
1906	38	20
		Dreadnoughts.
1908		58
1912		61

Under the operation of German naval legislation, it was determined to provide sixty-one large armoured ships of maximum power, all of them less than twenty years old. The Act did not specify the character of the vessels of the various classes to be laid down. It was elastic in this respect. It left to the Marine Office complete freedom in the matter of design; but, on the other hand, it tied effectually the hands of the Reichstag, and it could not, except it repealed the Navy Law, reduce in any year the number of keels to be laid down. There could be no reduction in the output of naval material until a new Navy Law had been passed. This is a point which was frequently forgotten in England.

But the notable feature of the Navy Act passed by the Reichstag in 1912 was not the additions to the shipbuilding programme, though these were notable, but the steps taken to increase the instant readiness of the fleet for war. Prior to the passage of this measure it had been the practice in the British Navy to maintain only about half the men-of-war of various classes on a war footing, relegating the remainder to reserves representing various stages of preparedness for action. The German Navy Act of 1912 set up an entirely new standard with a view to obtaining the maximum advantage from a conscript service, where the pay is low, in competition with a voluntary service, such as obtains in the British Fleet, with very much higher rates of pay. In the speech which he delivered in Committee in the House of Commons on July 22nd, 1912, Mr. Winston Churchill, the First Lord of the Admiralty, gave a lucid explanation of the essential features of this German Navy Act. He said:

"The main feature of that Law is not the increase in the new construction of capital ships, though that is an important feature. The main feature is the increase in the striking force of ships of all classes

which will be available, immediately available, at all seasons of the year. A third squadron of 8 battleships will be created and maintained in full commission as part of the active battle-fleet. Whereas, according to the unamended Law, the active battle-fleet consisted of 17 battleships, 4 battle or large armoured cruisers, and 12 small cruisers; in the near future that active fleet will consist of 25 battleships, 8 battle or large armoured cruisers, and 18 small cruisers; and, whereas at present owing to the system of recruitment which prevails in Germany, the German Fleet is less fully mobile during the winter than during the summer months, it will, through the operation of this Law, not only be increased in strength, but rendered much more readily available.

"Ninety-nine torpedo-boat destroyers—or torpedo-boats, as they are called in Germany—instead of 66, will be maintained in full commission out of a total of 144. Three-quarters of a million pounds had already been taken in the general estimate for the year for the building of submarines. The new Law adds a quarter of a million to this, and that is a provision which, so far as we can judge from a study of the finances, would appear to be repeated in subsequent years. Seventy-two new submarines will be built within the currency of the Law, and of those it is apparently proposed to maintain fifty-four with full permanent crews.

"Taking a general view, the effect of this Law will be that nearly four-fifths of the entire German Navy will be maintained in full permanent commission—that is to say, instantly and constantly ready for war. Such a proportion is remarkable, and so far as I am aware, finds no example in the previous practice of modern naval Powers. So great a change and development in the German Fleet involves, of course, important additions to their personnel. In 1898 the officers and men of the German Navy amounted to 25,000. To-day that figure has reached 66,000.

"Under the previous Laws and various amendments which have preceded this one, the Germans have been working up to a total in 1920, according to our calculations, of 86,500 officers and men, and they have been approaching that total by increments of, approximately, an addition of 3,500 a year. The new law adds a total of 15,000 officers and men, and makes the total in 1920 of 101,500.^[17] The new average

annual addition is calculated to be 1,680 of all ranks, but for the next three years by special provision 500 extra are to be added. From 1912 to 1914 500 are to be added, and in the last three years of the currency of the Law 500 less will be taken. This makes a total rate of increase of the German Navy personnel of about 5,700 men a year.

"The new construction under the Law prescribes for the building of three additional battleships—one to be begun next year (1913), one in 1916, and two small cruisers of which the date has not yet been fixed. The date of the third battleship has not been fixed. It has been presumed to be later than the six years which we have in view.

"The cost of these increases in men and in material during the next six years is estimated as £10,500,000 above the previous estimates spread over that period. I should like to point out to the Committee that this is a cumulative increase which follows upon other increases of a very important character. The Law of 1898 was practically doubled by the Law of 1900, and if the expenditure contemplated by the Law of 1900 had been followed the German estimates of to-day would be about £11,000,000. But owing to the amendments of 1906 and 1908, and now of 1912, that expenditure is very nearly £23,000,000. But the fact that the personnel plays such a large part in this new amendment, and that personnel is more cheaply obtained in Germany than in this country, makes the money go farther there than it would do over here.

"The ultimate scale of the new German Fleet, as contemplated by the latest Navy Law, will be 41 battleships, 20 battle or large armoured cruisers, and 40 small cruisers, besides a proper proportion—an ample proportion—of flotillas of torpedo-boat destroyers and submarines, by 1920. This is not on paper a great advance on the figures prescribed by the previous Law, which gave 38 battleships, 20 battle or large armoured cruisers, and 38 small cruisers. That is not a great advance on the total scale. In fact, however, there is a remarkable expansion of strength and efficiency, and particularly of strength and efficiency as they contribute to striking power. The number of battleships and large armoured cruisers alone which will be kept constantly ready and in full commission will be raised by the Law from twenty-one, the present figure, to thirty-three—that is to say, an addition of twelve, or an increase of about 57 per cent. The new fleet will in the beginning

include about twenty battleships and large cruisers of the older types, but gradually, as new vessels are built, the fighting power of the fleet will rise until in the end it will consist completely of modern vessels.

"This new scale of the German Fleet—organized in five battle squadrons, each attended by a battle or armoured cruiser squadron, complete with small cruisers and auxiliaries of all kinds, and accompanied by numerous flotillas of destroyers and submarines, more than three-fourths—nearly four-fifths, maintained in full permanent commission—the aspect and scale of this fleet is, I say, extremely formidable. Such a fleet will be about as numerous to look at as the fleet which was gathered at Spithead for the recent Parliamentary visit, but, of course, when completed it will be far superior in actual strength. This full development will only be realized step by step. But already in 1914 two squadrons will, so far as we can ascertain, be entirely composed of Dreadnoughts, or what are called Dreadnoughts, and the third will be made up of good ships like the *Deutschlands* and the *Braunschweigs*,^[18] together with five Dreadnought battle-cruisers. It remains to be noted that this new Law is the fifth in fourteen years of the large successive increases made in German naval strength, that it encountered no effective opposition in its passage through the Reichstag, and that, though it has been severely criticized in Germany since its passage, the criticisms have been directed towards its inadequacy."

Such is the evolution which German naval ambitions have undergone since the Reichstag in the early years of the Emperor's reign refused to believe that four relatively small battleships in full commission, with the same number of ineffective coast-defence ships of small size, did not represent the maximum naval power which Germany need provide, and that an expenditure of two and three-quarter millions sterling was not sufficient burden to impose annually upon the Teutonic peoples over and above the cost in money and service of the predominant army.

Nothing reveals the statesmanship of Admiral von Tirpitz so strikingly as the character of the naval legislation for which he has been responsible, and the manner in which he has bent every influence in Germany and every occurrence abroad to promote his ends. Prior to the introduction of the

Navy Act of 1898, the only example of a continuous naval policy was the Naval Defence Act of 1889, under which seventy ships of various types were added to the British Navy during a period of four years. Of these vessels only ten were of the armoured classes. This measure was confined to shipbuilding, and it made no provision for increasing the personnel or for setting up a fixed standard of commissioning. It merely provided a certain number of ships and left it to Parliament to provide or not to provide crews with which to man them, and, as a matter of fact, Parliament did not provide the necessary officers and men until long after the ships were at sea. Admiral von Tirpitz was not satisfied with so unmethodical and unstatesman-like a measure of procedure when he went to the Marineamt in 1897. He presented to the Reichstag a complete scheme of naval expansion, making provision not only for the construction of ships in specified numbers over a period of six years, but providing also for the due expansion of the personnel and for the attainment of a fixed establishment of ships first in full commission, secondly with nucleus crews, and thirdly in reserve. In obtaining the assent of the Reichstag to this measure, which to a great extent removed the naval expansion movement from the control which it had hitherto exercised annually on the presentation of the Estimates, the Minister of Marine achieved his first great triumph.

This Act was to have remained in operation for a period of six years, and was represented as an embodiment of German needs, quite independent of the naval preparations then being made by other Powers. During the next two years no development occurred in the naval programmes either of Great Britain or other foreign countries, but an Anglophobe wave passed over the Continent as a result of the South African War. German sympathies in particular were aroused, and Admiral von Tirpitz at once seized the opportunity to repeal the fixed and immutable Fleet Law of 1898, and to replace it by a new enactment providing a Battle Fleet of roughly twice the strength of that legalized in the establishment of the former measure. This measure was to have remained in force until 1917. Six years later—a Liberal Government, intent on disarmament, having assumed office in the United Kingdom—an amendment representing another expansion was passed; two years after that the fourth Fleet Law became operative, and in 1912 another measure was adopted by the Reichstag under the influence of a renewed Anglophobe movement in Germany. Experience has shown that German Fleet Laws are regarded as immutable and fixed when proposals in

the direction of a limitation of armaments are made, but as flexible as though no Fleet Law existed when political circumstances are favourable for making a further effort towards a higher standard of naval power.

Nor does this study exhaust the remarkable features of this naval legislation. An ordinary statesman, ignorant of naval matters, might have so framed the successive Naval Laws as seriously to tie the hands of the naval authorities in the development of the fleet, whereas Admiral von Tirpitz, with great skill, restricted the powers of interference on the part of the Reichstag, while leaving the Marine Office with almost complete freedom in shaping the naval machine in the process of expansion. This double end was achieved by the use of generic naval terms in the loose manner adopted by those unfamiliar with their significance. Admiral von Tirpitz made up his "paper" establishment in the Fleet Laws by styling every ship of slow speed but carrying an armoured belt "a battleship," and then, under the terms of the Law, he made provision for these dummy vessels to be replaced by veritable battleships of maximum power. Thus ships of 4,000 tons displacement have been replaced by Dreadnoughts of 25,000 tons, carrying the heaviest guns, and protected by thick armour. The establishment fixed by the Reichstag has not been exceeded, but by a simple process of conjuring, small coast-defence ships have been quietly converted into first class sea-going battleships, ranking in strategical and tactical qualities with the most formidable ships in the British Fleet. The naval authorities have by this means been able to prove to the uninitiated when challenged that they have kept within the four corners of the Law, that the number of battleships has remained fixed according to the establishment between the periods of each enactment, and at the same time they have been in a position to follow an active shipbuilding policy, while raising from year to year the necessary personnel for manning the new vessels. This is another notable feature of Admiral von Tirpitz's policy. The legislation has been so elastic as to enable him to raise the necessary number of officers and men to suit the requirements of the Fleet. When a Dreadnought, requiring 1,106 officers and men, has been completed for sea to take the place of a ship of the Hagen class, with a crew of only 306, the additional personnel has been instantly ready.

The same process has been adopted in increasing the cruiser squadrons of the German Navy. The Law has specified that a certain number of "large

cruisers" shall be built, and it has been left to the discretion of the naval authorities to interpret this elastic term in tons, guns, armour, knots of speed, and personnel. In accordance with the Law, Admiral von Tirpitz has thus been able to replace cruisers of negligible fighting value and of small size by Dreadnought battle cruisers mounting guns of immense power and attaining speeds hitherto without precedent. Similarly, small torpedo-boats have given way in the establishment of the Navy to torpedo-boat destroyers of large size, and step by step the naval strength of Germany has been increased by a process, the cleverness and ingenuity of which even the German people themselves have not realized.

Germany has immensely increased her resources of ships and men, but she has done more than that: she has forced other Powers to organize and train their squadrons on a standard of efficiency never attempted in the past. She has increased the strain and stress of peace until it resembles closely the actual conditions of war, and having determined year in and year out to keep nearly four-fifths of her fleet always on a war footing, always instantly ready for action, she has compelled other countries, in accordance with the dictates of ordinary foresight, to take similar action, however onerous the financial burden. It is on Great Britain and the United States that the weight of this burden has borne most heavily, for in those States alone is reliance placed on a voluntary system of manning, which is necessarily very costly.

FOOTNOTES:

[8] It is interesting to note, however, that even at this early date the German Admiralty made provision for the storage of oil in order to supplement the coal supply.

[9] *Cf.* Appendix I.

[10] The five later ships were given a belt with a thickness of 9·4 inches amidships, but otherwise their protection and armament closely resembled those laid down at an earlier date.

[11] See Appendix II.

[12] One of these "armoured cruisers" was not built.

[13] It has since become known that the Americans had designed an all-big-gun ship before the British Dreadnought was laid down.

[14] British naval opinion held from the first that these ships of the Nassau type vitiated the Dreadnought principle of simplicity of armament, and were so over-gunned as to be ineffective units. Sea-service has tended to confirm this view.

[15] In the three succeeding years, in accordance with the British Government's policy of a limitation of naval armaments, and as an example to other Powers, this advantage was partially lost, and hence the large programme of 1909-10.

[16] *Cf.* Appendix II.

[17] In his speech in the House of Commons on March 26th, 1913, the First Lord corrected this figure. He stated that the maximum to be attained under the new Fleet Law in 1920 was 107,000, apart from reserves.

[18] These two groups of ships are of practically the same design.

CHAPTER VI

GERMAN SHIPS, OFFICERS, AND MEN

In material, in the art of constructing and equipping ships of war, Germany at the beginning of the war ranked far above most of the Great Powers, and she was little, if anything, behind even Great Britain in workmanship, rapidity and cheapness. Her personnel also stood high, for she had succeeded in translating into naval terms the professional and disciplinary codes which have raised the German Army to a position of pre-eminence. Above all she had succeeded, in a degree never before attempted by any country, in keeping ships and men in constant association. The German naval authorities recognized that, while a conscriptive system of manning a fleet brings into the organization certain grave and ineradicable disadvantages, it did at least enable large numbers of officers and men to be borne for service at a relatively small annual cost. Realising this economic benefit of conscription, the Marineamt had no hesitation in increasing its personnel rapidly from year to year. The expansion of this element of naval power kept pace with the activity of the shipyards. This policy of simultaneous increase of ships and of men, accompanied as it was by the expansion of her shipbuilding and allied industries and of her dockyards, has been the secret of the rapid rise of Germany as a maritime Power wielding world-wide influence.

Within the memory of the present generation German ships of war, if not built in England, were constructed in Germany with materials obtained entirely or in part from England. Her earliest armoured ships of any account

—the *Deutschland*, the *Kaiser* and the *König Wilhelm*—were all constructed on the banks of the Thames at the old Samuda Yard. The great industry which Germany and other foreign nations helped to support is now dead, and on the other side of the North Sea is to be seen an activity more intense and on a far larger scale than the Thames establishments could boast even in the day of their greatest prosperity.

Though there are many shipbuilding yards and engine-making establishments in Germany, the naval authorities depend exclusively upon the vast establishment of Krupp for armour and guns, and the repute of the firm in both respects stands high. The vast establishment which supplies the German and many other Governments was founded in 1810 by Friedrich Krupp, who bought a small forge and devoted himself, with little commercial success, to the manufacture of cast steel. In this he was ahead of Germany's requirements, but on the basis thus laid by the father, the son built; and in 1851 a solid steel ingot which he exhibited at the Great Exhibition in London completely took the metallurgic world by surprise, and his fortune was made. He turned his energy and knowledge to the making of guns, armour, weldless steel rails, and other manufactures; and the modest works at Essen continued to expand until to-day they and the associated establishments give employment to about 70,000 men, not all of whom, of course, always are engaged on the manipulation of armaments.

For many years the Krupp process of armour manufacture was adopted in every country of the world, but later on the British Admiralty, it is common knowledge, adopted a superior process which produces a plate of greater resisting power, and the German cemented type of armour no longer holds the premier position which it occupied when its advantages over the Harvey plate were demonstrated. On the other hand, the Krupp firm still claim that their ordnance is not equalled by any in the world, and on the strength of this claim they have obtained most valuable orders, extending over a long series of years, from foreign Governments. British guns are made on the wire-wound system—that is, steel ribbon is wound under great pressure round the gun, and over this is placed an outer hoop; Krupp's, on the other hand, still remain faithful to the solid steel tube to resist the gas pressures exerted, arguing that their method of steel manufacture enables them to submit it to strains which other steel might not stand. There has been endless controversy as to the merits of the two systems; and the subject was

again discussed as recently as the end of 1912, when the Italian Minister of Marine laid a report before the Italian Parliament with reference to the armaments of the principal fleets. According to this statement the British, Italian, and Japanese are the only Navies to mount wire-wound guns; the probable life of the Italian and Japanese 12-inch guns was given at 80 rounds, whereas the English gun was good for only 60 rounds. On the other hand, the Austrian and German guns were given from 200 to 220 rounds, and the American 14-inch gun was estimated to have a probable life of 150 rounds. Particulars with reference to British and German guns were given as follows:

	British.		German.		
Calibre in inches	12	13·5	12	14	15
Length in calibres	50	45	45	50	50
Weight in tons	69	80	53	83	102
Weight of projectile in pounds	850	1,240	850	1,360	1,650
Initial velocity	2,950	2,800	3,000	3,000	3,000
Energy at muzzle in metric tons	16,540	22,150	17,520	27,650	33,910
Energy per kilogramme in kilometres	240	277	330	330	330
Probable life in rounds	60	60	200	200	200

The attention of the First Lord of the Admiralty was directed to these statements in the House of Commons, and he reiterated the assurance of former Ministers that the expert advisers were satisfied as to the wisdom of retaining the wire-wound system. He gave no data as to the foundation of this confidence, and in the German technical Press—no doubt with an eye

to foreign orders—the superiority of the German gun over the British was repeated with at least equal assurance.

The great advantage of the wire-wound system, it has always been claimed, is that after much use, when the rifling is worn, the gun can be given a new inner tube, a comparatively simple and cheap operation which results in practically a new gun being made available for sea service in a short time. All that can be said as to the two systems from practical experience is that the Japanese found the British-made weapons give eminently satisfactory results during the war with Russia, while the Krupp artillery guns used by the Turkish Army in the Balkan War of 1912 did not realize expectations.

Probably in naval material—in ships, their armour, armament, and engineering equipment—there is little difference as between the leading navies. One may be thought to have an advantage in some particular respect, but this may possibly be counterbalanced by the rival's superiority in another. Generally, the British ships mount fewer guns but of larger calibre, and to the experienced eye they look very workman-like; while the German ships carry smaller guns in greater number and have a crowded appearance which does not appeal to British naval opinion in its desire for simplicity of design and plenty of working room. Virtually, all the instruments for exerting naval power as they exist to-day are experimental, based upon the empirical knowledge. When the war between the United States and Spain occurred, it was anticipated that it would throw light upon these problems, but these anticipations were not realized, and even the struggle between Russia and Japan failed to satisfy fully the natural curiosity of the naval constructor and the naval officer owing to the inefficiency with which the Russian ships were handled, and the deplorable slackness of the administration.

It is the fashion to calculate the relative strength of fleets in tons and guns, but the probability is that on the day of trial in a great battle at sea these nice paper computations will be entirely upset by the course of events. Morale, as Napoleon observed, dominates war. This dictum is no less true to-day than it was in the past. Man is still greater than the instruments of his creation, and the experience of war on a grand scale will certainly confirm the teaching of history—that the important element in naval power is men rather than ships. On the eve of the Battle of St. Vincent, when Jervis, in command of fifteen ships, was pacing the quarter-deck of his flag-ship and

the Spanish Fleet was entering the field of vision, the numbers of the enemy were reported by the Captain of the Fleet to the Commander-in-Chief as they were counted. "There are eight sail of the line, Sir John," "Very well, sir," answered the Admiral. "There are twenty sail of the line, Sir John." "Very well, sir," Jervis responded. "There are twenty-five sail of the line, Sir John." "Very well, sir," the Admiral again replied imperturbably. "There are twenty-seven sail of the line, Sir John," the Captain of the Fleet at length reported, and when he had the temerity to remark on the great disparity between the British and Spanish Fleets, the Admiral, confident in the efficiency of his small fleet, replied: "The die is cast, and if there be fifty sail, I will go through them." We may be sure that the victor of the Battle of St. Vincent, who by stern but wisely directed measures created the fleet which Nelson used with such dramatic effect at Trafalgar, would have scorned and ridiculed an entire reliance on mere paper calculations of guns and tons, realizing that victory or defeat depends mainly upon the personal element and morale.

It is in respect of officers and men that there is the greatest contrast between the British Fleet and the Navies of the Continent of Europe. The British service is organized on a voluntary system, while the Continental fleets are manned mainly by conscripts; the former serve for many years, while the latter for the most part submit to only the short period of duty required by law and then pass into the reserve. In the matter of officers, however, the German Fleet is certainly not worse served than the British Navy; though the cadets begin their training at a somewhat later age, a thoroughly good sea officer is produced. The marked distinction between the two services is that, whereas under the White Ensign special duties are assigned to special classes of officers—gunnery, torpedo, navigation, signalling and physical training—in the German Navy no hard-and-fast lines are drawn. It is held that the British system would entail a larger number of officers than are available on the other side of the North Sea. However this may be, the German authorities can certainly pride themselves upon a corps of executive officers which in many respects is not excelled in any country. As in the British service, special lines of officers are trained for engineering, medical, and accountant duties and these have no executive standing.

The method of training executive officers for the German Fleet differs in some important respects from that which obtains in England. In the British

service the cadets, who enter when they are, on the average, thirteen and a half years of age, have not completed their general education, and consequently spend four years at the Naval Colleges at Osborne and Dartmouth respectively before they go afloat in a training ship. The German naval officer receives much the same general education as any other boy before he enters the navy, whereas the British cadet, after entering, is submitted to an educational course specially devised with a view to his future naval career; his studies embrace physical science and practical engineering, and emphasis is laid upon athletics and as much sea experience as can be obtained in small craft. When the four years ashore are completed he goes afloat at about the same age as the average German cadet and makes a six months' cruise. Which is the better system? Who shall say? This is certain, however, that British naval officers have always held that lads for the sea service cannot be caught, broken in, and inoculated, so to speak, too early.

Throughout the years of naval expansion the German authorities have been struggling to eliminate as far as possible the disadvantages of conscription in its application to naval conditions. The War Department is responsible for putting in force the conscription law, and periodically the navy sends in its requisition, stating the number of recruits who will be needed, and where and when they are to join. The men selected are passed direct into the fleet without preliminary training each October. Under the British system boys are entered at about sixteen years of age, and receive a short training first in one of the shore or stationary sea establishments, and are subsequently drafted into one of the ships of the Training Squadron, thence joining the sea-going fleet. A certain number of youths are also entered at an average age of about seventeen and a half years, and these recruits dispense with the preliminary course, but are also drafted to the Training Squadron before joining the fleet. Nearly all the men of the fleet sign on for twelve years' active service, and the best of these are permitted to re-engage for another ten years in order to earn pensions. A relatively small number of men, not boys, join the British Navy for a term of only five years, with the obligation to remain in the Reserve for seven years. Five years, consequently, is the minimum in the British Navy, and applies to only a relatively small number of men; but three years is the maximum period of German conscripts, and during this time the officers and warrant officers have to do their best to transform the raw material provided by the State into skilled seamen.

It is easy to imagine the difficulties which assail the administration in Germany in these circumstances. Every year one-third of the naval conscripts complete their period of active service and are passed into the Reserve, and their places are taken by an equivalent batch of raw recruits. The result is that in the winter months the officers and petty officers of the fleet are occupied in licking into shape these embryo sailors, and from October until May the fighting ships of the Empire become practically training vessels.

If this were a complete representation of the conditions in the German Fleet its efficiency would be of a low order. The Navy is, however, stiffened by a proportion of conscripts who re-engage voluntarily, and by a certain number of volunteers who enter as boys. These lads engage at ages ranging from fifteen to eighteen years. They agree to undergo an apprenticeship of two years followed by seven years of active fleet service. Volunteers are not trained ashore or in fixed naval establishments as in the United Kingdom, but are drafted to sea-going training ships, which cruise in home waters during the summer months and pass into the Mediterranean during the winter. By these two expedients the German naval authorities have been able to secure about 25 per cent. of the German personnel on what passes in Germany for a long-service system. The boy volunteers and the conscripts who re-engage constitute the class from which petty officers are drawn, and these men are the backbone of the naval organization ashore and afloat, and it is to their efforts that the high standard of efficiency which Germany's Navy has attained may in a large measure be traced.

Year by year, in order to provide crews for the larger number of ships passed into the fleet, the Marine Office has been compelled to increase the number of conscripts required for sea service, and thus the task of training the Navy has been increased in advance of the expansion of the material, because men must begin training before their ships are ready for sea. The officers and petty officers have had not only to train raw recruits embarked to take the place of conscripts at the end of their three years' term, but to find means also of training additional recruits entered as net additions to the naval strength. When it is added that in 1894 the number of officers and men in the Navy was less than 21,000, whereas it is now nearly 80,000, and under the Navy Act of 1912 is to be raised to 107,000, some conception may be formed of the character of the problem which has presented itself,

not only to the central administration ashore, but to the officers afloat, intent upon attaining the highest standard of efficiency at sea. Admission of these difficulties was made by Admiral von Tirpitz in the explanatory Memorandum which accompanied the last Navy Bill presented to the Reichstag and which directed attention to "two serious defects" in the organization of the fleet:

"The one defect consists in the fact that in the autumn of every year the time-expired men—*i.e.* almost one-third of the crew in all ships of the battle fleet, are discharged and replaced mainly by recruits from the *inland population*. Owing to this, the readiness of the battle fleet for war is considerably impaired for a prolonged period."

When it is recalled that the maritime population of Germany amounts only to 80,000, and that compulsory service in the active fleet lasts for only three years, it will be realized that most of the recruits taken for the German Navy must necessarily be landsmen. The personnel in 1914 numbered roughly over 70,000, after deducting from the total the executive officers, engineers, cadets, and accountants. If approximately 17,000 of these are regarded as long-service men there remain roughly 54,000 conscripts, one-third of whom pass annually into the Reserve, and are replaced by raw hands. Under the new Navy Law it was intended to strengthen the personnel in the next few years by 6,400 annually. While the average period of service in the British Navy, including the relatively small number of five years' men entered for short service, is about ten years, the average in the German Fleet does not amount to as much as half this period.

It is possible to attach too much importance to the fact that the German Navy is recruited "mainly by recruits from the inland population." The inherited sea habit counts for less to-day than at any time since men attempted to navigate the seas. Ships of war have become vast complicated boxes of machinery, and naval life requires the exercise of qualities different from those it demanded in the sail era. Then brute courage, endurance, and familiarity with the moods of the sea were the main attributes of sailors, but to-day a large proportion of the crews must be experts in the handling of complicated mechanical appliances. In these changed conditions the compulsory system of education in Germany has proved of the greatest advantage in providing recruits of a high standard of

intelligence, who probably acquire in six months as complete a familiarity with their work as it would have taken a seaman of the old school as many years to attain. At the same time, while resisting the temptation to place too great importance upon the inherited sea habit, it would be no less a mistake to ignore entirely its influence upon naval efficiency. Familiarity breeds contempt for the terrors of the sea and for the horrors of a naval action, and it is reasonable to expect that in the hour of trial the long-service men of the British Navy will exhibit a moral standard when projectiles are falling fast and thick far higher than that of the conscript. A modern Dreadnought is intended to fire its guns in broadsides and not in succession, and when it is borne in mind that at one discharge these guns will deliver on an enemy's ship, if they are fired accurately, between five and six tons of metal, it will be realized that at such a moment the calibre of men will count more than the calibre of guns.

When the Act of 1900 was introduced the Reichstag was informed by Admiral von Tirpitz in a Memorandum that "as, even after the projected increase has been carried out, the number of vessels in the German Navy will still be more or less inferior to that of other individual Great Powers, our endeavours must be directed towards compensating this superiority by the individual training of the crews and by tactical training by practice in larger bodies.... Economy as regards commissioning of vessels in peace time means jeopardizing the efficiency of the fleet in case of war." Never since navies existed have a body of officers and men been worked at higher pressure than those of Germany; drill has never ceased; no effort has been spared to obtain the last ounce of value out of every one on board the ships. The promotion of officers rests with the Emperor, and he is unsparing in his punishment of anything like slackness; an officer who is not enthusiastic, alert, and competent, stands no chance of rising in rank. The German Navy has no use for anything but the best which the Empire can provide, and in order that the highest expression of the *esprit de corps* which has contributed to German influence on shore may be instilled into the Navy, no officer, however influential or brilliant, can enter either the executive or engineering branch unless his claims are endorsed by all his contemporaries; one black ball—if the term may be used—is sufficient to disqualify an aspirant, though he may have passed all the prescribed examinations brilliantly.

The German Fleet has its limitations, but within those limitations it probably has no superior in the world: the ships are well built, the officers are capable sailors, and the men are raised to the highest pitch of efficiency possible under a short-service system.

CHAPTER VII

WILLIAM II. AND HIS NAVAL MINISTER

The German Fleet, as it is to-day, may be regarded as the work of two men—the Emperor William II. and Admiral von Tirpitz.

Even for those who have lived long in Germany, it is difficult to form a judgment as to the aims and motives of the Emperor William's naval policy, and of the part which he has played in its carrying out. With regard to their sovereign, Germans are inclined to fly to one of two extremes; according to the class to which they belong, they represent him either as a heaven-born genius of universal gifts, or as a busybody whose meddlesomeness is rendered specially mischievous by mediæval delusions as to the functions of monarchs and their relations to the Deity. Everything that he does or says is set down as quite right by the one party and as quite wrong by the other. Moreover, the opinions of those brought into closest contact with him are vitiated by the prevalence of a type of sycophancy which is fortunately becoming extinct in other countries.

The patriotic German, who is familiar with his country's history, knows that, five or six hundred years ago, his forefathers monopolized the markets and policed the seas of Northern and Western Europe. He realizes keenly that Germany's maritime and industrial progress was first checked, and then retarded for centuries, by political division and internecine and foreign wars. Possibly he still remembers that great crescendo of victory in which Prussia smothered Denmark, then overthrew Austria in a single battle, and finally, at the head of the kindred Teutonic States, humbled France in the dust, and welded Germany together in one indivisible whole. Even if he does not remember it as part of his own personal experience, all its vivid and stimulating episodes have been a thousand times impressed upon his

mind by schoolmaster, politician, historian, and journalist. That after this tremendous martial achievement he should regard his country as the mistress of the continent of Europe is no matter for surprise. But he sees, too, that the Germany of Luther and Goethe, of Ranke, Liebig, Helmholtz, and Mommsen, of Bismarck and Moltke, has become also the Germany of Krupp, Siemens, Rathenau, Ballin, and Gwinner; that the products of German industry, the fruits of an unexampled application of the discoveries of science to the processes of manufacture, have been carried by German ships to the remotest ends of the earth; that the material prosperity of his country has been advancing in every direction by leaps and bounds. And he thus believes Germany to be strong, wise, and wealthy, and in every way fitted to stand at the head of mankind. But in one respect he has felt, to his bitter mortification, that she is powerless. Wherever he goes on the world's oceans, he is confronted by those iron walls of Great Britain, which mean that he is there only by the sufferance of one who is immeasurably stronger than himself.

The German patriot has never realized that no efforts on the part of Germany could materially alter the balance of sea-power to her advantage as against Great Britain, and that she would be compelled to fight for her pretensions long before she was in a position to give battle on anything like equal terms. He has believed that the British nation is unnerved and effete, that it has lost both its martial and industrial vigour, that its energies have been sapped by too much wealth and prosperity, and that it is rapidly following the downward path. Finally, he is convinced that the British Parliament, under the influence of an aggressive democracy, exclusively concerned with its own immediate material needs, is losing the capacity to realize and grapple with the larger problems of international politics, and that the Cabinets proceeding from it will, in timorous anxiety, procrastinate and vacillate till it is too late to strike. In this idea he has been only confirmed by the pacifist movement in Great Britain, by the British agitation for disarmament by international agreement, and by the well-meant but unfortunate attempt of Sir Henry Campbell-Bannerman to effect by example what much amiable precept had done nothing to accomplish. These phenomena he has looked upon not as evidence of good-will and peaceableness, but as symptoms of physical, moral, and financial exhaustion.

Such was the view of many in Germany to whom we cannot fairly deny the name of "patriot" if we are to claim it for an analogous disposition among ourselves. It was the view almost universally held by the officers of the German Army and Navy, and, with certain qualifications and reservations, it may be said to have been the view of the Emperor William. This will be evident if, with the help of his many spoken and written utterances, we attempt to follow the main lines which, with many sudden and violent deviations, his thought has taken on this subject. He has, for example, in his speeches repeatedly dwelt on the power and renown of the Hanse League—"one of the mightiest undertakings that the world has ever seen," which "was able to raise fleets such as the broad back of the sea had probably never borne up to that time," which "won such high prestige for the German name abroad," which "created markets for the German industrial regions," and which "only failed because it lacked the support of a strong united Empire obedient to a single will." At Hamburg, in June, 1911, he used these words: "I have only acted historically, for I said to myself on my accession, that the tasks which the Hansa attempted to solve by itself, and which it could not solve because the strong Empire was not at its back, and the defensive and executive power of the Empire did not exist, must unquestionably at once fall on the shoulders of the newly-arisen German Empire; and it was simply the obligations of old traditions that had to be resumed." It was in one former Hanse town that the Emperor spoke the familiar words, "Our future lies on the water"; in another that he declared "The trident should be in our hand"; in a third that he uttered the appeal, "We have bitter need of a strong German Fleet."

Again, he has repeatedly extolled the Great Elector—"the one among my ancestors for whom I have the most enthusiasm, who has from my earliest youth shone before me as a bright example," who, "looking far ahead, carried on politics on a large scale, as they are carried on to-day." In his great speech at Bremen in 1905, the Emperor said: "When as a youth I stood before the model of Brommy's ship, I felt with burning indignation the outrage that was then done to our fleet and our flag"; and these words undoubtedly referred to the injudiciously-phrased note in which Palmerston threatened that vessels which undertook belligerent operations under the colours of that greater German Empire, which then was not and was never to be, would render themselves liable to be treated as "pirates." The present realities of sea-power had been early revealed to him when, as he told the

officers on board a British flag-ship in the Mediterranean, he "was running about Portsmouth Dockyard as a boy"; and, as he said in a speech made during the visit of King Edward to Kiel in 1904, "the stupendous activity on the sea at the headquarters of the greatest navy in the world impressed itself indelibly on his youthful mind," and made him, "as Regent, endeavour to realize on a scale corresponding to the conditions of his country what he had seen as a young man in England."

How far the Emperor has helped to realize his own naval ambitions, and how far his efforts have actually told against them, it is very difficult to determine with anything like exactitude. His agitation for a bigger fleet has been open and unwearying, and outside Germany the idea is very prevalent that he not only contrived the naval policy of the Empire, but also, almost single-handed, generated the degree of popular support without which it could not have been carried out. This idea will be seen to be erroneous. The Emperor's influence upon his own people is very greatly overrated in other countries, and even the crisis of 1908, in which the storm of discontent which had long been gathering burst with full force upon his head, does not seem to have been properly understood outside Germany. On that occasion, the Imperial Parliament listened without a protest, without a murmur, as a Liberal deputy, slowly, deliberately, and with dramatic emphasis, spoke the following words: "In the German Reichstag not a single member has come forward to defend the actions of the German Emperor." The incident was without a parallel in the history of parliaments. Even the Conservative party, which has always gloried in being the chief prop of the throne, passed and published a resolution expressing the wish that the Emperor should "in future exercise a greater reserve in his utterances," and declaring that "arrangements must be made to prevent with certainty a recurrence of such improper proceedings." It may be remarked, in passing, that this blow fell upon William II. because he had confessed to having had Anglophile sentiments, and to having performed friendly services to Great Britain, at a time when the general feeling of the German people was one of hostility to this country. Nor was it without significance that when, after holding aloof from public affairs for several weeks, he at last emerged from the solitude of his palace at Potsdam, it was in England that he sought the recuperation and rest of which he stood in need.

The dismissal of Bismarck and the subsequent attempts of the Emperor to depreciate the life-work of the man to whom he owed the Imperial crown, were, of course, the principal causes of the spirit of opposition which flared up with such startling suddenness in 1908. The popularity of William I. was in no small measure due to his absolute trust and confidence in his Chancellor, and the abrupt ejection of this incomparable statesman from his office will never be forgotten or forgiven till the generation of his contemporaries has passed away.

These things go far to explain why it was that, in spite of the vigorous naval agitation of the Emperor, the German Fleet, as was pointed out in the Memorandum attached to the Bill of 1898, became weaker instead of stronger during the first ten years of his reign. From the day of his accession he had lost no opportunity of manifesting his interest in the fleet and his desire that it should be largely increased. Among his earliest acts as monarch was his unheralded appearance in admiral's uniform at a parliamentary luncheon given by Bismarck, to decorate one of the guests who had displayed sympathies and wishes with regard to the Navy similar to his own. Year after year, tables of diagrams, showing the disparity between the fleet of Germany and those of the leading naval Powers, and prepared, it is said, by the Emperor's own hand, were sent out over his signature to the Reichstag, the Government departments, and all public institutions where it was thought they might meet the gaze of appreciative eyes. At a *soirée* given at the New Palace at Potsdam in 1895, he assembled round him a group of members of the majority parties of the Reichstag, and lectured them for two-and-a-half hours on Germany's need of sea-power. Bismarck's eightieth birthday was then approaching, and the Emperor concluded his remarks by urging upon his hearers that they should seize the opportunity of "doing the founder of our colonial policy the pleasure of passing the sum absolutely required for the Navy." A couple of years later, he delivered a similar address after a dinner given to members of the Reichstag by the Finance Minister, von Miquel, illustrating his arguments with the diagrams of warships mentioned above. About the same time, an English illustrated paper published a picture of the foreign war vessels on the East Asian station. Among them, as the sole representative of Germany, was a small gunboat, which, as was pointed out in the accompanying text, was "under sail only." Against these words the Emperor wrote, "What mockery lies therein," and the picture, with this comment, was laid before

the Budget Commission of the Reichstag, then engaged in the discussion of the naval estimates. Moreover, the monarch had himself recourse to the paint-brush, and exhibited in the Berlin Academy of Arts a picture of an attack by a flotilla of torpedo craft on a squadron of ironclads. No doubt he hoped in this way to arouse sympathy for his ideas in some who were not accessible to the ordinary methods of political persuasion. The "Song to Aegir," the Scandinavian Neptune, of which he composed the music, was probably also intended to have a similar operation.

But all these pleas and cajoleries had little or no positive result. Indeed, taken in conjunction with other phrases of the Imperial activity, they seem rather to have excited opposition in the breasts of the members of the Reichstag, who possibly considered themselves just as well qualified as the monarch to estimate the degree and appreciate the needs of Germany's maritime interests, and at any rate half-suspected that his efforts directly to influence their deliberations involved an encroachment on their constitutional privileges. The first naval estimates submitted in the new reign, which provided for the laying down of the unusually large number of four battleships, were got through the Reichstag without much difficulty, but when Admiral von Hollmann became Minister of Marine in the following year, he found that quite a different temper had taken possession of the Parliament. It was not only that the Emperor's general governmental acts had begun to stir up opposition; his oratorical flights in praise of sea-power and world-empire had also generated strong suspicions that he was urging Germany along a path which would lead her to ruin at home and disaster abroad. Hollmann's by no means exorbitant demands were branded both in the Reichstag and the press as "unconscionable," his programme as "boundless," and on every side were heard contemptuous and impatient references to "the awful fleet." For a decade the naval estimates were ruthlessly and recklessly cut down to, on an average, not far short of half their original figure, and finally, in 1897, the ministerial career of Hollmann was terminated by the unceremonious rejection of three out of the four cruisers which, in a special Memorandum, he had sought to prove were indispensable for the protection of the Empire's stake on the seas. And all this time the Emperor had never ceased to agitate, by word and deed, for the ideas which he had so much at heart and to which the Reichstag nevertheless showed itself so completely indifferent, if not actually hostile.

The change that came with the appointment of Admiral von Tirpitz to the Ministry of Marine was as complete as it was sudden, and it is to this very able man that we must look if we wish to find not only the intellectual author of German naval legislation, but the statesman who devised and directed the means by which it was popularized and passed through the Reichstag. The transformation which he effected was one both of policy and of method. The three rejected vessels which brought about Hollmann's fall represented a principle—that of "cruiser warfare." At that time the imperfectly-thought-out strategy of the German Naval Ministry was based on the two ideas of coastal defence and commerce destruction. Pitched battles between ships of the line on the high seas played a very secondary part in its calculations. In the programme which he submitted to the Reichstag, Hollmann laid it down that fifteen battleships would be sufficient for Germany's purposes, and those who are best qualified to form a judgment of the Empire's naval policy at that epoch are of opinion that this number was intended to be not merely a provisional, but a final estimate of the country's requirements in this type of vessel.

There are good reasons for supposing that in the Hollmann era no clear idea existed as to the problems with which Germany might be confronted in a naval war, and that his programmes were the product rather of vague general principles than of calculated odds and chances. In fact, one of his main difficulties with the Reichstag was his inability to justify his estimates by numerical demonstrations.

On the other hand, Admiral von Tirpitz's strength always lay chiefly in this, that he knew exactly what he wanted and why he wanted it. When he came into office, it was generally stated that he had years previously already laid before the Emperor a Memorandum embodying his conception of Germany's maritime needs, and how they could be satisfied, and it is certain that the main outlines of his policy were at any rate clearly sketched out in his head long before he was given an opportunity of carrying it out. He was recalled from the command of the East Asian Squadron to take charge of the Naval Ministry, and he seems to have employed his leisure on the homeward voyage in drafting a programme, which he had worked out in all its details before he took over his portfolio. In its very fundamental principles it was a reversal of that of his predecessor, for it was based on the idea, probably adopted from Mahan, that battleships alone are the decisive

factors in naval warfare. As he himself put it in the Reichstag: "If we have a strong battle fleet, the enemy will have to defeat it before he can blockade our coasts. But in such circumstances he will, before he declares war on Germany, consider very carefully whether the business will cover its expenses and justify the risk." It was this principle of risk which he took as his standard of the Empire's naval requirements. From the literature which he inspired it is evident that he was one of those who believe that Germany was destined to occupy the position on the seas which now belongs to Great Britain. It was, however, impossible for a Minister of State to argue this belief in public, for the open confession of it would have at once produced incalculable complications in international affairs which would certainly not have contributed to its realization. Besides, the consummation which he wished for could in any case only be reached by gradual stages over a long period of years. The defensive formula which he invented was quite as effective for his immediate domestic purposes, and, as the sequel showed, was not appreciated abroad in its true and full significance. It was that "the German Fleet must be so strong that not even the greatest naval Power will be able to enter upon a war with it without imperilling its position in the world."

It was only after a good deal of hesitation, and some resistance, in high quarters that Admiral von Tirpitz was able to make his view prevail. Even courtly panegyrists admit that at the commencement of his term of office deep-seated differences of opinion existed between him and the Emperor on cardinal points of naval policy. The monarch was then a firm adherent of the cruiser-war theory, and no doubt had been responsible for its adoption by his Ministry of Marine. It may be regarded as his most substantial contribution to the present strength of the German Fleet that he finally yielded to Admiral von Tirpitz's arguments.

In one other very essential respect the new Minister revolutionized the policy of his predecessor. In the Memorandum already referred to, Hollmann defined the needs of the navy only for the three succeeding years, and in the course of the debate on the estimates, he used these words: "Neither the Federated Governments nor the Reichstag will ever agree to be bound to a formal programme for years in advance. That is quite impossible, and even if both factors desired it, impossible, for the very simple reason that the art of war is changeable on sea just as it is on land,

and that to-day no Naval Ministry can prophesy what we shall need ten years hence. It can only tell you what are our immediate requirements, and if the circumstances change, then our demands will change too. As to that there is no doubt whatever." Here again, Admiral von Tirpitz not merely modified, but diametrically reversed the policy of his predecessor, and, it may be added, of the Emperor. Starting from the conclusion that the main types of war vessel and their respective functions remain unaltered in principle throughout the ages, he induced the Reichstag to commit itself statutorily to a fixed warship establishment, a building programme of nearly twenty years' duration, and an automatic renewal of the units of the fleet when they had reached a prescribed age. This is the one absolutely new feature of German naval legislation, and it was undoubtedly the idea of the new Minister.

Admiral von Tirpitz has, in fact, been the Bismarck of German naval policy, and just as the Iron Chancellor fulfilled the hopes of the men of the Frankfort National Assembly, so the smiling and urbane Minister of Marine has gone far towards realizing the dreams of Friedrich List and Prince Adalbert of Prussia. It may be questioned whether he would not have done this work quite as effectually without the Emperor's loud and tempestuous advocacy of his schemes on the open stage of the world. The trumpet tones in which William II. proclaimed his dreams of world-wide rule and maritime dictatorship, not only exercised a disquieting effect in foreign countries, but conjured up in the minds of many Germans unpleasant visions of provocative and perilous adventure. Other nations were anything but delighted at the prospects of being swallowed up in a universal Teutonic Empire, however peaceful its conquests and however beneficent its rule, and they took steps by which the successive moves of German naval policy were successively counteracted.

If we may judge from the discretion which he has shown by keeping as far as possible in the background, Admiral von Tirpitz would, if left to himself, have built up the German Fleet with the same silent and systematic persistency with which Bismarck, Roon, and Moltke prepared to crush France, and to some extent he combines in his character the qualities of these three. He is at any rate the adroitest politician, the ablest organizer, and the most far-sighted strategist in the Imperial service. Long before he was thought of as Naval Minister, he had won for himself among his

colleagues, by the skill and thoroughness with which he grappled with every problem allotted to him, the title of "The Master." It was he who, against the ignorant protests of the older school of naval officers, chiefly concerned for the smartness of their paint, the cleanness of their decks, and the brightness of their brasswork, forced the torpedo upon them, and brought the service of this weapon up to the high pitch of efficiency which it has to-day attained in the German Fleet. As Chief of the Staff to the General Command of the Navy, he evolved fresh rules of strategy and new tactical formations, and insisted upon Man[oe]uvres being carried out in such a way as to test the value of both. He has been no less successful as statesman, politician, and diplomatist. Here, too, he deserves the name of "Master" among his contemporaries, for what he has done has been the greatest ministerial achievement of our day. It is true that he was favoured by an extraordinary run of luck that was vouchsafed to none of his forerunners, and that he would never have been able to drive his machine but for the energy generated by a series of international dissensions, but at the same time it must be conceded that he took advantage of his opportunities with rare promptitude and address.

He at once took the measure of the Reichstag, and saw how he could make it obedient to his will. It is traditional in the higher ranks of the German official hierarchy to despise popular assemblies, and to treat them with an air of pedagogic superciliousness. Hollmann had become so impatient at the continual mutilation of his estimates that at last he thumped his fist menacingly on the table. That precipitate action sealed his fate. Admiral von Tirpitz recognized that it would be better for him if he disguised his contempt, and smothered his anger in his beard. In one of Rostand's plays, a lady is asked how she passed the sentries who were posted round a jealously guarded camp, and she replies: "I smiled at them." If the Naval Minister were to be asked how he induced the parties who had been so obdurate to his predecessor's demands to pass his own so much more expensive projects, he, too, might have replied: "I smiled at them." Completely breaking with the tradition of schoolmasterly superiority, he was all complacency and urbanity to the ignorant mediocrities who had it in their power to frustrate his designs. His beaming rubicund countenance was ever the brightest and most ingratiating feature in the debates on his bills and estimates. His good humour was inexhaustible, his courtesy unflagging, his patience undisconcertable. He knew exactly what he wanted, and

thought only of that. His mind was not clouded, like those of so many of his ministerial colleagues, by religious or political prejudices. He was ready to accept ships from the hands of Catholics or Socialists. Whether they ranked the Pope above the Emperor, or preferred a republic to a monarchy, was quite indifferent to him, if only they would grant him the ships and the men he asked for.

In one of his many veiled conflicts with the Foreign Office, Admiral von Tirpitz is understood to have exclaimed: "Politics are your affair—I build ships!" and it was precisely because he attended strictly and conscientiously to his own business that he was able to do it so well. It was incumbent upon him as administrator of the Navy to make it as strong and efficient as possible, and it lay with the Chancellors to decide whether the line he was following was consistent with the general policy of the Empire. That, against their own convictions and what they conceived to be Germany's foreign interests, they allowed him to have his own way, only proved their weakness and his strength.

While he was amiable and polite to all parties and persons who could assist him in the carrying out of his ideas, flattered their vanity by pretended confidences from the region of high politics, took them for cruises in war vessels, and had them deferentially escorted round Imperial shipyards, the Admiral was quick to appreciate the importance of winning the good graces of the Catholics, without whose favour, as party relationships stood and were likely to stand, he could hope to effect little. Young and active members of the Centre party, who showed a particular interest in the details of naval policy, were singled out for special attention, and soon were numbered among his most devoted champions. He likewise realized the value of popular support, and this was secured through the instrumentality of the Press Bureau of the Ministry of Marine. This institution was administered in the same spirit which gained the Admiral his parliamentary triumphs. The naval officers by whom it is manned have always received all journalists, domestic and foreign, with open arms, and, according to the objects and nationality of their visitors, furnished them with ideas, information and directions. No German writer on naval affairs could afford to dispense with official assistance so profusely and willingly supplied. The Press Bureau placed at his disposal all the historical and statistical data which could be used to demonstrate Germany's need of a big fleet, all the

articles from the foreign press which were likely to have a stimulating effect upon his readers, all the details of ship and gun types which could safely be made public, all the rules of naval strategy and tactics which might be of service to him in the formulation of his themes. If diffidence or a spirit of independence prevented him from coming to the Press Bureau, the Press Bureau went to him, as will be seen from the following document which found its way into print:

"IMPERIAL MINISTRY OF MARINE,
"NEWS OFFICE.

"BERLIN,
"——, 1907.

"It has become known here that, some time ago, you published in —— articles of a maritime nature. For this reason the News Office gladly takes the opportunity of enquiring whether you would care to receive occasional batches of service material and press comments for possible use in further articles. In view of the impending Navy Bill, your support in the Press might be particularly valuable in the immediate future.

"Your most obedient servant,
"BOY-ED."

By such means the Admiral succeeded in obtaining a control, gentle, persuasive, and veiled, but none the less effective, over practically the entire body of writers on naval topics in the German Press.

The unanimity of view on naval subjects which the Bureau imported into the German Press was naturally most effective. When the simple citizen found that all the papers to which he had access spoke with one voice, simultaneously adopting an identical attitude to a fresh situation or propounding a novel theory, he could only assume that they must be in the right. The proposal that Great Britain should abandon her Two-Power standard and accept in its stead a ratio of three to two, which appeared almost at the same moment in a score of different papers while the 1912 Navy Bill was under process of dilution, is an instance in point. Up till then all naval writers in Germany had been unanimous in protesting that

agreements to fix a naval ratio between two countries were in their very nature impossible, and the suddenness and simultaneity of their conversion must have been due to the intervention either of Providence or the Marine Minister. Indeed, the Minister's statement a year later in the Reichstag Budget Commission definitely set at rest any doubt that might have existed as to the original source of the proposal. Since Bismarck, no one has shown such adroitness as Admiral von Tirpitz in the management of the Press.

In addition to controlling the naval views of independent publications, the Press Bureau also makes important direct contributions of its own to periodical literature with the annual *Nauticus* and the monthly magazine *Die Marine Rundschau*. Both these publications are further testimonies to the energy with which the Admiral performs the duties of his office.

But with all his cleverness, perseverance, and patience, Admiral Tirpitz would never have reached his goal had not Germany been swept by successive waves of Anglophobia. Both speeches in the Reichstag and articles in the Press make it quite evident that the motive uppermost in the minds of most deputies when they voted for the Navy Bills was the desire to impress, annoy, or terrify Great Britain. The truth is that, but for the Boer War, the Bill of 1900 could never have been so much as introduced; but for the perpetual international friction over Morocco and the fantastic legend of King Edward's designs against Germany, the Bills of 1906 and 1908 would have had but small chance of acceptance; and but for Mr. Lloyd George's speech and Captain Faber's indiscretions—and, it should be added, the misrepresentations of both of them by Admiral von Tirpitz's Press—the Ministry of Marine would never have been able to win its last victory against the opposition of the Treasury and the misgivings of the Chancellor. The lesson of 1848 cannot be too thoroughly learnt. The naval movement of that year was almost entirely popular in its character. It arose out of a sense of wounded dignity, and fits of national temper, blind to all the prudential considerations of domestic and international politics, have given Germany to-day the second largest fleet and the largest Socialist party in the world. It may seem almost like a contradiction in terms to suggest that a national sentiment has contributed to swell German Socialism to its present dimensions. But this is—for Germany, at any rate—no paradox, for in no other country does so small a proportion of the population constitute what is in practice and in effect the "will of the people."

It should have become clear that the part which the Emperor William has played in the formulation and carrying out of Germany's naval policy has been quite insignificant in comparison with that played by his Minister. The really effective work which the monarch has done for his fleet has been that of which the wider public has heard least. The Emperor's brain is not an originating or creative one, but it is keenly apprehensive, appreciative, and assimilative, and its owner was quick to perceive the value of many of the forces and institutions which have made the British Fleet supreme, not only in numerical strength but also in *esprit de corps* and organization. From his visits to England he took back much useful information as to the construction and handling of ships, and in many other respects he found British models which he considered worthy of imitation in his own country. Thus the Institution of Naval Architects was provided with a German counterpart in the Schiffbau-technische Gesellschaft, the ideals of self-discipline of sport were fostered in the Imperial Navy, and when the temperance movement in the British Fleet had developed sufficient strength to attract attention, the Emperor inaugurated a similar propaganda among his crews. As has already been seen, William II. has generously admitted the debt of the German Fleet to its British sister, and beyond all doubt he has done more than anyone else to incur it.

The Emperor has also been able to do a good deal towards the propagation of his naval ideas through his autocratic control over the official machinery of Prussia, which constitutes more than three-fifths of the area, and nearly that proportion of the population of Germany. In a country where the tentacles of the central authority reach to the remotest village this control means a great deal. In particular, through the Ministry of Education, the rising generation has been initiated into the mysteries of "world-policy" and sea-power. The teaching of history and geography has been used to impress upon susceptible minds the importance of colonies and fleets, and to suggest with more or less precision and emphasis that Great Britain is the jealous rival who chiefly obstructs Germany's path to that "place in the sun" which is her due. The process, commenced in the schools, has been continued at the universities. Indeed, here as elsewhere, Germany's professors have been the pioneers of her progress, and were putting forward her claim to sea-power long before the Emperor was born. Friedrich List, the father of German economics, urged, in 1840, that Denmark and Holland should be taken into the Germanic Confederation, which "would then

obtain what it at present lacks—namely, fisheries and sea-power, ocean-borne trade, and colonies." In another passage he said:

"What intelligent citizens of those seaports (Hamburg and Bremen) can rejoice over the continual increase of their tonnage, when he reflects that a couple of frigates, putting out from Heligoland, could destroy inside twenty-four hours the work of a quarter of a century."

List also maintained that Germany was "called by nature to place herself at the head of the colonizing and civilizing nations," and "that the time had come for the formation of a Continental alliance against the naval supremacy of England." Treitschke, writing of the European situation in the later thirties, said:

"Against so absolutely ruthless a commercial policy, inciting and making mischief all over the world, all other civilized nations seemed natural allies. England was the stronghold of barbarism in international law. To England alone was it due that, to the shame of humanity, naval warfare still remained organized piracy. It was the common duty of all nations to restore on the seas that balance of power, long existing on the Continent, that healthy equipoise which permitted no State to do exactly as it liked, and consequently assured to all a humane international law. The civilization of the human race demanded that the manifold magnificence of the world's history, which had once commenced with the rule of monosyllabic Chinese, should not end in a vicious circle with the empire of the monosyllabic Britons. As soon as the Eastern Question was reopened a far-sighted statesmanship was bound to attempt at least to restrict the oppressive foreign rule which the English Fleet maintained from Gibraltar, Malta, and Corfu, and to restore the Mediterranean to the Mediterranean peoples."

At the same time the Professor was teaching his students at the Berlin University that "the settlement with England will be the most difficult of all," and that "the result of our next war must be, if possible, the acquisition of some colony."

The modern schoolmasters and professors of Germany have worked to produce a race inspired with the ambitions of List and the rancours of

Treitschke, and imbued with the idea that an unexampled destiny awaits their nation. That the Emperor William early recognized what schools and universities might be made to do in this direction is clear from the speech with which he opened the Educational Conference convened by him in 1890, and in which he complained that the traditional curriculum "lacked a patriotic basis." "We should," he exclaimed, "rear patriotic Germans and not young Greeks and Romans." It was also with a political purpose that he recommended a reversal of the usual order in which history was taught—that is to say, that the most recent periods should be taken first, and the student led back step by step to the events of antiquity.

While the Emperor is not omnipotent in legislation, he is, in Prussia, at any rate, practically unfettered in administration—that more extensive and equally important branch of government—and so the impulsions of his will can be forced down through the reticulations of the bureaucratic system till they are felt by the humblest official. He thus has at his disposal a large body of zealous co-operators anxious to comply with his desires even if they should have no direct relation to their official duties.

To appreciate the operation of this force, it is only necessary to turn over the pages of the German Navy League Handbook and notice how prominent a part the provincial agents of the central authority and subordinate members of the official body have played in the propaganda of that organization. It is impossible to avoid the conclusion that, wherever difficulty has been experienced in forming a local branch of the League, gentle pressure has been brought to bear on the stationmaster, postmaster, or gymnasium-director of the town, and has compelled him to take the initiative. In numerous cases such persons have, of course, come forward and founded branches of the League without any prompting, knowing well that their zeal would be in accordance with the "wishes of the Emperor," and would be rewarded by preferment when a suitable opportunity arose.

The Navy League is the only instrument the Emperor possesses for systematically and persistently propagating his ideas on world-policy and sea-power among the German people as a whole. It was founded in 1898, at his personal instance, but in all probability at Admiral von Tirpitz's suggestion, with the assistance of funds principally furnished by the Krupp family, which, as the chief material beneficiary from any increase in the German Fleet, could well afford to invest a little money in this way. Even in

Bismarck's time the head of the Krupp firm had been induced to start a number of newspapers to advocate the augmentation of those armaments from which he had derived a considerable proportion of his vast wealth, and it is one of the least edifying features of modern Germany that those of its citizens who show the most bellicose spirit have a direct personal interest in the waging of war. The financial founders of the Navy League included other prosperous manufacturers who were anxious to deserve decorations or titles, and who, in some instances, went so far as to compel their employees to join the organization and so help to swell its membership.

Three weeks before the League was constituted, the first Navy Bill had already received the Emperor's signature, and the order of these events is a plain demonstration that even then the measure was intended to be merely the thin end of the wedge. It is an interesting and significant fact that almost all the ruling houses of Germany have been induced to identify themselves with the League, though it is nominally an absolutely independent and unofficial organization. The Emperor's brother, Prince Henry of Prussia, has assumed the general protectorate, and among the protectors of the affiliated State federations are Prince George of Bavaria, the Kings of Saxony and Württemberg, the Grand Dukes of Baden, Hesse, the two Mecklenburgs, Oldenburg, and Saxe-Weimar, the Dukes of Anhalt, Saxe-Altenburg, and Saxe-Coburg-Gotha, the Princes of the two Lippes, Waldeck-Pyrmont, and the two Reusses, the Statthalter of Alsace-Lorraine, the Regent of Brunswick, and the Burgomasters of Hamburg and Bremen. Thus the State governments have a direct interest in the League, are under a moral obligation to promote its work, and, it may be added, bear a certain amount of responsibility for the manner in which its agitation is carried on. The purposes of the organization are defined in the statutes as follows:

"The German Navy League regards a strong German Fleet as necessary—principally in order to ensure the sea frontiers of Germany against the danger of war, to maintain the position of Germany among the Great Powers of the world, and to support the general interests and commercial communications of Germany as well as the safety of her citizens at work in oversea countries. Accordingly, it is the aim of the German Navy League to awaken, cultivate, and strengthen the interest of the German people for the importance and functions of the fleet."

The members of the League are divided into two classes—"individual" and "corporative." The latter are members of branches of other societies which enrol themselves in the League *en masse*. The most fruitful sources of support of this kind are those kindred bodies, the Pangerman Federation and the Colonial Association. On December 31st, 1911, the corporative members numbered 756,000, the individual members 298,000. The qualifications for individual membership are the attainment of the sixteenth year and a money contribution, which, if not fixed by the branch, is left for the member to determine for him or herself. The pecuniary contribution of a corporation joining the League is fixed by special arrangement in each case. From the accounts published it would appear that the average annual member's subscription falls a good deal short of sixpence. A considerable number of the members are young persons of both sexes who send in their names because it is a cheap and easy method of gratifying the association instinct, so strong in Germans, or for the sake of the dances and other purely social entertainments which are arranged by the branches.

A monthly paper, *Die Flotte*, which is published in an edition of 350,000 copies, is the League's chief organ in the Press, but the Central Office also issues immense quantities of pamphlets and leaflets. These are largely distributed with newspapers owned or controlled by the iron and steel and shipbuilding industries—what the Socialists call the "armour-plate Press"—but naturally find their way to all quarters to which Government influence can give them access. Under the name of "Communications," items of naval news and controversial paragraphs are sent out about once a week to all the papers, and though little notice is taken of them in the metropolitan Press, struggling provincial journals are very glad to have their columns filled up with topical matter by expert and authoritative pens. The League also publishes a profusely illustrated *Naval Album*, of which the Emperor every year buys 600 copies for distribution as prizes in the schools of Prussia—a typical example of the inter-action of the wheels of the naval agitation and the Government machine. Lecturing, too, occupies a prominent place in the League's activity, and the Central Office keeps a stock of magic-lanterns and slides, which it lends out free of charge to the local branches. It also supplies uniforms, badges, and bunting for local festivities.

By far the most effective department of the League's activity is, however, the excursions to the German naval ports, which it arranges for the benefit

of schoolmasters and their classes. The participants in these outings are, as far as possible, selected from the inland states and districts, in which it is most difficult to arouse enthusiasm for the sea and the fleet. They are taken to Kiel or Wilhelmshaven, received with effusive courtesy by the naval officers delegated to look after them, and escorted through the streets by a ship's band to the dockyards of war vessels, over which they are conducted by amiable guides, who supply them with all the information likely to stimulate their interest in what they have seen. If the distance they have travelled makes it impossible for them to return home the same day, naval barracks or storehouses which happen for the moment to be vacant are placed at their disposal as night quarters. So much official complaisance and amenity, especially in a country where neither of these qualities is particularly common in the public services, arouses in those on whom it is expended a flattering sense of their own and their national importance, and schoolmasters thus captivated naturally, in due time, convey their impressions to their pupils. Though the numbers of persons thus dealt with are inevitably somewhat limited, the League unquestionably gains more ground in this way than it can hope to win by pamphlets which are read and lectures which are listened to mainly by the already convinced.

The Emperor is the real director of the Navy League, and it puts forward no demand that has not already received his approval, in principle if not in detail. The League is, in short, little more than a Government department, the function of which is to carry on an agitation for more warships. It must, however, always be remembered that the League's demands represent not what the Government desires or expects to get, but what it wants to be asked for. In order that it may keep up the pretence that it is an unofficial and independent organization, the League must naturally avoid too close a correspondence between its own programme and that of the Ministry of Marine, and it is also guided by the principle that it is necessary to ask much in order to get little. Occasionally it makes a show of hurrying and worrying the Naval Minister, and of being positively objectionable to the Government, but no one suffers less than Admiral von Tirpitz from these "attacks" upon him.

APPENDIX I

GERMANY'S NAVAL POLICY

The key to the naval policy of Germany is to be found in the Memorandum which was appended to the Navy Act of 1900. It is the most illuminating of State documents and is of peculiar interest in view of the war at sea which opened on August 4th, 1914.

Only the more salient passages of this Memorandum need be recalled to illustrate how far the performances of the German Fleet have fallen short of the high hopes which were entertained for it.

In the opening passages of the Memorandum, it was explained why "the German Empire needs peace at sea":

For the German Empire of to-day the security of its economic development, and especially of its world-trade, is a life question. For this purpose the German Empire needs not only peace on land but also peace at sea—not, however, peace at any price, but peace with honour, which satisfies its just requirements.

A naval war for economic interests, particularly for commercial interests, will probably be of long duration, for the aim of a superior opponent will be all the more completely reached the longer the war lasts. To this must be added that a naval war which, after the destruction or shutting-up of the German sea fighting force, was confined to the blockade of the coasts and the capture of merchant ships, would cost the opponent little; indeed he would, on the contrary, amply cover the expenses of the war by the simultaneous improvement of his own trade.

An unsuccessful naval war of the duration of even only a year would destroy Germany's sea trade, and would thereby bring about the most disastrous conditions, first in her economic, and then, as an immediate consequence of that, in her social life.

Quite apart from the consequences of the possible peace conditions, the destruction of our sea trade during the war could not, even at the close of it, be made good within measurable time, and would thus add to the sacrifices of the war a serious economic depression.

The Memorandum then proceeded to justify the abandonment of the Navy Law passed as recently as 1898:

The Navy Law (of 1898) does not make allowance for the possibility of a naval war with a great naval Power, because, when it was drafted in the summer of 1897, the first consideration was to secure the carrying out in modern ship material of the 1873 plan for the founding of the fleet, limiting the increase to the small number of battleships which was necessary to establish, at least for a double squadron, the organization demanded by tactical exigencies.

The Justificatory Memorandum to the Navy Law (of 1898) left no doubt as to the military significance of the Battle Fleet. It is therein expressly stated:

"Against greater sea-powers the Battle Fleet would have importance merely as a sortie fleet."

That is to say, the fleet would have to withdraw into the harbour and there wait for a favourable opportunity for making a sortie. Even if it should obtain a success in such a sortie, it would nevertheless, like the enemy, suffer considerable loss of ships. The stronger enemy could make good his losses, we could not. In war with a substantially superior sea-power, the Battle Fleet provided for by the Navy Law would render a blockade more difficult, especially in the first phase of the war, but would never be able to prevent it. To subdue it, or, after it had been considerably weakened, to confine it in its own harbour would always be merely a question of time. So soon as this had happened, no great State could be more easily cut off than Germany from all sea intercourse worthy of the name—of her own ships as also of the ships of neutral Powers. To effect this it would not be necessary to control long stretches of coast, but merely to blockade the few big seaports.

In the same way as the traffic to the home ports, the German mercantile ships on all the seas of the world would be left to the mercy of an enemy who was more powerful on the sea. Hostile cruisers on the main trade-routes, in the Skager-Rack, in the English Channel, off the north of Scotland, in the Straits of Gibraltar, at the entrance to the Suez Canal, and at the Cape of Good Hope, would render German shipping practically impossible.

Also with regard to this the Justificatory Memorandum to the Naval Law (of 1898) speaks unambiguously. In it is observed:

"Protection of sea trade on all the seas would occur principally in time of peace. In case of war it would be the task of the foreign service cruisers to afford their own mercantile ships the 'utmost possible protection.'"

That is to say, the ships would do the "utmost possible." What would be possible in this respect is clear when it is realized that the Navy Law provides altogether for forty-two cruisers, whilst the greatest Naval Power, for example, to-day already possesses 206 cruisers (finished or under construction), and, moreover, has at its disposal bases and coaling stations on all the chief trade-routes.

To protect Germany's sea trade and colonies in the existing circumstances there is only one means—Germany must have a battle fleet so strong that even for the adversary with the greatest sea-power a war against it would involve such dangers as to imperil his position in the world.

For this purpose it is not absolutely necessary that the German Battle Fleet should be as strong as that of the greatest naval Power, for a great naval Power will not, as a rule, be in a position to concentrate all its striking forces against us. But even if it should succeed in meeting us with considerable superiority of strength, the defeat of a strong German Fleet would so substantially weaken the enemy that, in spite of the victory he might have obtained, his own position in the world would no longer be secured by an adequate fleet.

In order to attain the goal which has been set, the protection of our sea trade and of our colonies by ensuring a peace with honour, Germany requires, according to the standard of the strength-relationships of the great Sea-Powers, and having regard to our tactical formations, two double squadrons of efficient battleships, with the necessary cruisers, torpedo-boats, and so on, pertaining thereto. As the Navy Law (of 1898) provides for only two squadrons, the building of a third and fourth squadron is contemplated. Of these four squadrons two will form a fleet. The second fleet is to be organized in its tactical composition in the same way as the first fleet provided for in the Navy Law.

For the scope of the maintenance in commission in time of peace the following consideration has been decisive: As the ship-establishment of the German Navy, even after the carrying out of the projected increase, will still be more or less inferior to the ship-establishments of some other great Powers, compensation must be sought in the training of the personnel and in tactical training in the larger combinations.

A trustworthy training of the separate ships' crews, as well as an adequate training in the larger tactical combinations, can be ensured only by permanent maintenance in commission in time of peace. To economize in commissioning in time of peace would mean to jeopardize the efficiency of the fleet for the event of war.

The minimum of commissioning is the permanent formation of that fleet which comprises the newest and best ships as an active combination—that is to say, a combination in which all battleships and cruisers are in commission. This fleet would form the school for tactical training in double squadron, and in case of war would bear the first shock. For the second fleet, which will comprise the older battleships, it must suffice if only half of the ships are permanently in commission.^[19] For training in the larger combination some further ships must then, it is true, be placed temporarily in commission during the man[oe]uvres. In case of war this second fleet—the Reserve Battle Fleet—will have to make up its arrears in the training of the separate ships' crews and the deficiency of training in the larger combination behind the protection afforded by the Active Battle Fleet.

If Germany possesses four squadrons of efficient battleships, a coast squadron composed of small armoured ships is less important.

Besides the increase of the home Battle Fleet, an increase of the foreign service ships is also necessary. In consequence of the occupation of Kiauchow and the great enhancement of our oversea interests in the last two years, it has already become necessary, at the cost of the scouting ships of the Battle Fleet, to send abroad two large ships more than were provided for by the plan of the Navy Law. Indeed, for an effective representation of our interests it would have been necessary to send out even more ships, if such had only been available. In order to form a judgment of the importance of an increase of the foreign service ships, it must be realized that they are the representatives abroad of the German defence forces, and that the task often

falls to them of gathering in the fruits which the maritime potency created for the Empire by the home Battle Fleet has permitted to ripen.

Moreover, an adequate representation on the spot, supported on a strong home Battle Fleet, in many cases averts differences, and so contributes to maintain peace while fully upholding German honour and German interests.

A numerical demonstration of the additional requirements cannot be given for a considerable time in advance in the same manner as for the Battle Fleet, which rests upon an organic foundation.

If the demand is made that the foreign service fleet shall be in a position (1) energetically to uphold German interests everywhere in time of peace, (2) to be adequate for warlike conflicts with oversea States without navies deserving of the name, an increase of at least five large and five small cruisers, as well as of one large and two small cruisers as material reserve, seems called for. The Navy Law foresees as ready for use three large and ten small cruisers, and as material reserve three large and four small cruisers.

A distribution of the foreign service fleet among the foreign stations cannot be given, as this distribution depends upon the political circumstances, and these can only be estimated from case to case.

FOOTNOTES:

[19] This principle was abandoned under the Law of 1912, and a standard of greater instant readiness for war was substituted, with three squadrons fully manned and two with nucleus crews.

APPENDIX II

BRITISH AND GERMAN SHIPBUILDING PROGRAMMES.

The following table shows the British and German ships laid down between 1897 and 1914 and the programmes of subsequent years—the British figures for 1915-18 being based on the Admiralty forecast, and the German on the latest German Fleet Law:

		Great Britain.					Germany.				
		Battle ships.	Armoured Cruisers.	Protected Cruisers.	Destroyers.	Torpedo Boats.	Battle ships.	Armoured Cruisers.	Protected Cruisers.	Destroyers	
Mixed armament period	{1897-1898	4	4	3	6	—	1	—	—	—	
	{1898-1899	7	8	1	12	—	2	1	2	6	
	{1899-1900	2	2	1	—	2	3	—	2	6	
	{1900-1901	2	6	1	5	2	2	—	2	6	
	{1901-1902	3	6	2	10	5	2	1	3	6	
	{1902-1903	2	2	6 ^[20]	9	4	2	1	3	6	
	{1903-1904	5	4	4 ^[20]	15	—	2	1	2	6	
	{1904-1905	2	3	—	—	—	2	1	3	6	
	{1905-1906	—	—	—	—	—	2	1	3	6	
	Totals	27	35	18	57	13	18	6	20	48	
Dreadnought period	{1905-1906	4	—	—	6	12 ^[21]	—	—	—	—	
	{1906-1907	3	—	—	2	12 ^[21]	2	1	2	12	
	{1907-1908	3	—	1	5	12 ^[21]	3	—	2	12	
	{1908-1909	2	—	6	16	—	4	—	2	12	
	{1909-1910	8	—	6	20	—	4	—	2	12	
	{1910-1911	5	—	5	20	—	4	—	2	12	
	{1911-1912	5	—	4	20	—	4	—	2	12	
	{1912-1913	4	—	8 ^[22]	20	—	2	—	2	12	
	{1913-1914	5	—	8	16	—	3	—	2	12	
	{1914-1915	4	—	4	12	—	2	—	2	12	
	{1915-1916	4	—	— ^[23]	— ^[23]	— ^[23]	2	—	2	12	
	{1916-1917	4	—	— ^[23]	— ^[23]	— ^[23]	3	—	2	12	
	{1917-1918	4	—	— ^[23]	— ^[23]	— ^[23]	2	—	2	12	
		Totals authorised (1905-1914) (Dreadnought period)	43	—	42	137	36	28	1	18	108

As is explained elsewhere, Germany has remained faithful to the policy outlined in the Memorandum, but by successive Navy Acts she greatly increased the means for giving effect to it—one legislative measure succeeding another in quick succession, always making an increase in the naval establishment.

FOOTNOTES:

[20] Included in these two figures are eight scouts—small cruisers—which were laid down in 1902 and 1903.

[21] The cruisers of 1912-14 were designated "light armoured cruisers."

[22] These thirty-six craft are small destroyers, and were built as such.

[23] No programme of British cruisers or torpedo craft announced.

The forty-three British battleships exclude the two Colonial vessels—*Australia* and *New Zealand*—and the battleship given by the Federated Malay States, and ordered early in 1913. With these the number of Dreadnought vessels is increased to forty-six.

END.

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